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AWS Compute Cloud and Serverless

AWS Cloud Practitioner

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Outline

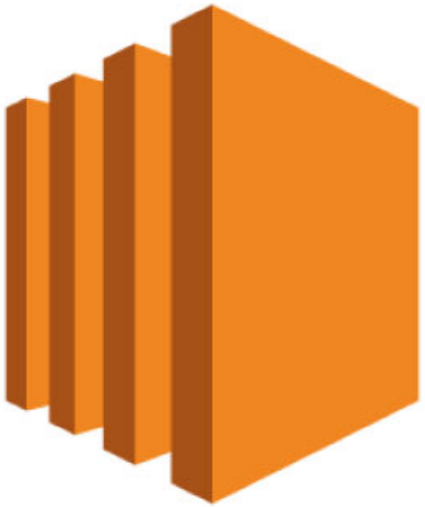
- AWS EC2
- Scalability & High Availability
- AWS Serverless
- Questions

Compute Cloud

Compute Cloud

❖ Getting Started

Google Compute Engine

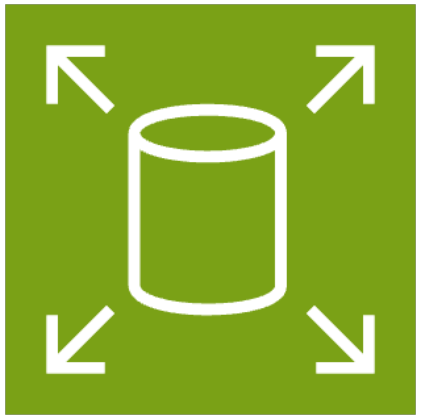


amazon EC2

Compute Cloud

❖ Amazon Elastic Compute Cloud

- EC2 is one of the most widely used services in AWS.
- **Amazon EC2** provides scalable **virtual servers in the cloud** (Infrastructure as a Service – IaaS).
- It mainly offers the ability to:
  - Launch and manage virtual machines (EC2 Instances)
  - Attach persistent block storage (Amazon EBS)
  - Distribute traffic using Elastic Load Balancing (ELB)
  - Automatically scale resources using Auto Scaling Groups (ASG)

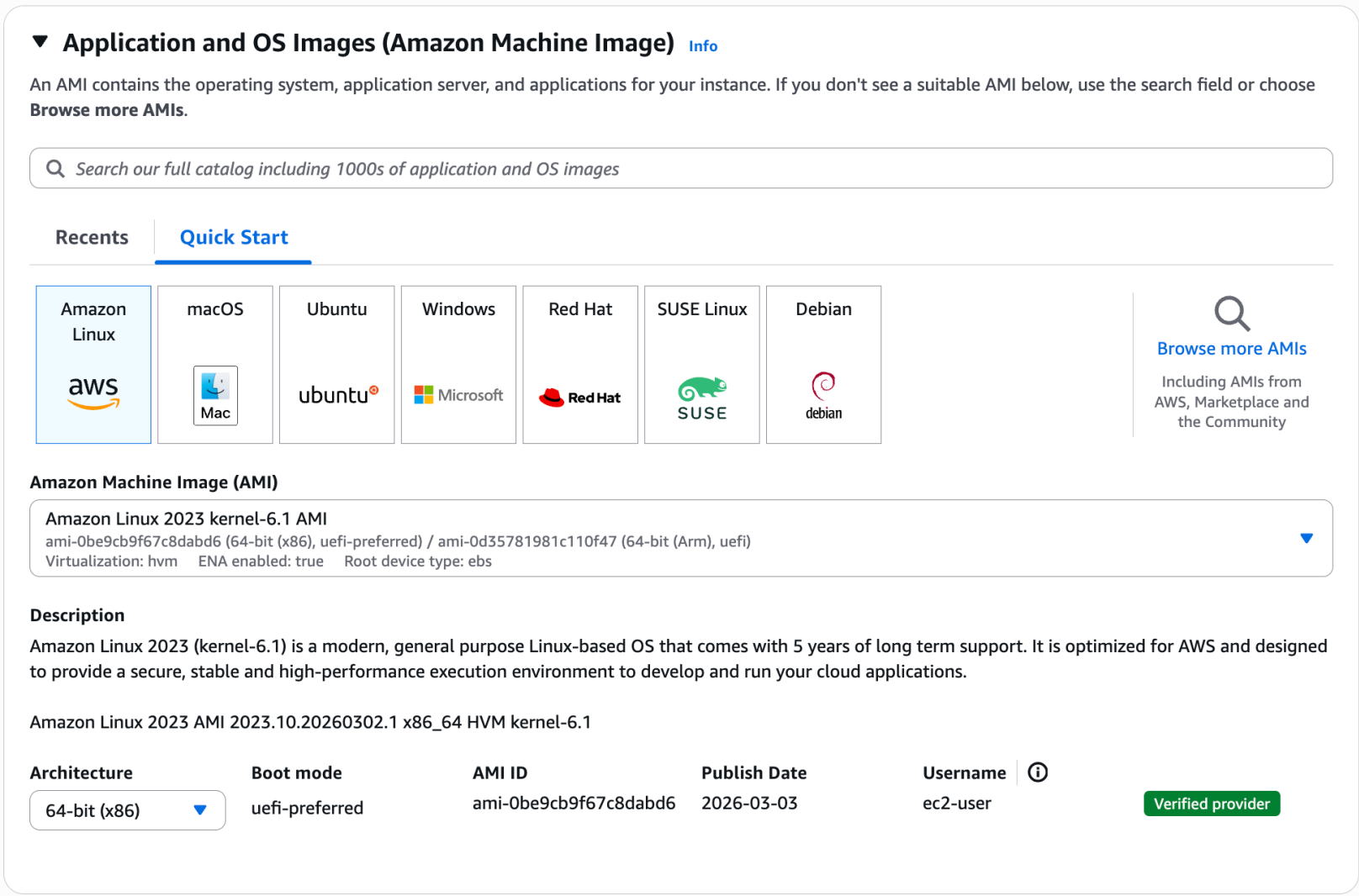


Ref: https://aws.amazon.com/ec2

Compute Cloud

❖ Amazon Machine Image (AMI)

AMI (Amazon Machine Image) is a **preconfigured template** used to launch EC2 instances.



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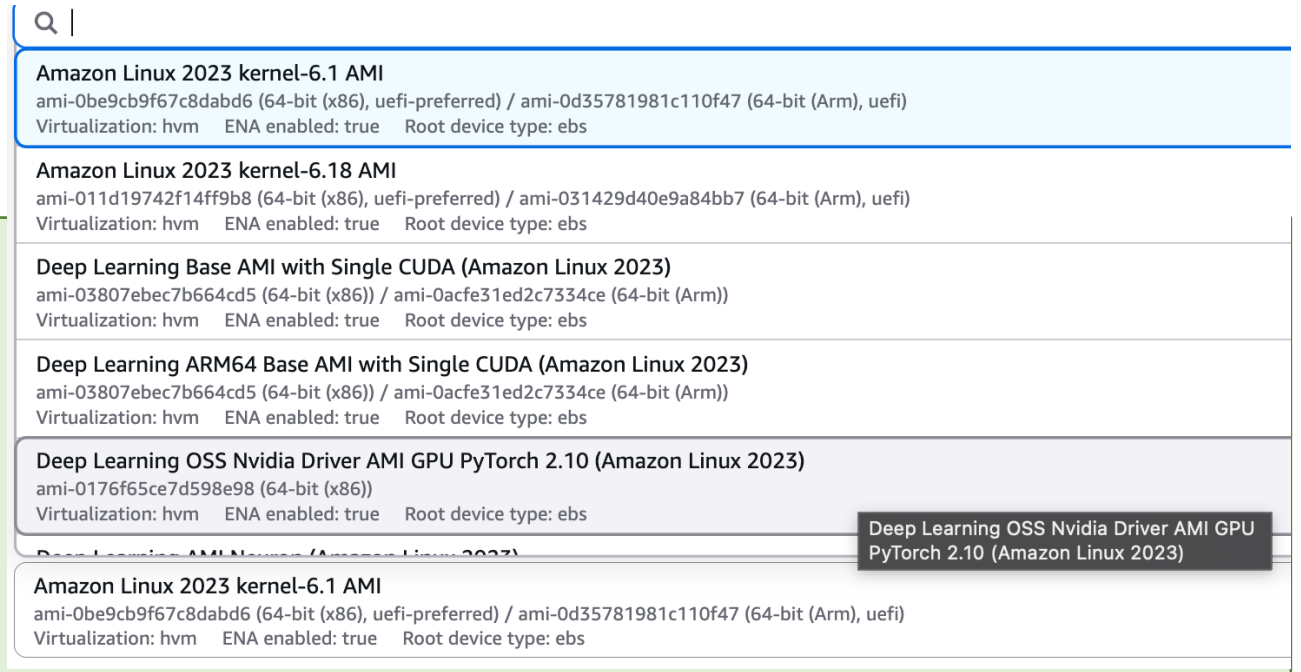
❖ Amazon Machine Image (AMI)



An AMI includes:

- Operating system
- Installed software
- Application configuration
- Monitoring or security settings

Using an AMI enables **faster instance launch and setup**, since the required software is **pre-packaged**.

Regional scope

AMIs are **created for a specific AWS region**, but they can be **copied to other regions** if needed.

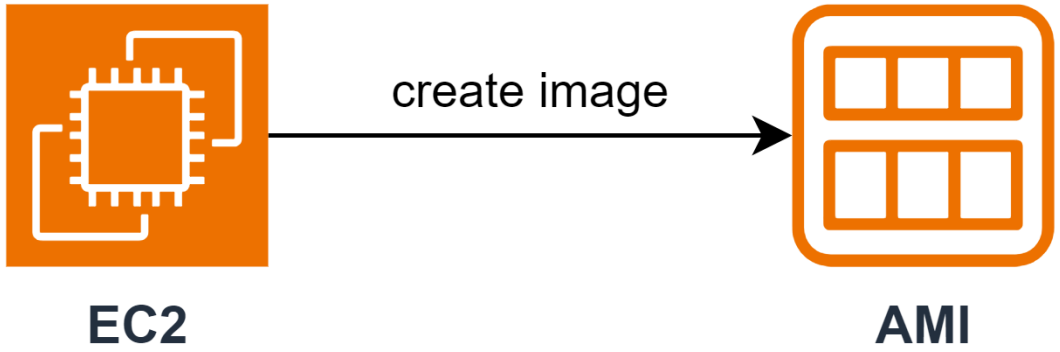
Public AMIs – provided by AWS or the community

Private AMIs – created and maintained **by your**

AWS Marketplace AMIs – provided by **third-party vendors**

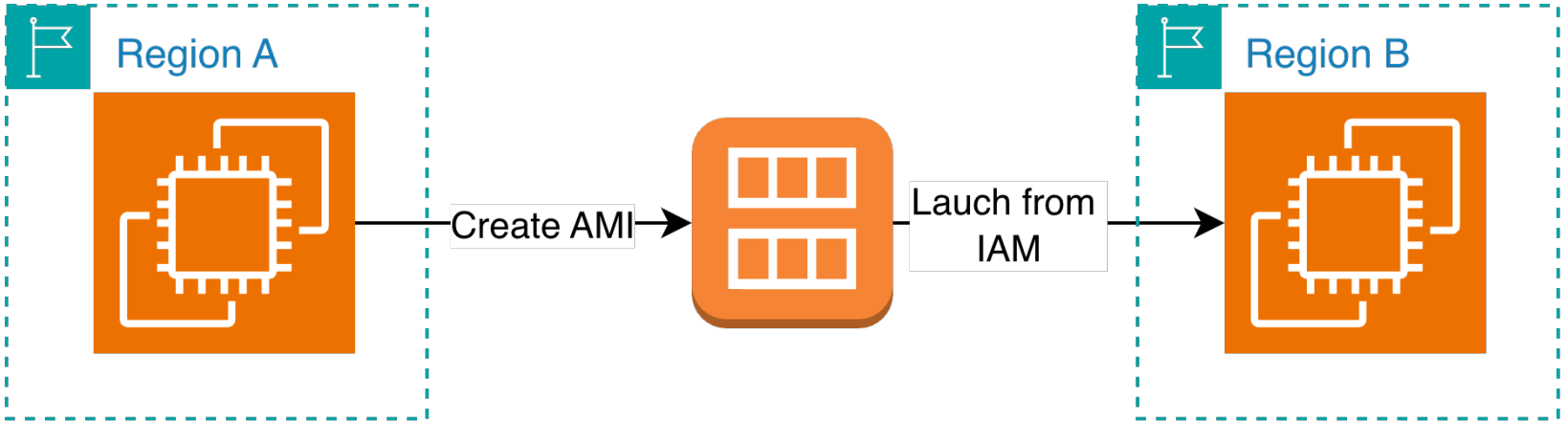
Compute Cloud

❖ Amazon Machine Image (AMI)



1. Launch an EC2 instance and configure it with the required OS, software, and settings.
2. Stop the instance to ensure data integrity before creating the image.
3. Create an AMI from the instance

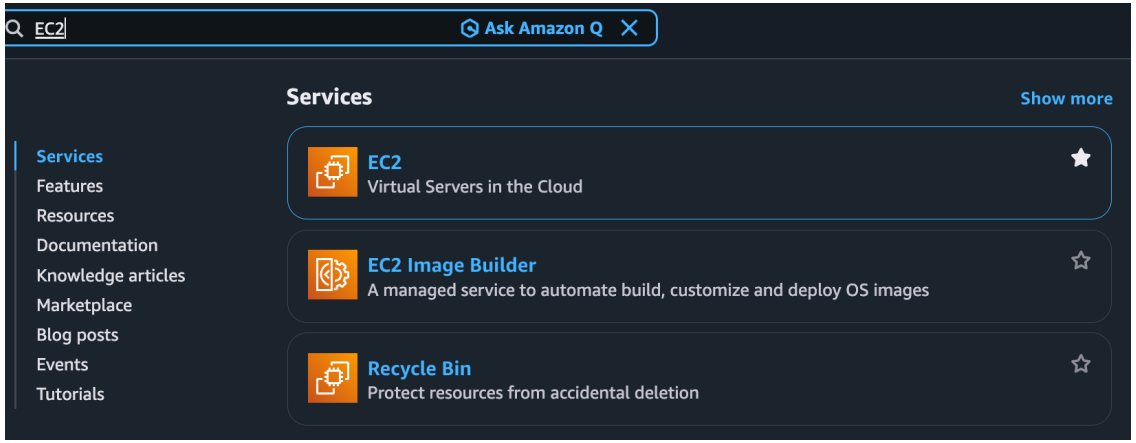
- Deploy the same application infrastructure in multiple regions
- Support disaster recovery (DR)
- Reduce latency for users in different geographic locations



Compute Cloud

❖ EC2 – Sizing & Configuration

- Operating System (OS):
  - ➢ Choose the OS: **Linux, Windows, or macOS**.
- Compute Resources (CPU):
  - ➢ Determines the number of **vCPUs**
- Memory (RAM):
  - ➢ Determines how much **RAM** is needed



- Storage Options:
  - ➢ Network-attached storage: **Amazon EBS, EFS**
  - ➢ **Instance storage**
- Networking:
  - ➢ Subnet
  - ➢ Option to assign a public IPv4 address or Elastic IP
  - ➢ Firewall **(Security Groups)**
- Instance Initialization (User Data):
  - ➢ Configure at first launch

Compute Cloud

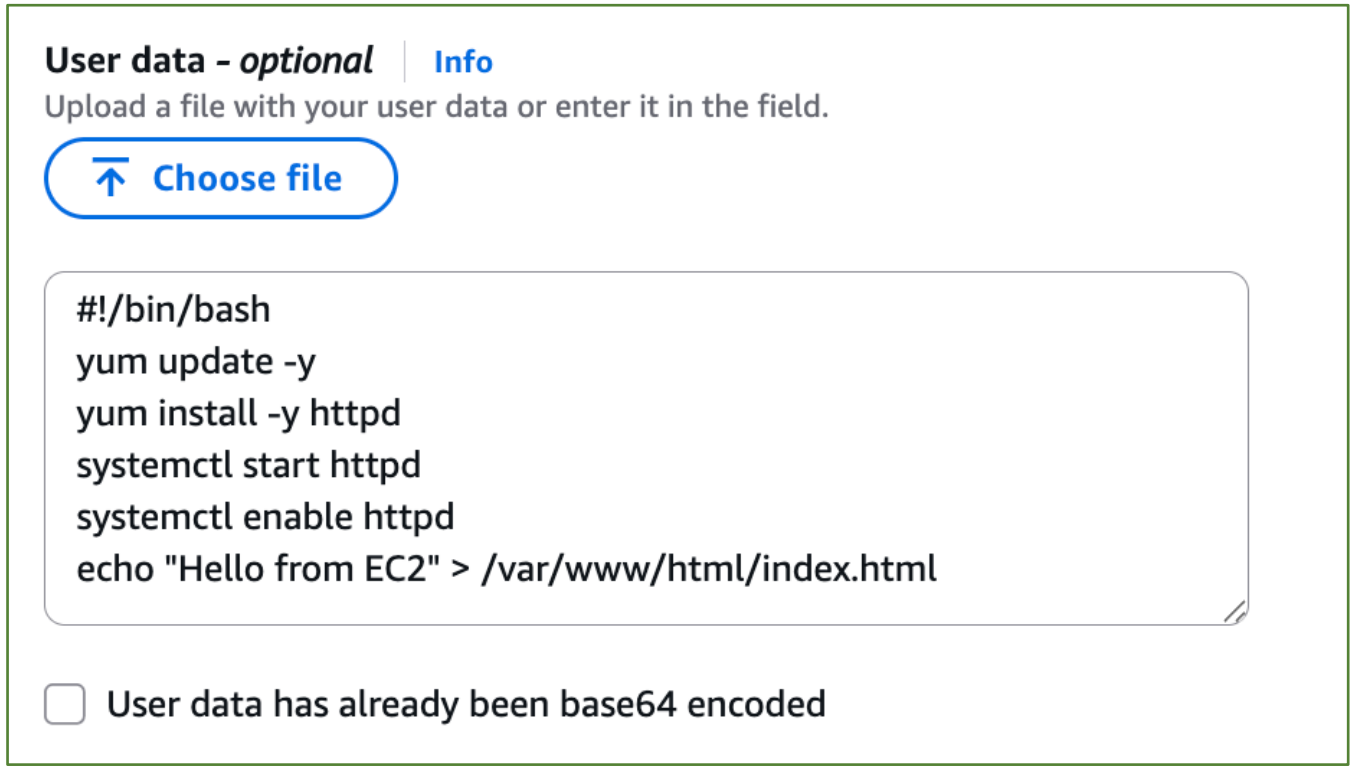
❖ EC2 - User Data

- **EC2 User Data** allows you to **bootstrap** an EC2 instance by running a script automatically when the instance starts.

Common use cases:

- Installing system updates
- Installing software packages
- Downloading files or application code from the internet
- Configuring services or environment settings
- Running any startup automation tasks

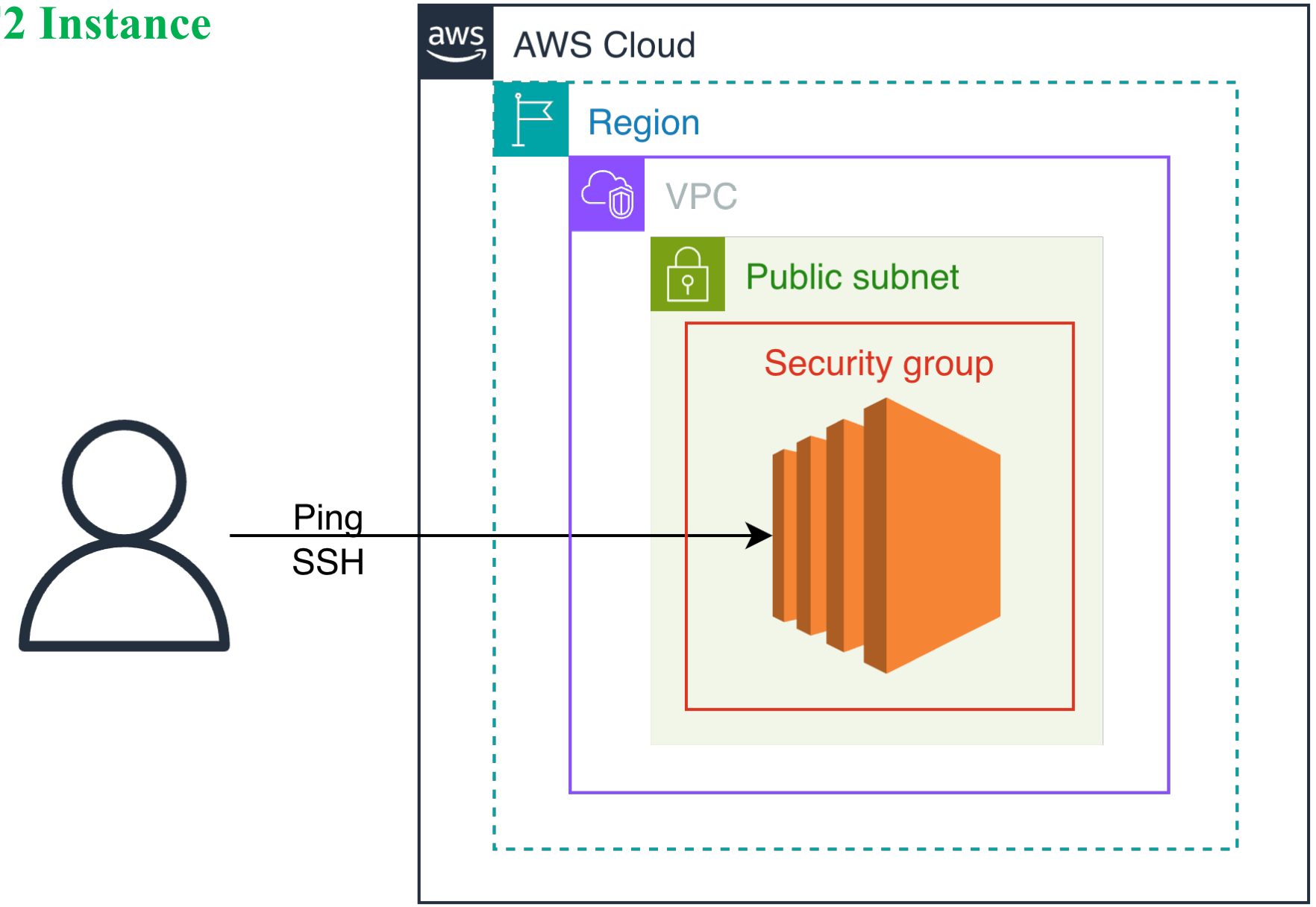
Bootstrapping means executing commands automatically during the **initial startup of a machine**.



Lab Session

Lab Session

❖ Create a EC2 Instance



Instance Types

Instance Types

❖ Getting Started

- Amazon EC2 provides many instance types optimized for **different workloads**.
- Each instance type offers a different balance of **CPU, memory, storage, and networking performance**.

General Purpose

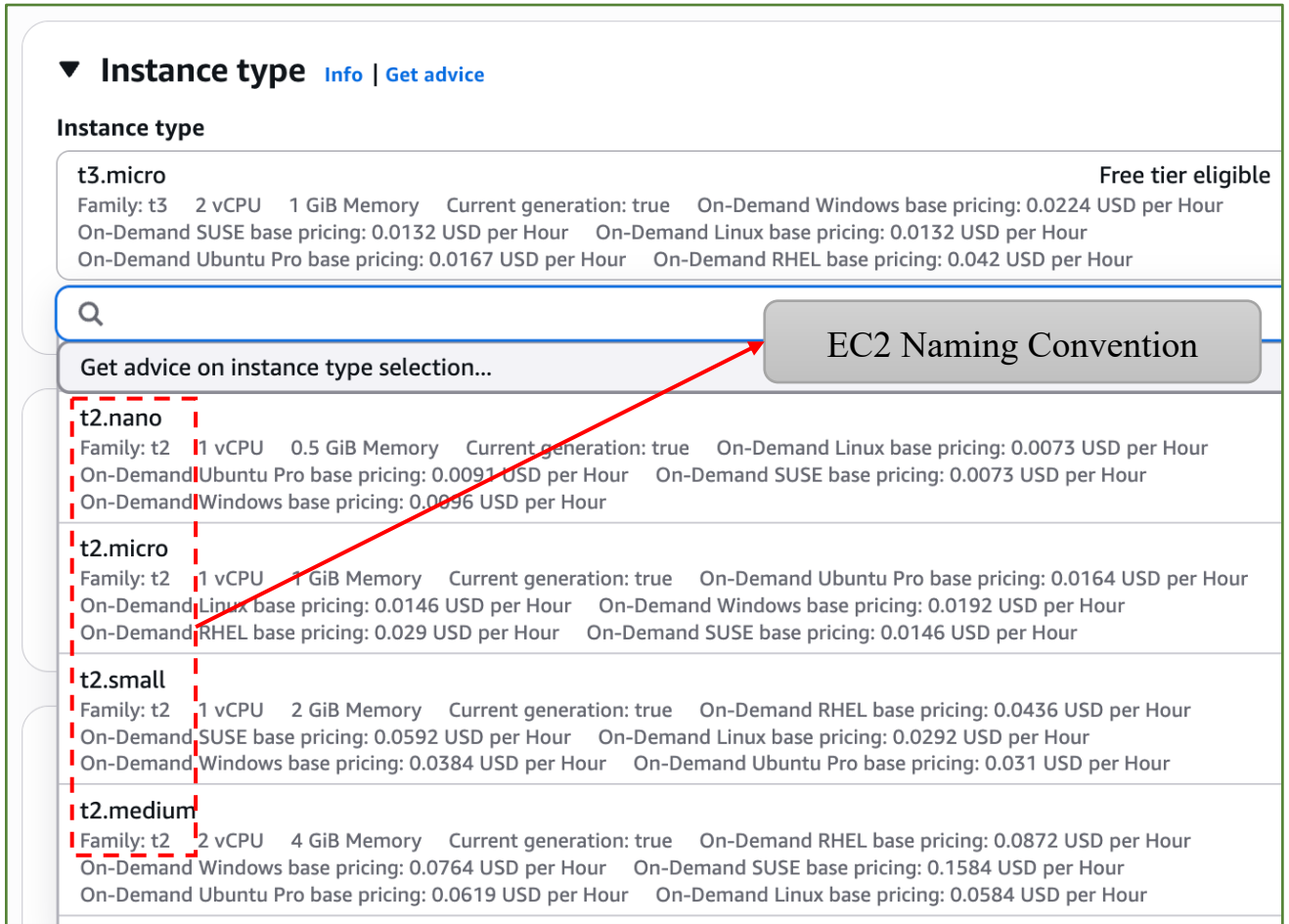
Compute Optimized

Accelerated Computing

Memory Optimized

HPC Optimized

Storage Optimized



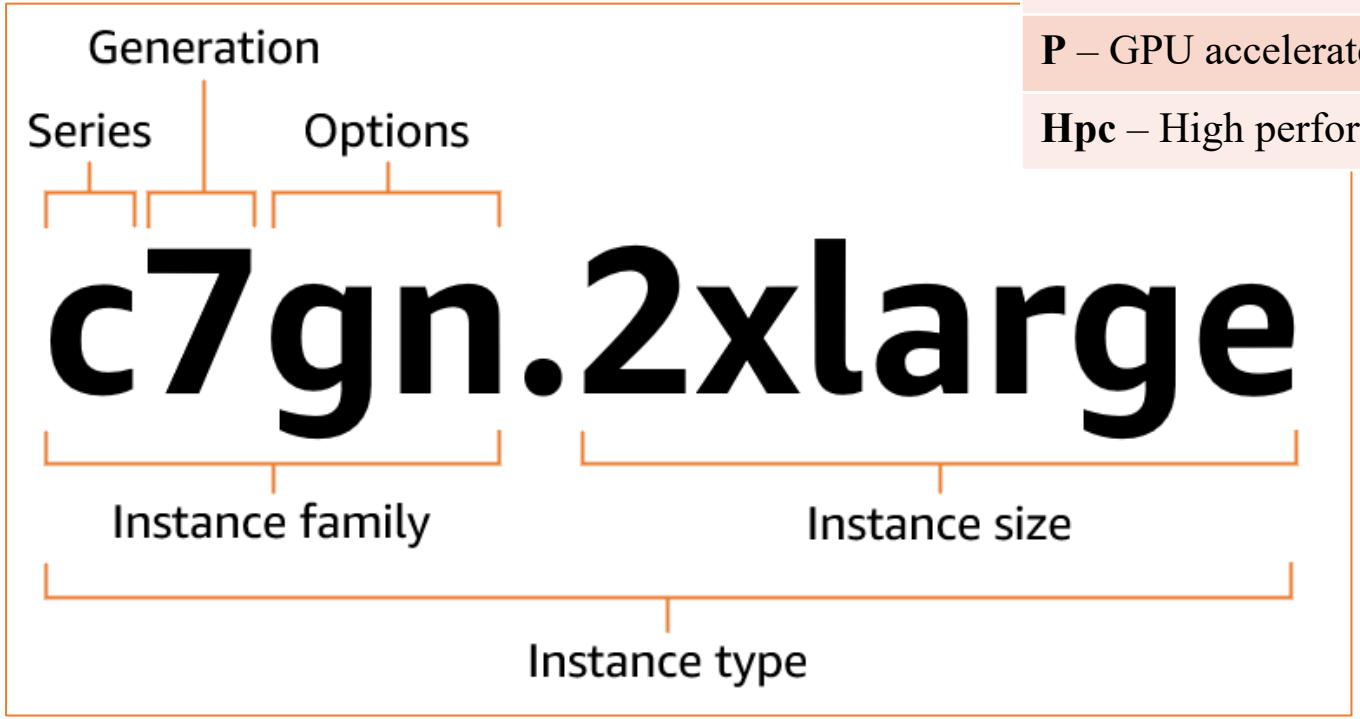
Ref: https://aws.amazon.com/ec2/instance-types

Instance Types

❖ Naming Convention

Amazon EC2 provides many instance types optimized for **different workloads**.

Series	Options
M – General purpose	**a** – AMD processors
C – Compute optimized	**b** – Block storage optimization
D – Dense storage	**d** – Instance store volumes
I – Storage optimized	**n** – Network and EBS optimized
P – GPU accelerated	**z** – High CPU frequency
Hpc – High performance computing	**m*** | **m*pro**– Apple chip



Each instance type offers a different balance of **CPU, memory, storage, and networking performance**.

Ref: https://docs.aws.amazon.com/ec2/latest/instancetypes/instance-type-names.html

Instance Types

❖ General Purpose

General purpose instances provide a balanced combination of:

- Compute (CPU)
- Memory (RAM)
- Networking performance

Typical use cases:

- Web servers and application servers
- Development environments
- Small to medium databases
- Microservices and containerized applications

Instance size	vCPU	Memory (GiB)	Instance storage (GB)	Network bandwidth (Gbps)	Amazon EBS bandwidth (Gbps)
m8gn.medium	1	4	EBS-only	Up to 25	Up to 10
m8gn.large	2	8	EBS-only	Up to 30	Up to 10
m8gn.xlarge	4	16	EBS-only	Up to 40	Up to 10

Ref: https://aws.amazon.com/ec2/instance-types/general-purpose

Instance Types

❖ Compute Optimized

Compute optimized instances are designed for compute-bound workloads that require **high-performance processors** and strong CPU performance.

Typical use cases:

- ➢ Compute-intensive applications
- ➢ Batch processing workloads
- ➢ High-performance web servers
- ➢ Machine learning workloads

Instance size	vCPU	Memory (GiB)	Instance storage (GB)	Network bandwidth (Gbps)	Amazon EBS bandwidth (Gbps)
c8g.medium	1	2	EBS-only	Up to 12.5	Up to 10
c8g.large	2	4	EBS-only	Up to 12.5	Up to 10
c8g.xlarge	4	8	EBS-only	Up to 12.5	Up to 10
c8g.2xlarge	8	16	EBS-only	Up to 15	Up to 10

Ref: https://aws.amazon.com/ec2/instance-types/compute-optimized

Instance Types

❖ Memory Optimized

Memory optimized instances are designed for workloads that require **large amounts of RAM** and **high memory performance**.

Typical use cases:

- ➢ Relational and non-relational databases
- ➢ Distributed caching systems (in-memory cache)
- ➢ Real-time analytics and big data processing

Instance size	vCPU	Memory (GiB)	Instance storage (GB)	Network bandwidth (Gbps)	EBS bandwidth (Gbps)
r8g.medium	1	8	EBS-only	Up to 12.5	Up to 10
r8g.large	2	16	EBS-only	Up to 12.5	Up to 10
r8g.xlarge	4	32	EBS-only	Up to 12.5	Up to 10
r8g.2xlarge	8	64	EBS-only	Up to 15	Up to 10

Ref: https://aws.amazon.com/ec2/instance-types/memory-optimized

Instance Types

❖ Storage Optimized

Storage optimized instances are designed for workloads that require **high, sequential read and write performance** on **large datasets stored on local instance storage**.

Typical use cases:

- Relational and NoSQL databases
- Data warehousing workloads
- Big data processing applications
- Distributed file systems

Instance Size	vCPU	Memory (GiB)	Instance Storage (GB)	Network Bandwidth (Gbps)	EBS Bandwidth (Gbps)
i8g.large	2	16	1 x 468 GB = 468 GB	Up to 10	Up to 10
i8g.xlarge	4	32	1 x 937 GB = 937 GB	Up to 10	Up to 10
i8g.2xlarge	8	64	1 x 1,875 GB = 1,875 GB	Up to 12	Up to 10
i8g.4xlarge	16	128	1 x 3,750 GB = 3,750 GB	Up to 25	Up to 10
i8g.8xlarge	32	256	2 x 3,750 GB = 7,500 GB	25	10

Ref: https://aws.amazon.com/ec2/instance-types/storage-optimized

Instance Types

❖ Accelerated Computing

- **Accelerated computing instances** use specialized **hardware accelerators** to perform certain tasks more efficiently than standard CPUs.
- These accelerators significantly improve performance for workloads that require **parallel processing or specialized computation**.

Typical use cases:

- ➢ Machine learning training and inference
- ➢ Video rendering and media processing
- ➢ Graphics and visualization workloads

P6e - Instance

GPU Memory (GB)	Network Bandwidth	GPUDirect RDMA	GPU Peer to Peer	Instance Storage (TB)	EBS Bandwidth (Gbps)
740	1600	Yes	1800	22.5	60

Ref: https://aws.amazon.com/ec2/instance-types/accelerated-computing

Instance Types

❖ HPC Optimized

HPC (High Performance Computing) optimized instances are designed for workloads that require **very high computational performance at large scale**.

Typical use cases:

- ➢ Engineering and computational modeling
- ➢ Financial risk modeling and analytics
- ➢ Visual effects and rendering workloads

Hpc8a

Instance size	Physical cores	Memory (GiB)	Instance Storage	EFA bandwidth (Gbps)	Network bandwidth (Gbps)
hpc8a.96xlarge	192	768	EBS only	300	75

Ref: https://aws.amazon.com/ec2/instance-types/hpc-optimized

Compute Cloud

❖ EC2 - Private vs Public IP (IPv4)

Private IP (IPv4)

- Assigned to instances inside a VPC.
- ***Not reachable*** from the public Internet.
- Used for communication between resources within the same VPC, such as EC2 instances, databases, or internal services.
- Private IP addresses remain associated with the instance during its lifetime.

Public IP (IPv4)

- ***Reachable*** from the Internet.
- Used for communication between an EC2 instance and external networks.
- A public IP is typically assigned automatically at launch.
- When an instance is stopped and started, the public IP may change.

Compute Cloud

❖ EC2 - Private vs Public IP (IPv4)

Instance summary for i-026ca405ae0c762e8 (d-clf-slide) Info

Updated 1 minute ago

Connect | Instance state | Actions

Instance ID
i-026ca405ae0c762e8

IPv6 address
–

Hostname type
IP name: ip-172-31-21-170.ap-southeast-1.compute.internal

Answer private resource DNS name
IPv4 (A)

Auto-assigned IP address
18.136.120.129 [Public IP]

IAM role
–

IMDSv2
Required

Operator
–

Public IPv4 address
18.136.120.129 | open address

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-31-21-170.ap-southeast-1.compute.internal

Instance type
t3.micro

VPC ID
vpc-01c8c9369d9da8c06

Subnet ID
subnet-06c1fffc1e9fd0bf9

Instance ARN
arn:aws:ec2:ap-southeast-1:882465234668:instance/i-026ca405ae0c762e8

Private IPv4 addresses
172.31.21.170

Public DNS
ec2-18-136-120-129.ap-southeast-1.compute.amazonaws.com | open address

Elastic IP addresses
–

AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name
–

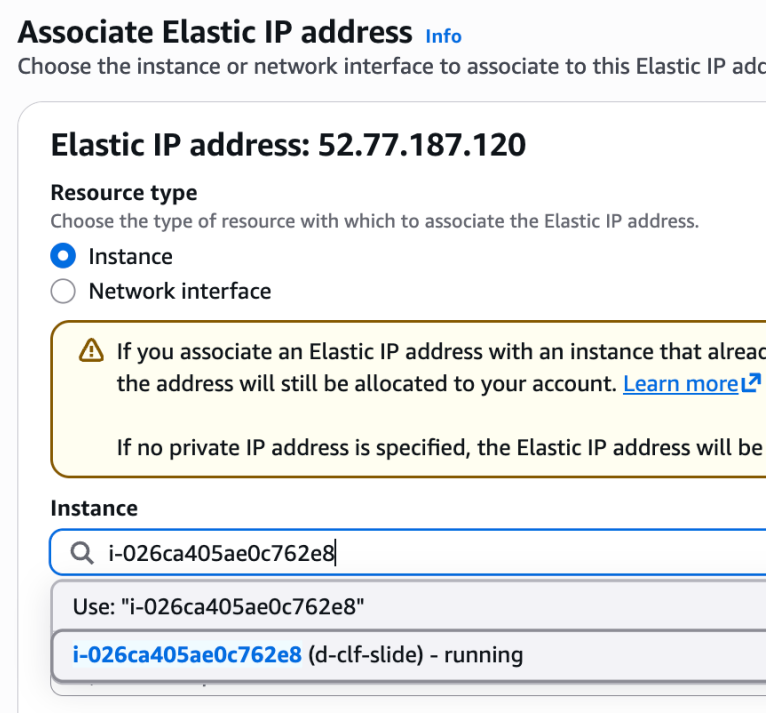
Managed
false

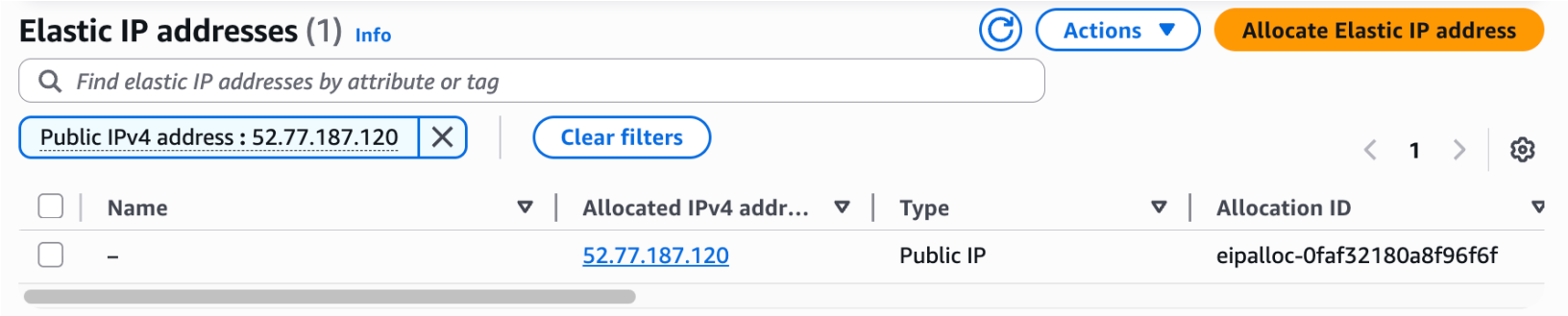
Compute Cloud

❖ EC2 - Elastic IPs

Elastic IP (EIP)

- A static public IPv4 address allocated from AWS.
- Can be associated with an EC2 instance and remains the same even if the instance is stopped or restarted.
- Useful for applications that require a fixed public IP address.
- Can be remapped to another instance if needed

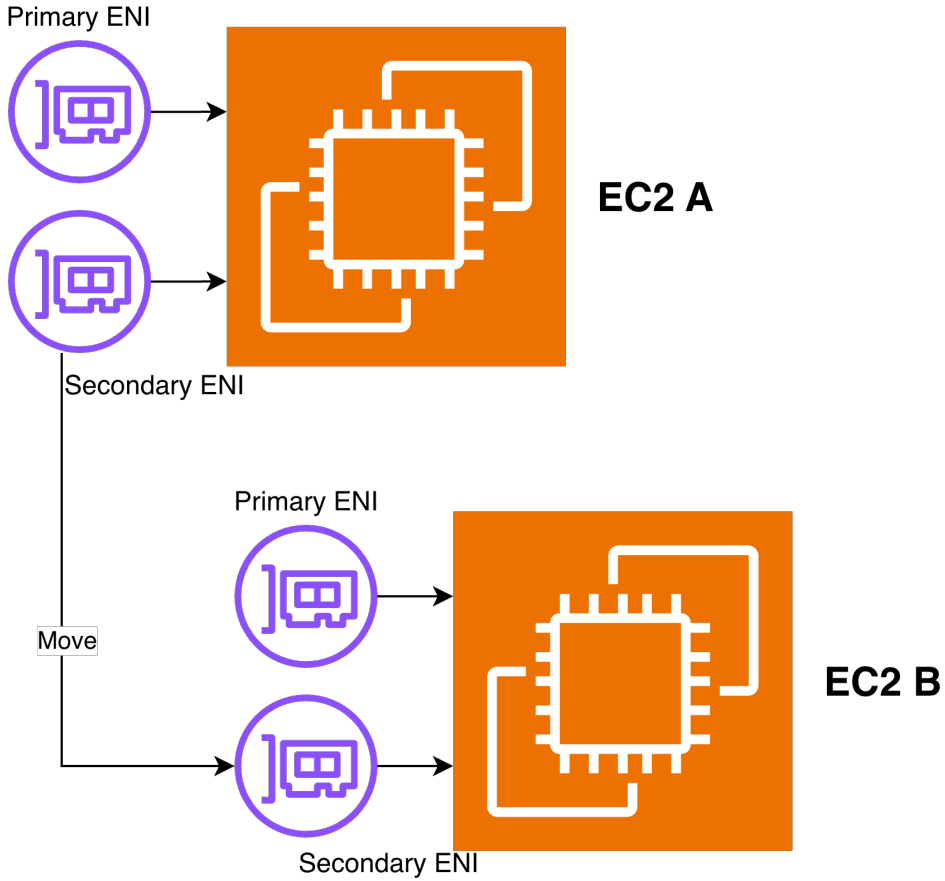




Compute Cloud

❖ EC2 - Elastic Network Interfaces (ENI)

An Elastic Network Interface (ENI) is a virtual network card that can be attached to an EC2 instance within a VPC.



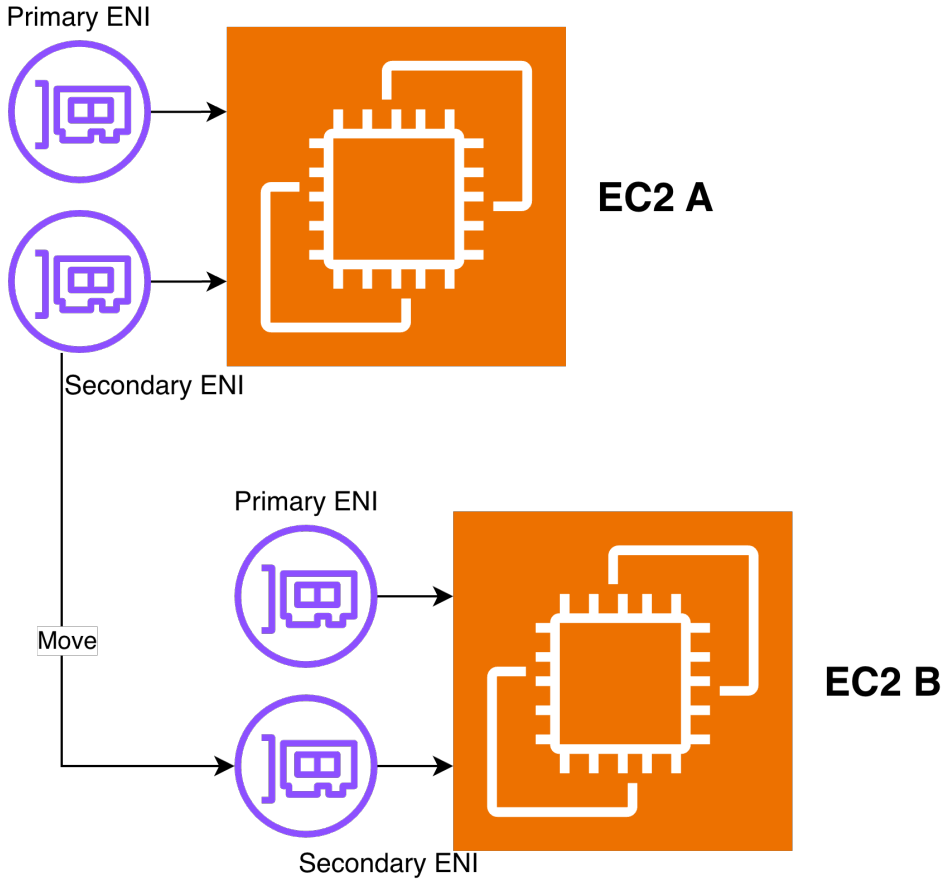
- Primary private IPv4 address
- One or more secondary private IPv4 addresses
- Elastic IP addresses (can be associated with private IPs)
- Optional public IPv4 address One or more security groups
- A unique MAC address

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-eni.html

Compute Cloud

❖ EC2 - Elastic Network Interfaces (ENI)

An Elastic Network Interface (ENI) is a virtual network card that can be attached to an EC2 instance within a VPC.



- ENIs can be created independently and attached or detached from EC2 instances.
- This enables moving network configurations between instances for failover or high availability.
- Each ENI is limited to a specific Availability Zone (AZ) and can only attach to instances in that AZ.

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-eni.html

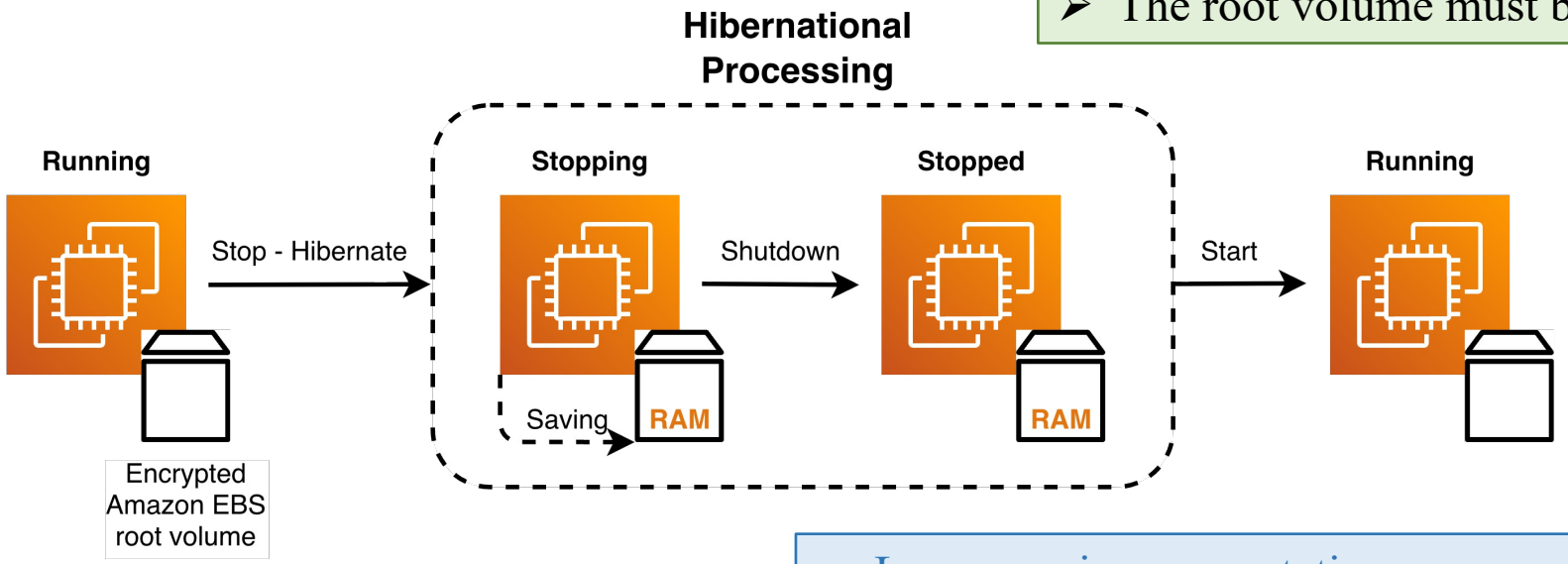
Compute Cloud

❖ EC2 - Hibernate

- Hibernate stops an EC2 instance and saves the contents of RAM to the root EBS volume.
- When the instance resumes, the **system state and applications are restored**, so startup is **much faster than a normal reboot**.

Requirements:

➢ The root volume must be an encrypted Amazon EBS volume.



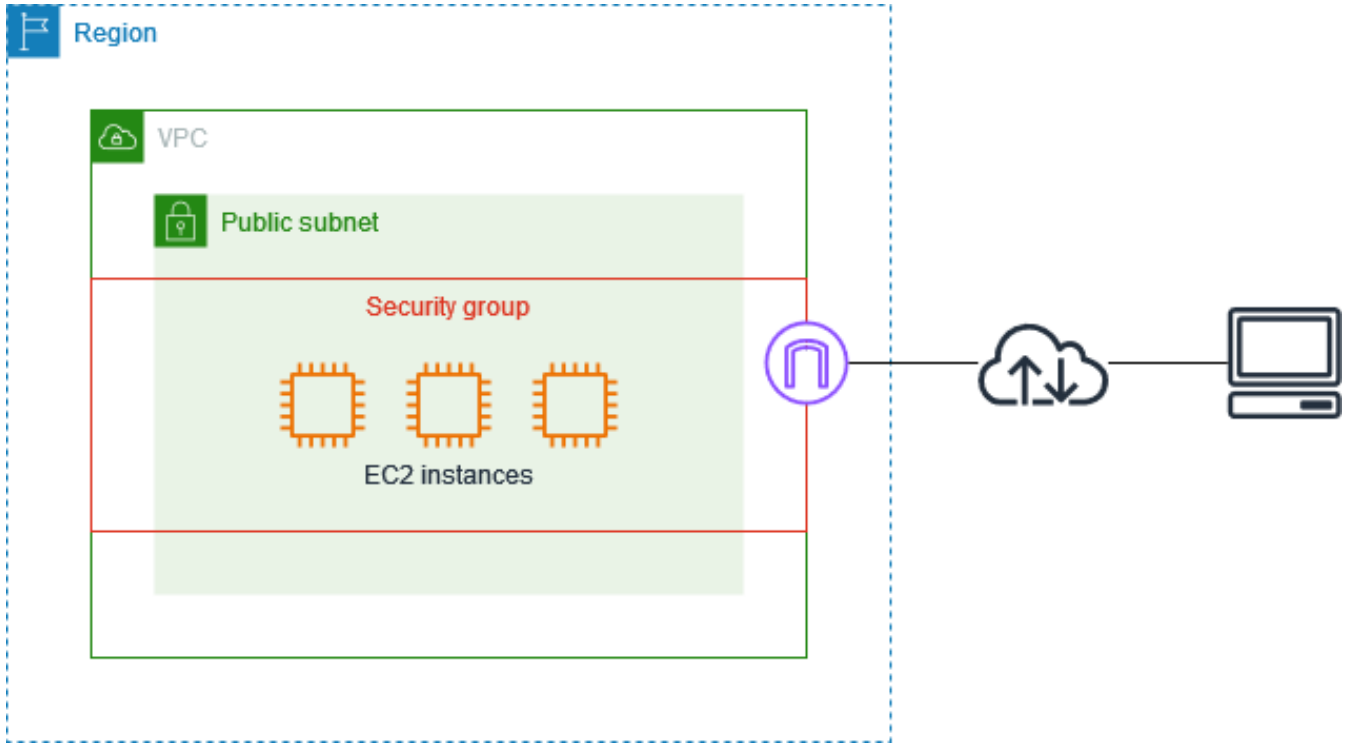
- Long-running computations or processing jobs
- Preserving the in-memory state (RAM)
- Applications or services that take a long time to initialize

Compute Cloud

❖ EC2 - Security Groups

- **Security Groups** control the **network traffic allowed to reach or leave EC2 instances**.
- They act as a **virtual firewall** at the instance level within a **VPC**.

- An EC2 instance can be associated with **multiple security groups**.
- A security group can also be associated with **multiple instances**.
- Each security group contains rules that allow **inbound or outbound traffic**.
- When rules are added or modified, the changes are applied **immediately** to all associated instances.

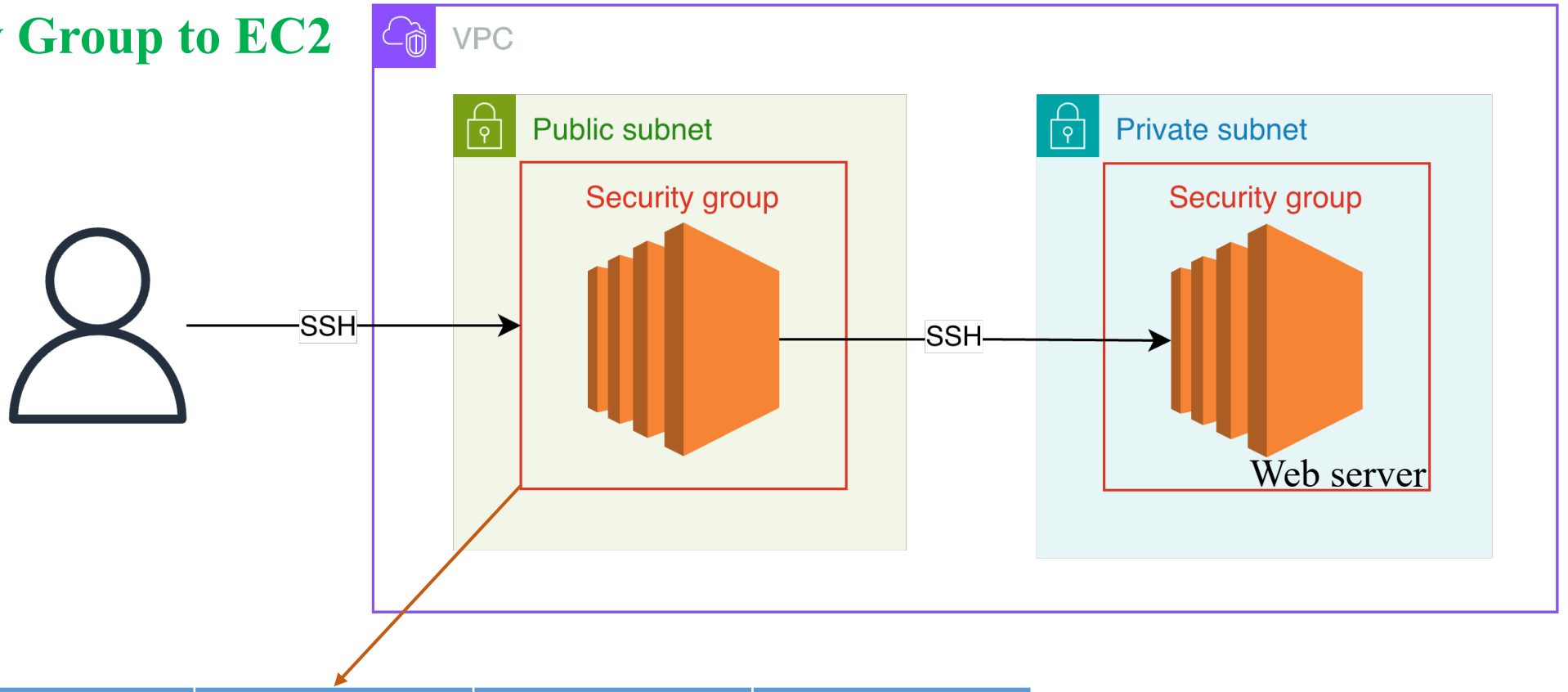


Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-security-groups.html

Lab Session

Lab Session

❖ Add Security Group to EC2

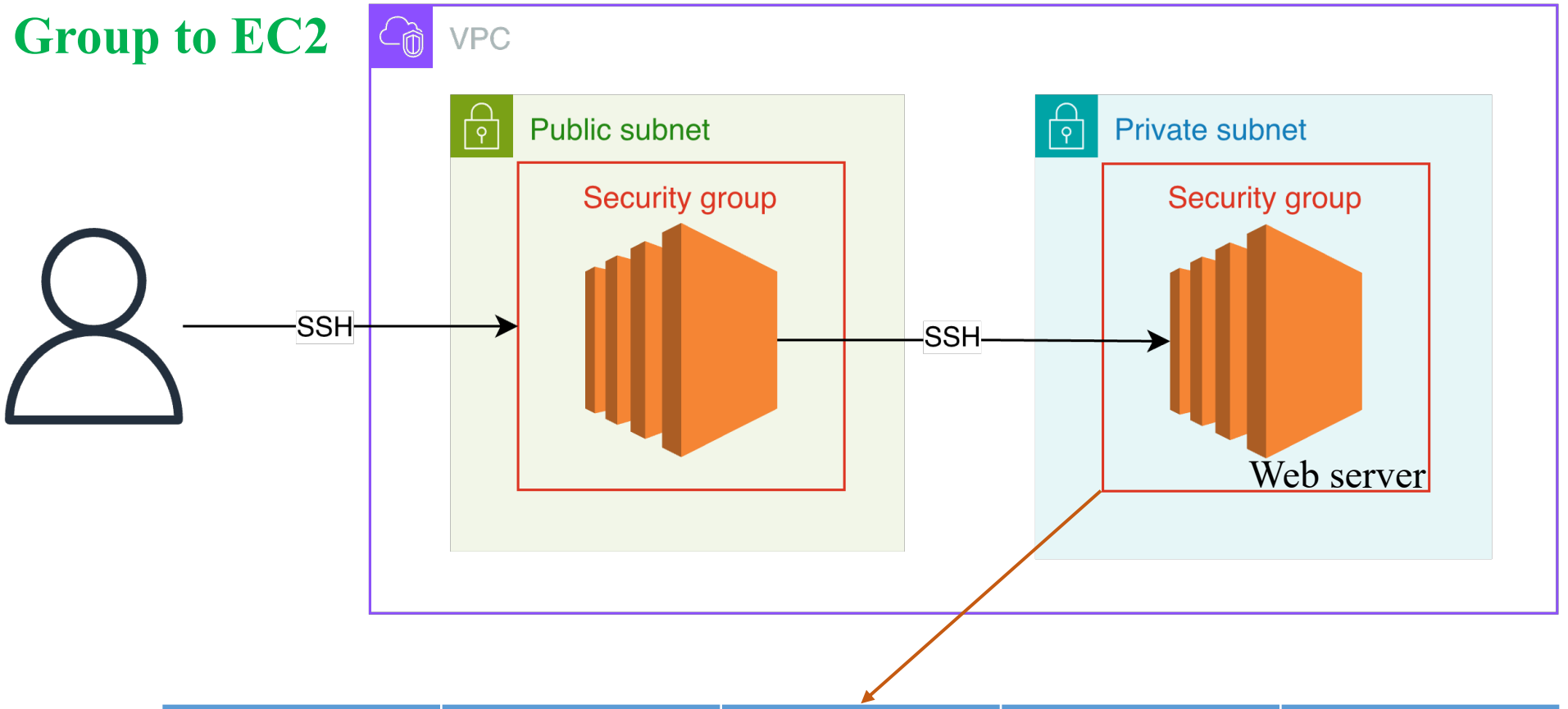


	Type	Protocol	Port range	Source
Inbound	SSH	TCP	22	0.0.0.0/0
Outboud	All traffic	All	All	0.0.0.0/0

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-security-groups.html

Lab Session

❖ Add Security Group to EC2



	Type	Protocol	Port range	Source
Inbound	SSH	TCP	22	`SG cua EC1`
Outboud	All traffic	All	All	0.0.0.0/0

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-security-groups.html

Purchasing Options

Purchasing Options

❖ Getting Started

On-Demand Instances	Pay-as-you-go (per second) pricing with no commitment, ideal for short or unpredictable workloads.
Savings Plans	Commit to a consistent compute usage for 1 or 3 years to receive flexible discounts.
Reserved Instances	Long-term commitment (1 or 3 years) for significant cost savings on steady workloads.
Spot Instances	Very low-cost instances using spare AWS capacity, but they can be **interrupted**.
Dedicated Hosts	Entire **physical server** dedicated to your use for licensing or compliance needs.
Dedicated Instances	Instances run on **hardware dedicated** to a single customer.
Capacity Reservations	Reserve compute capacity in a **specific Availability Zone** to ensure availability.

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/instance-purchasing-options.html

Purchasing Options

❖ On-Demand Instances

- No upfront payment and no long-term commitment
- Pay only for the compute time used (per second for most instances)
- Typically higher cost compared to other purchasing options

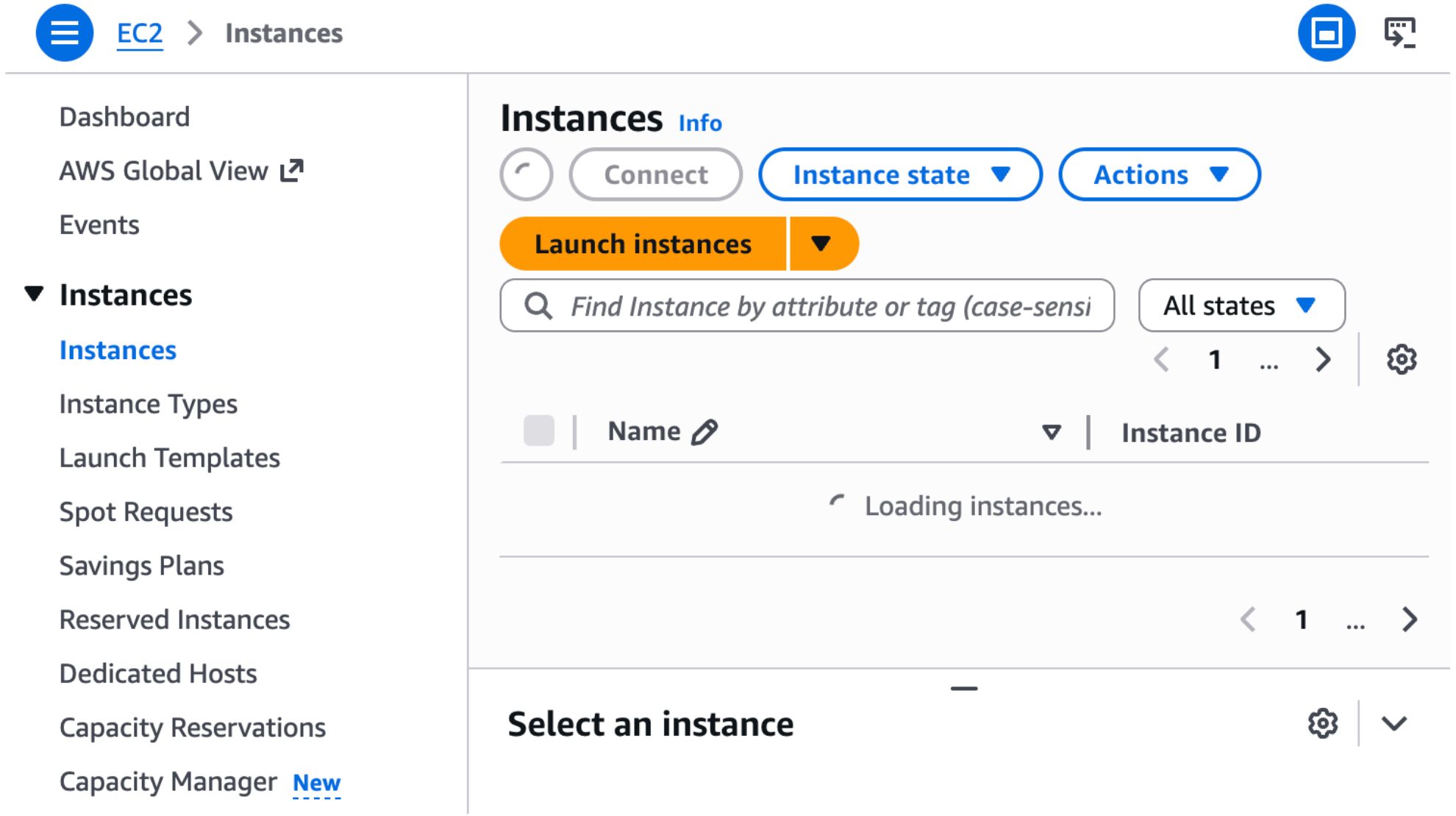
- Short-term or unpredictable workloads
- Applications that require reliable, uninterrupted compute capacity

- Development, staging, and testing environments
- Critical batch processing jobs
- Systems that run only during specific hours (e.g., business hours: 8AM–5 PM)
- Applications with frequent scaling or variable demand

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-on-demand-instances.html

Purchasing Options

❖ On-Demand Instances



Purchasing Options

❖ Savings Plans

- Commit to a consistent amount of compute usage ($/hour) for 1 or 3 years.
- Can provide up to ~72% cost savings compared to On-Demand pricing.

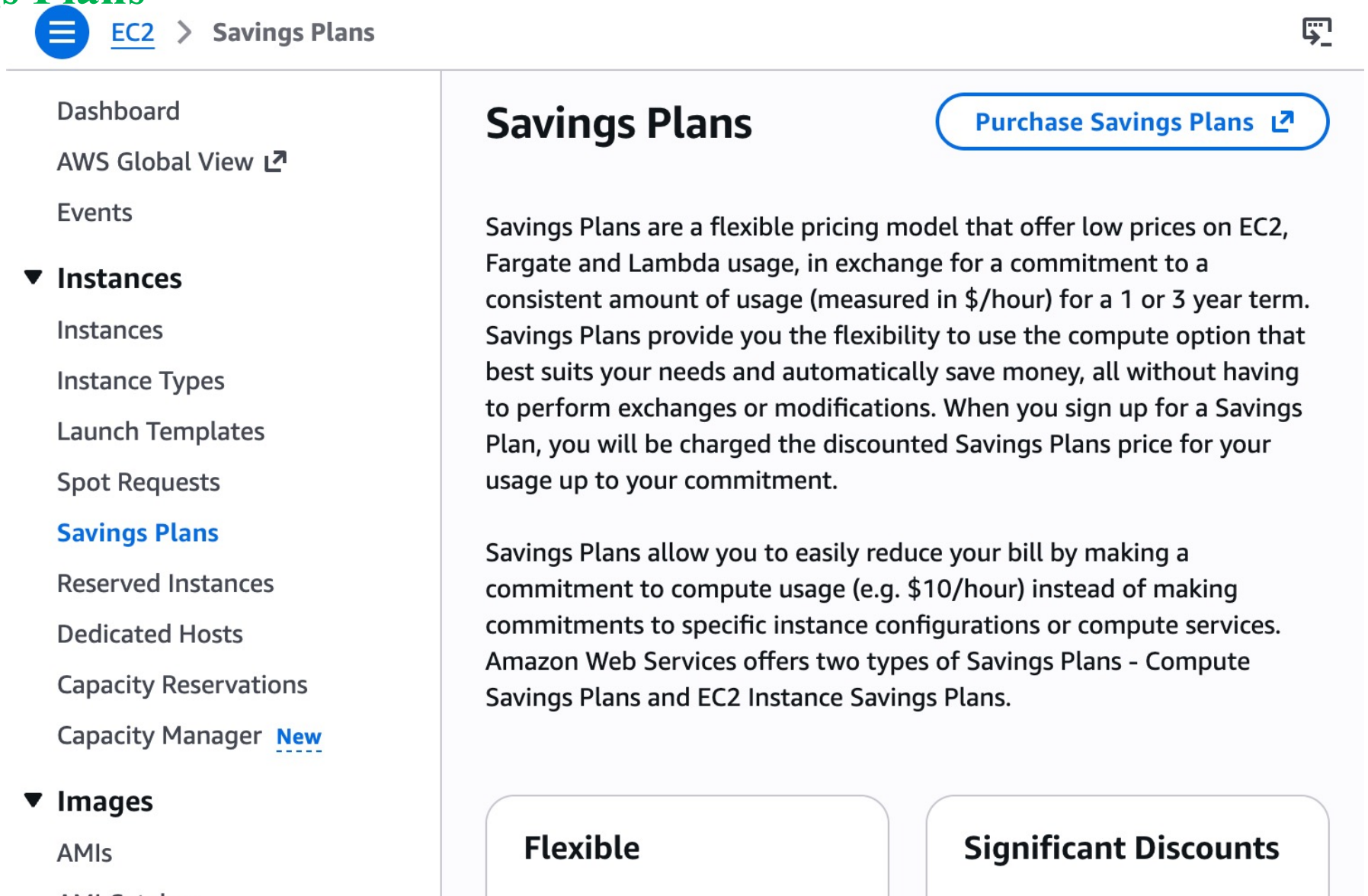
- Savings Plans provide flexibility across:
  - **Instance sizes** within the same family (e.g., *m5.xlarge → m5.2xlarge*)
  - **Operating systems** (Linux, Windows)
  - **Tenancy options** (Default, Dedicated, or Host)

- Long-running workloads with predictable compute usage.
- Systems running 24/7 in production.
- Workloads that may change instance sizes or configurations over time while maintaining cost savings.

Ref: https://docs.aws.amazon.com/savingsplans/latest/userguide/what-is-savings-plans.html

Purchasing Options

❖ Savings Plans



Purchasing Options

❖ Reserved Instances

- Reserved Instances provide significant cost savings (up to ~72%) compared to On-Demand pricing
- In exchange, you commit to using specific instance attributes such as instance type, region, OS, and tenancy for a fixed term.
- **Reservation term: 1 year** or **3 years** (longer term → larger discount).

Payment options:
- No Upfront
- Partial Upfront
- All Upfront (largest discount)

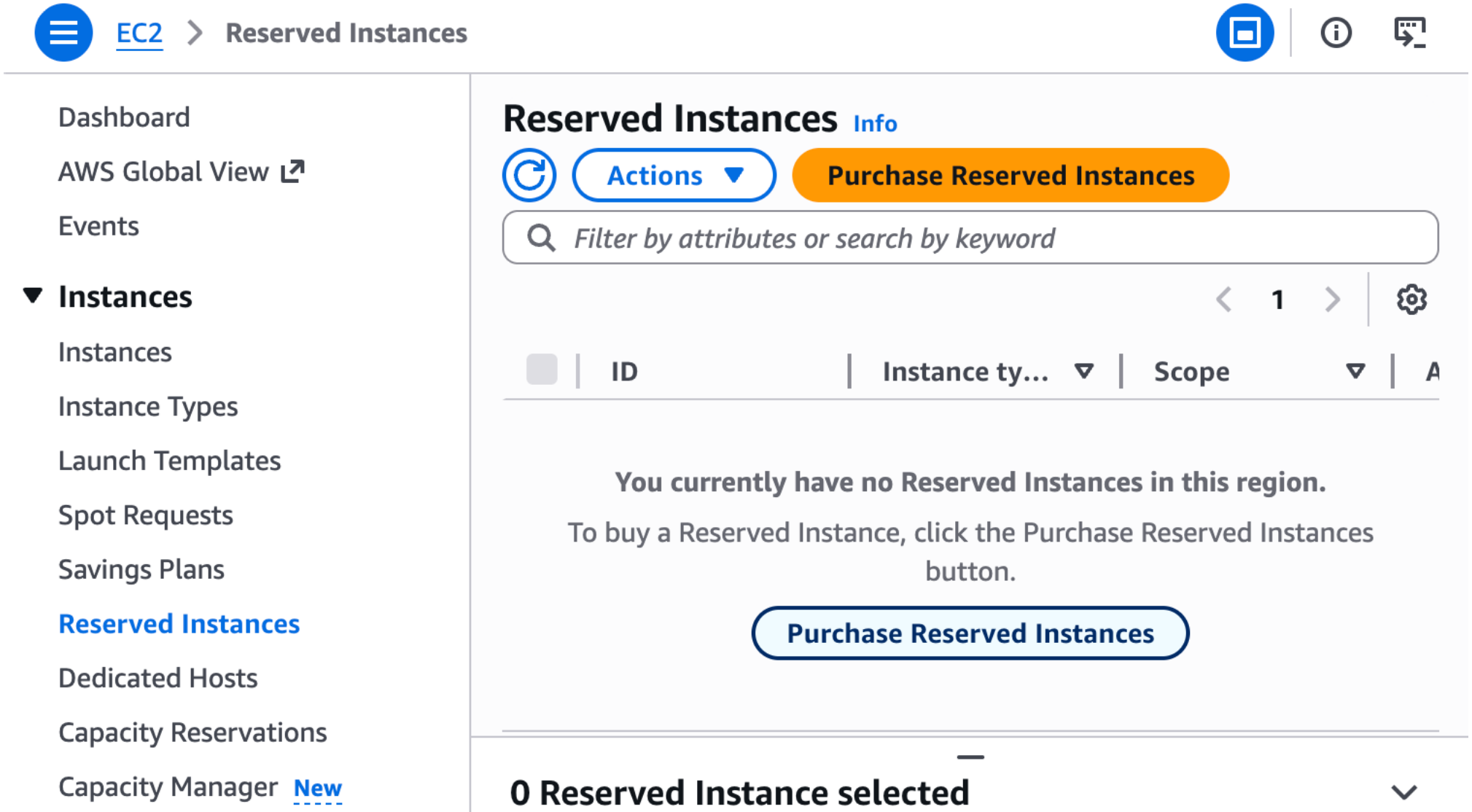
- Regional RI:
  - Discount applies to instances in any Availability Zone within a region.
  - Does not reserve capacity.
- Zonal RI
  - Applies to a specific Availability Zone.
  - Provides capacity reservation in that AZ.

- Systems that run continuously (24/7)
- Stable workloads that rarely change configuration
- Long-running infrastructure such as databases or backend services.

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/ec2-reserved-instances.html

Purchasing Options

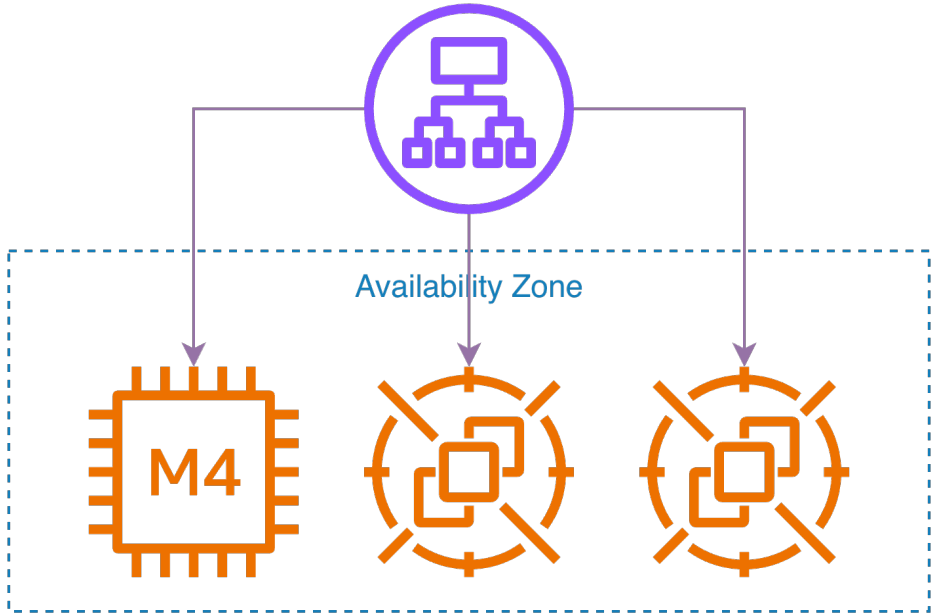
❖ Reserved Instances



Purchasing Options

❖ Spot Instances

- Stable workloads that rarely change configuration
- Long-running infrastructure such as databases or backend services.



- Spot instances may be interrupted by AWS when capacity is needed.
- When interruption occurs, AWS provides a 2-minute notice before the instance stops or terminates.

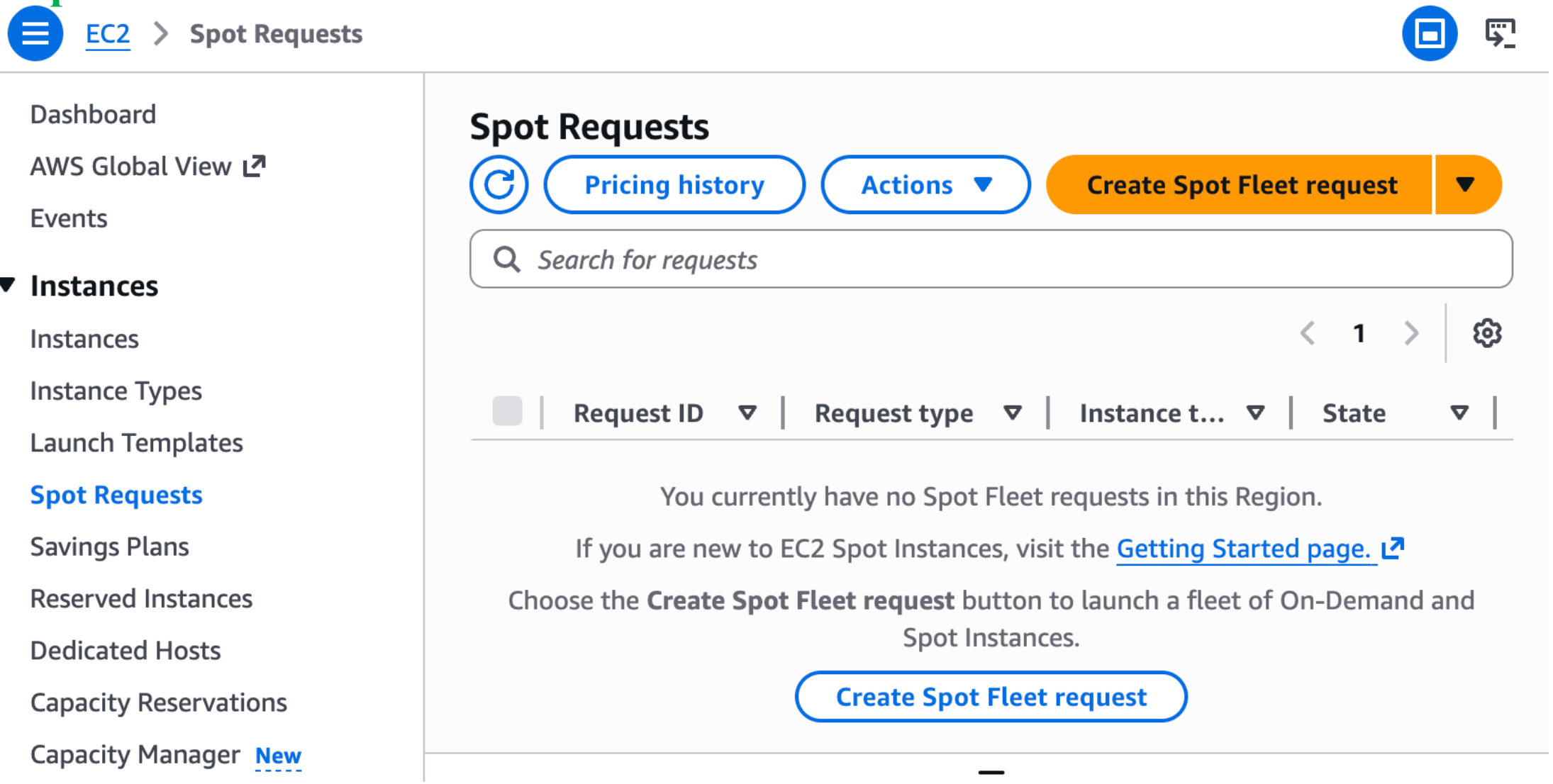
Workloads that are fault-tolerant and can handle interruptions, such as:

- ➢ Batch processing jobs
- ➢ Big data or data analysis workloads
- ➢ CI/CD pipelines
- ➢ Flexible workloads with adjustable start and end times

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/using-spot-instances.html

Purchasing Options

❖ Spot Instances



Purchasing Options

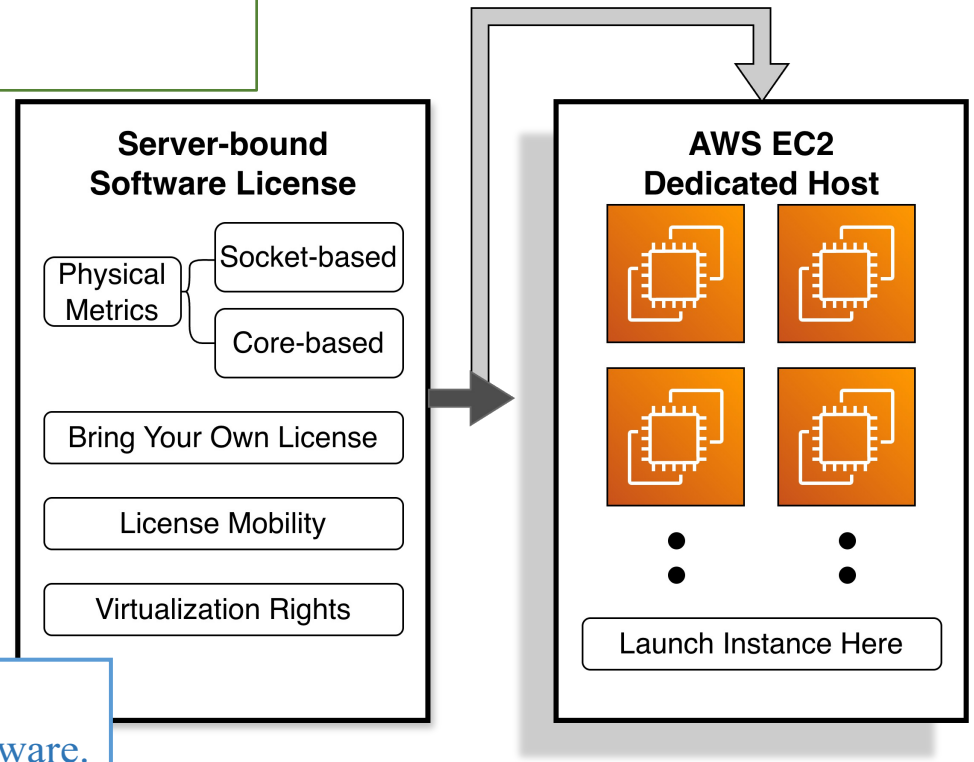
❖ Dedicated Hosts

- Dedicated Hosts are physical servers fully dedicated to a single AWS customer.
- You can launch EC2 instances on these hosts while maintaining control over instance placement on the physical server.

Enables use of existing server-bound software licenses, such as:

- ➢ Per-socket licenses
- ➢ Per-core licenses
- ➢ Per-VM licenses

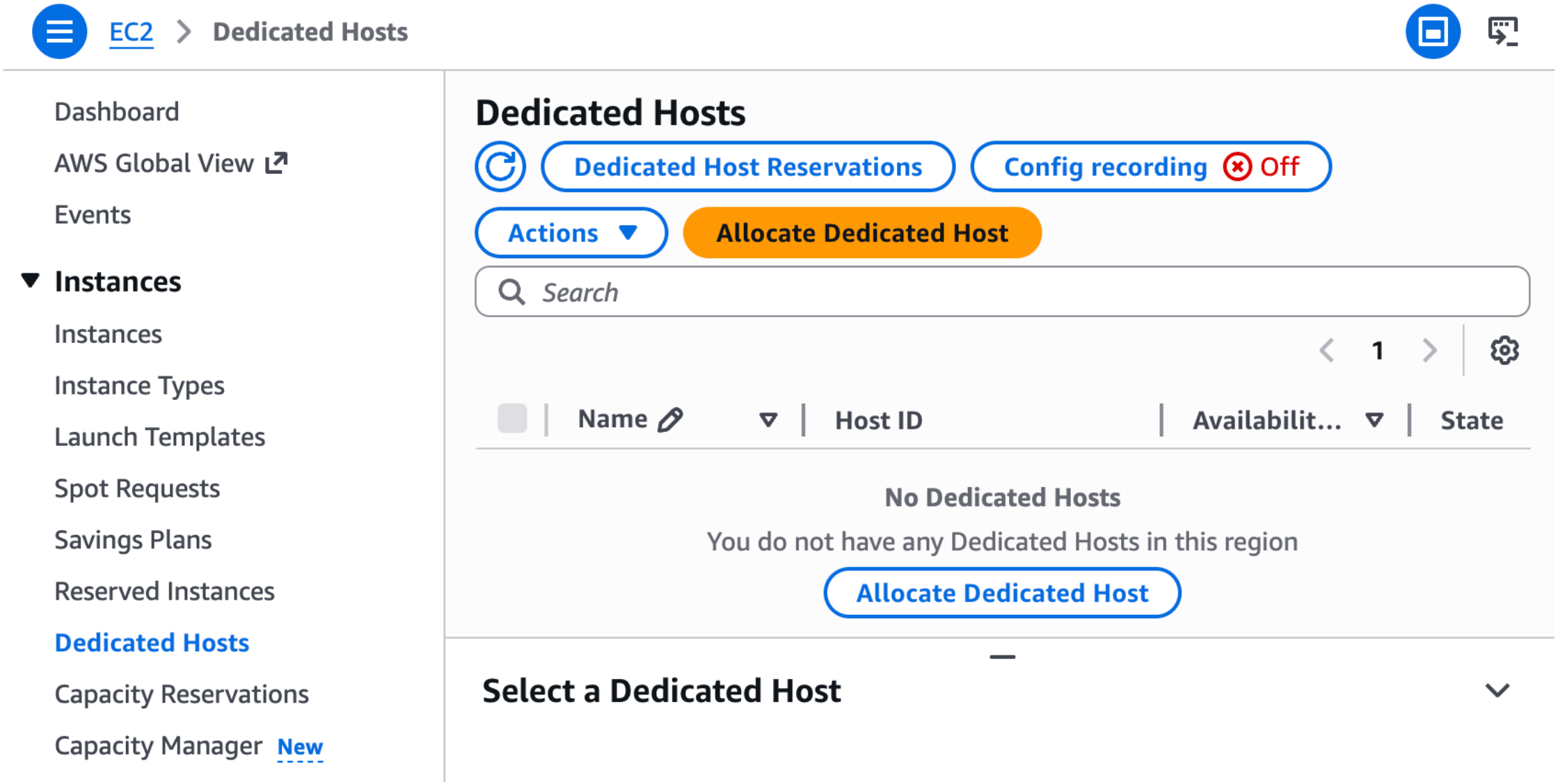
- Applications with strict licensing requirements (e.g., enterprise software).
- Compliance or regulatory requirements that require dedicated physical hardware.
- Organizations migrating on-premises licensed software to AWS.



Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/dedicated-hosts-overview.html

Purchasing Options

❖ Dedicated Hosts



Purchasing Options

❖ Dedicated Instances

- Dedicated Instances run on hardware dedicated to a single AWS customer.
- Instances may share the same physical host with other instances in the same AWS account.

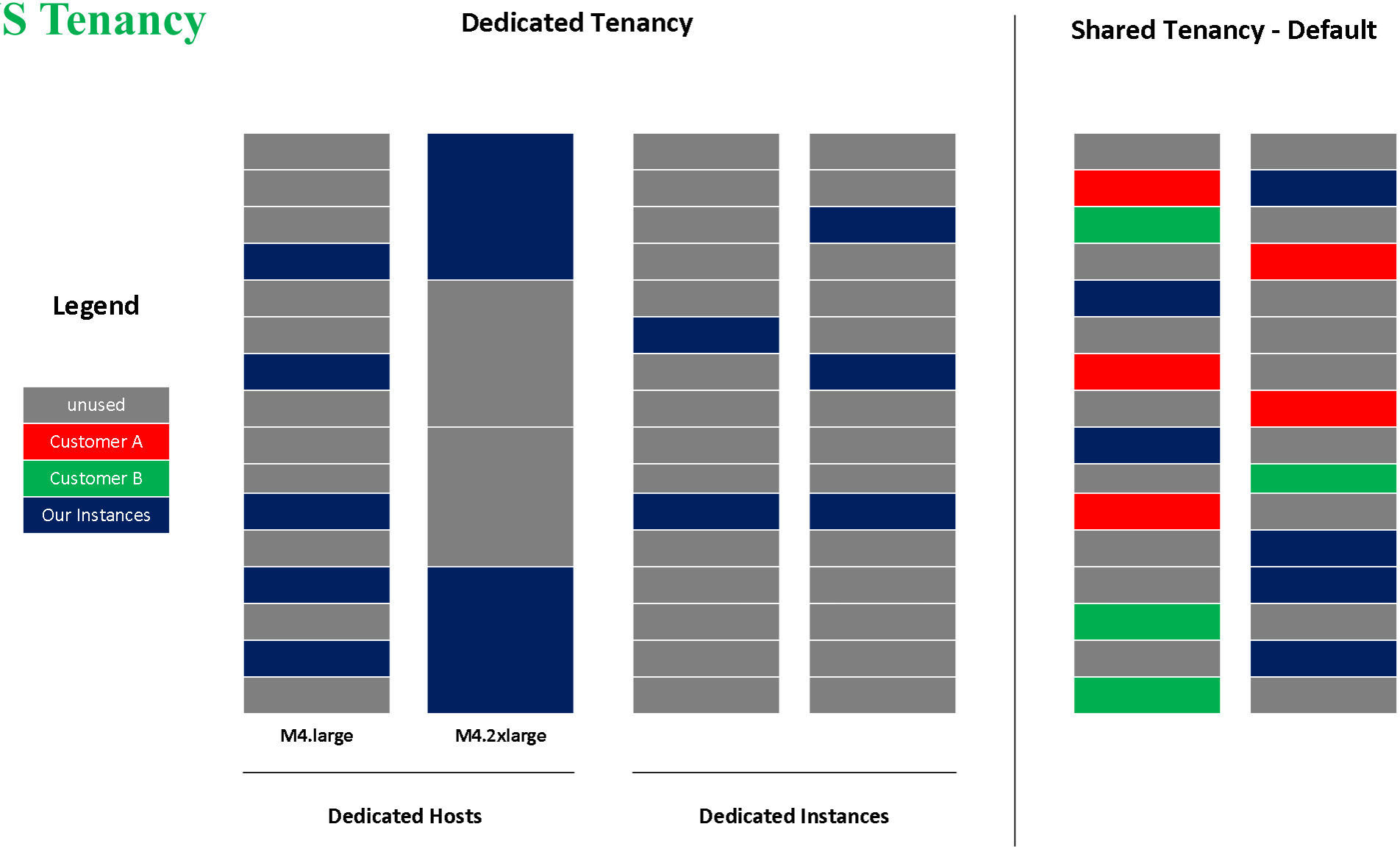
- Provide instance-level hardware isolation from other AWS customers.
- AWS manages host placement, so you do not control the physical server.
- Instances may move to different hardware after a stop/start operation.

- Workloads requiring isolated infrastructure for security or compliance.
- Systems with strict regulatory or policy requirements, such as:
  - Banking systems
  - Healthcare systems
  - Sensitive enterprise applications

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/dedicated-instance.html

Purchasing Options

❖ AWS Tenancy



Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/dedicated-change-tenancy.html

Purchasing Options

❖ Capacity Reservations

- Capacity Reservations allow you to reserve On-Demand EC2 capacity in a specific Availability Zone (AZ).
- Ensures you always have compute capacity available when needed.

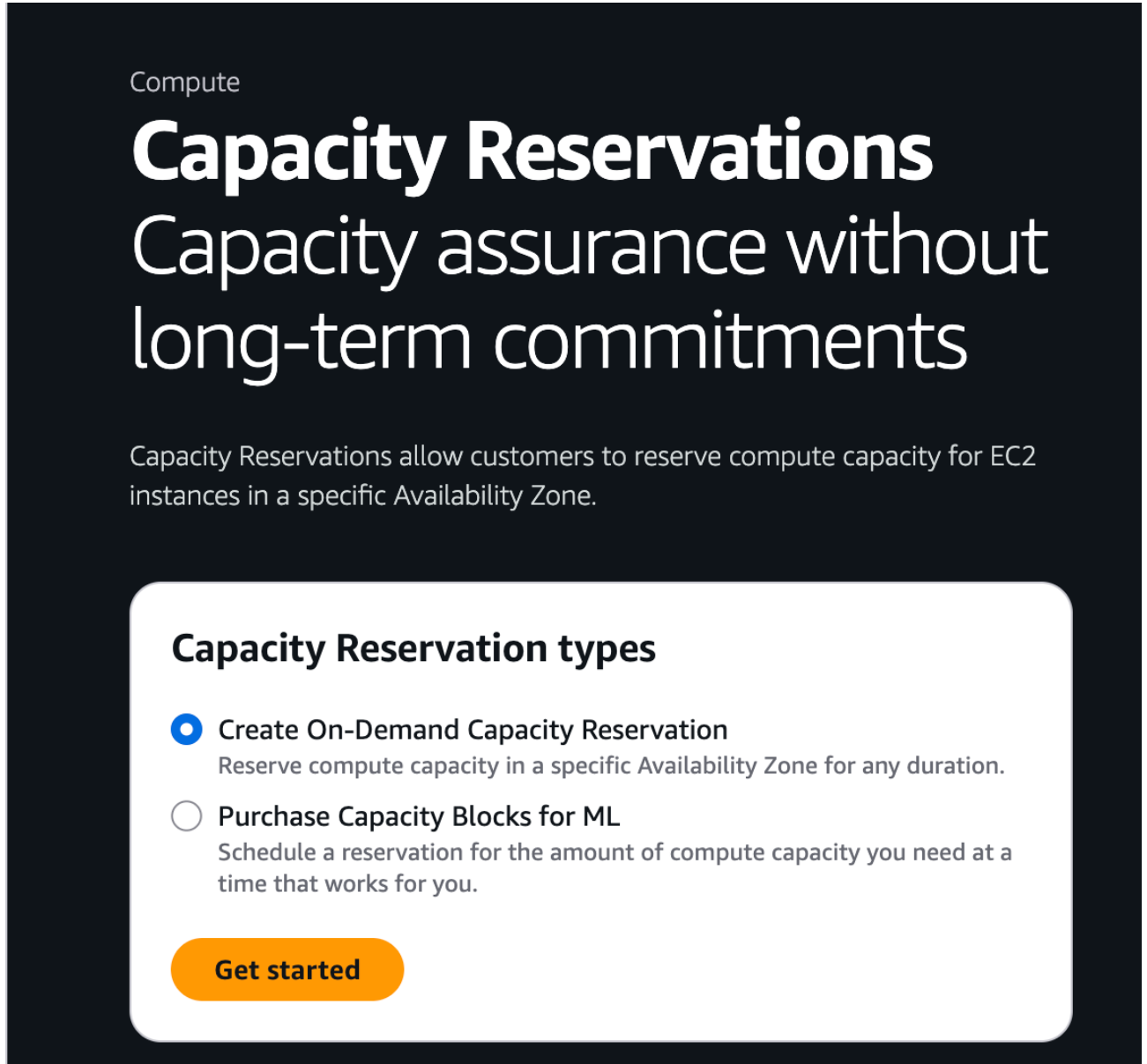
- No long-term commitment – can be created or cancelled at any time.
- No pricing discount; instances are billed at the standard On-Demand rate.
- You are charged whether instances are running or not while the reservation is active.

- Applications that require guaranteed capacity in a specific AZ.
- Short-term but critical workloads that must not be interrupted.
- Situations where capacity availability must be guaranteed during **peak demand**.

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/capacity-reservation-overview.html

Purchasing Options

❖ Capacity Reservations



Purchasing Options

❖ Summary

Purchasing Option	Pricing Model	Reliability	Typical Use Case
On-Demand Instances	Pay-as-you-go, no commitment	Very reliable	Short-term, unpredictable workloads, dev/test
Reserved Instances	1–3 year commitment with significant discount	Very reliable	Stable workloads running **24/7** (e.g., databases)
Savings Plans	Commit to **$/hour usage** for 1–3 years	Very reliable	Long-running workloads needing **flexibility in instance type**
Spot Instances	Up to ~90% cheaper than On-Demand	Can be interrupted	Batch jobs, data processing, fault-tolerant workloads
Dedicated Hosts	Highest cost, dedicated hardware	Very reliable	Licensing requirements, compliance workloads
Capacity Reservations	On-Demand price, no discount	Guaranteed capacity	Critical workloads needing **capacity in a specific AZ**

Placement Groups

Placement Groups

❖ Getting Started

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-groups.html

- Placement Groups allow you to control how EC2 instances are placed on the underlying AWS hardware.
- They are used to optimize performance, availability, or fault tolerance depending on the workload.

Cluster Placement Group

Spread Placement Group

Partition Placement Group

Placement Groups

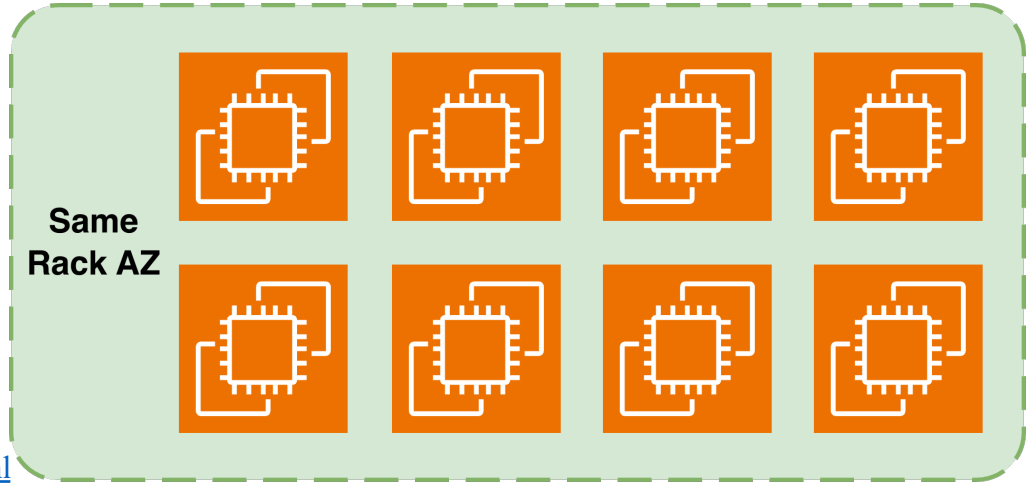
❖ Cluster

- Places EC2 instances close together within a single Availability Zone (AZ). Optimized for very low latency and high network throughput between instances.

❑ Advantages
- High network performance (up to tens of Gbps depending on instance type).
- Ideal for workloads requiring fast inter-instance communication.

❑ Limitations
- All instances are located within the same AZ.
- If the Availability Zone fails, all instances in the cluster may fail together.

- Big data processing jobs that need to complete quickly.
- Applications requiring extremely low latency and high network throughput, such as HPC workloads.



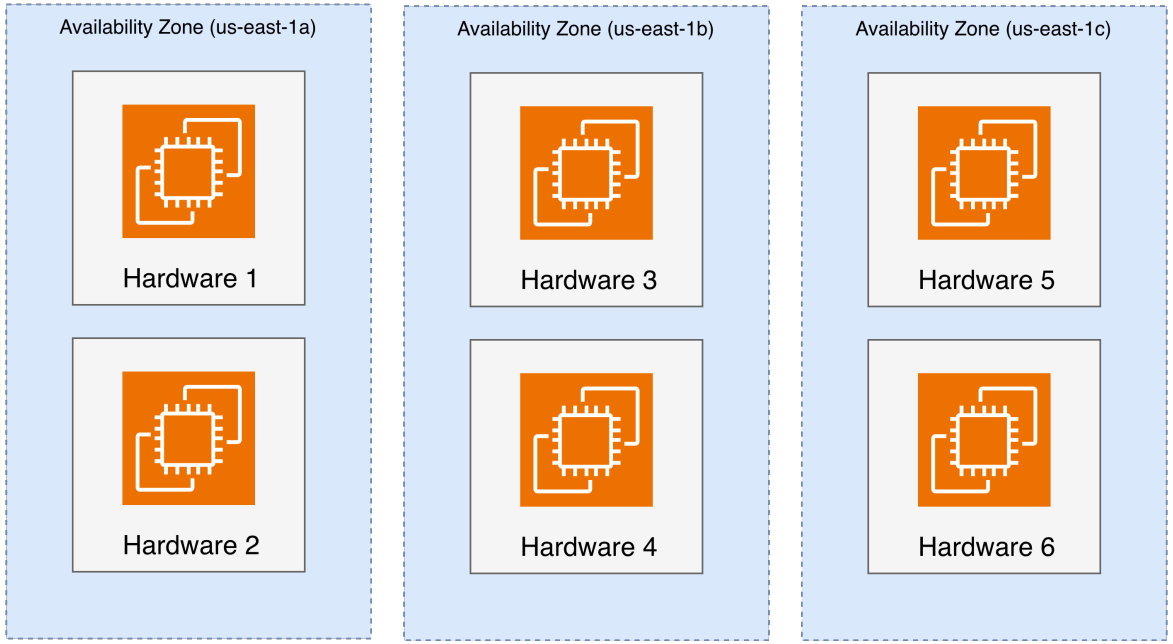
Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-strategies.html

Placement Groups

❖ Spread

Distributes EC2 instances **across different physical hardware** to reduce the risk of correlated failures

- ❑ Advantages
  - Can span multiple Availability Zones (AZs).
  - Minimizes the risk of simultaneous failure.
  - Ensures instances run on separate underlying hardware.
- ❑ Limitations
  - Maximum of 7 instances per Availability Zone per placement group.



- Applications that require high availability.
- Critical systems where each instance must be isolated from hardware failures of other inst

Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-strategies.html

Placement Groups

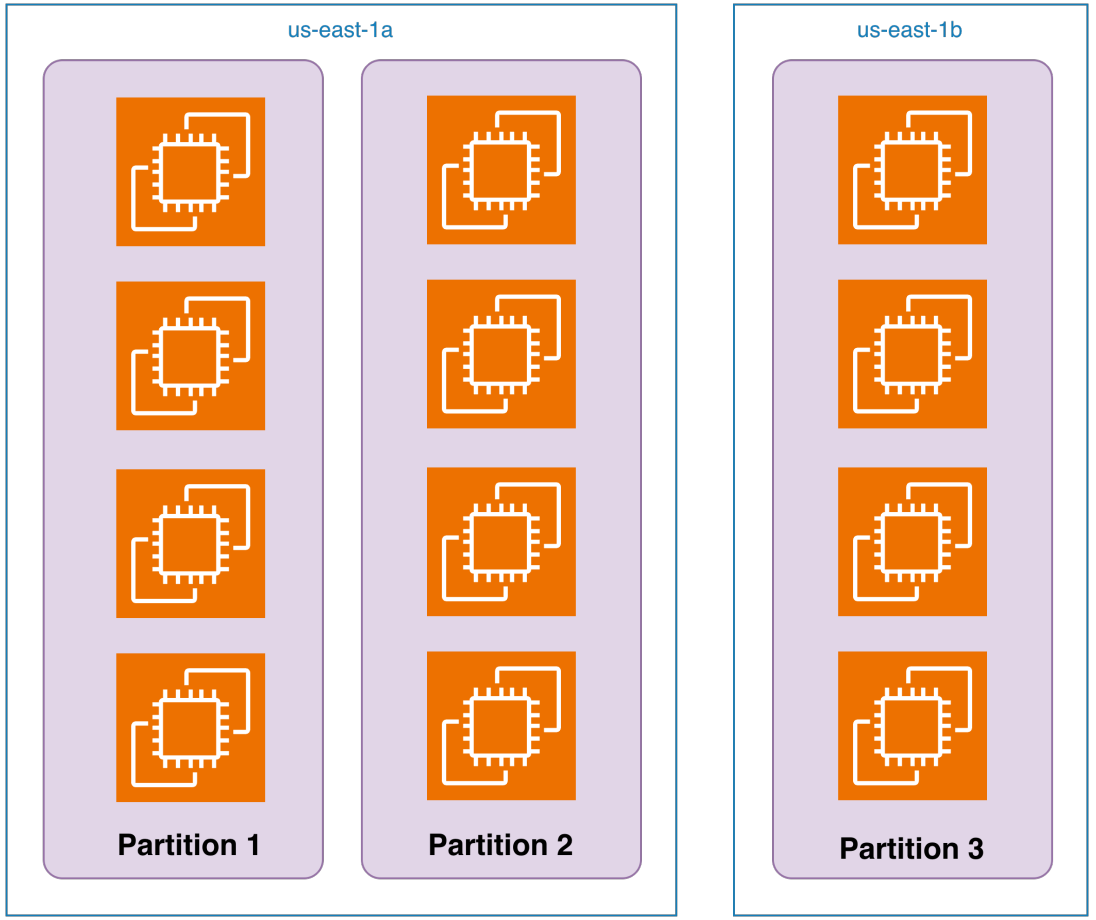
❖ Partition

Divides EC2 instances into **multiple partitions**, where each partition runs on **separate underlying hardware racks**.

❑ Advantages
- Can span multiple Availability Zones (AZs).
- Failure of one partition does not affect other partitions.
- Supports hundreds of EC2 instances in a group, making it suitable for large distributed systems.

❑ Limitations
- The number of partitions per Availability Zone is limited (up to 7).

Distributed data systems such as:
- HDFS / Hadoop
- HBase
- Cassandra
- Kafka



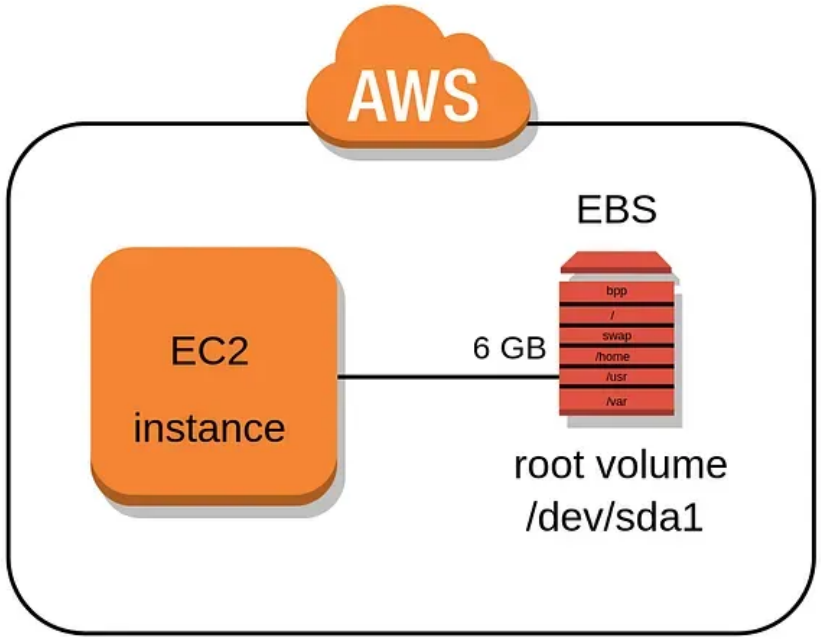
Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/placement-strategies.html

EC2 Storage

EC2 Storage

❖ What is Elastic Block Store (EBS)?

- Amazon EBS is a persistent block storage service for EC2 instances.
- It acts like a network-attached disk that can be attached to a running EC2.
- When you launch an **EC2 instance**, AWS automatically attaches an **EBS volume called the Root EBS Volume**.



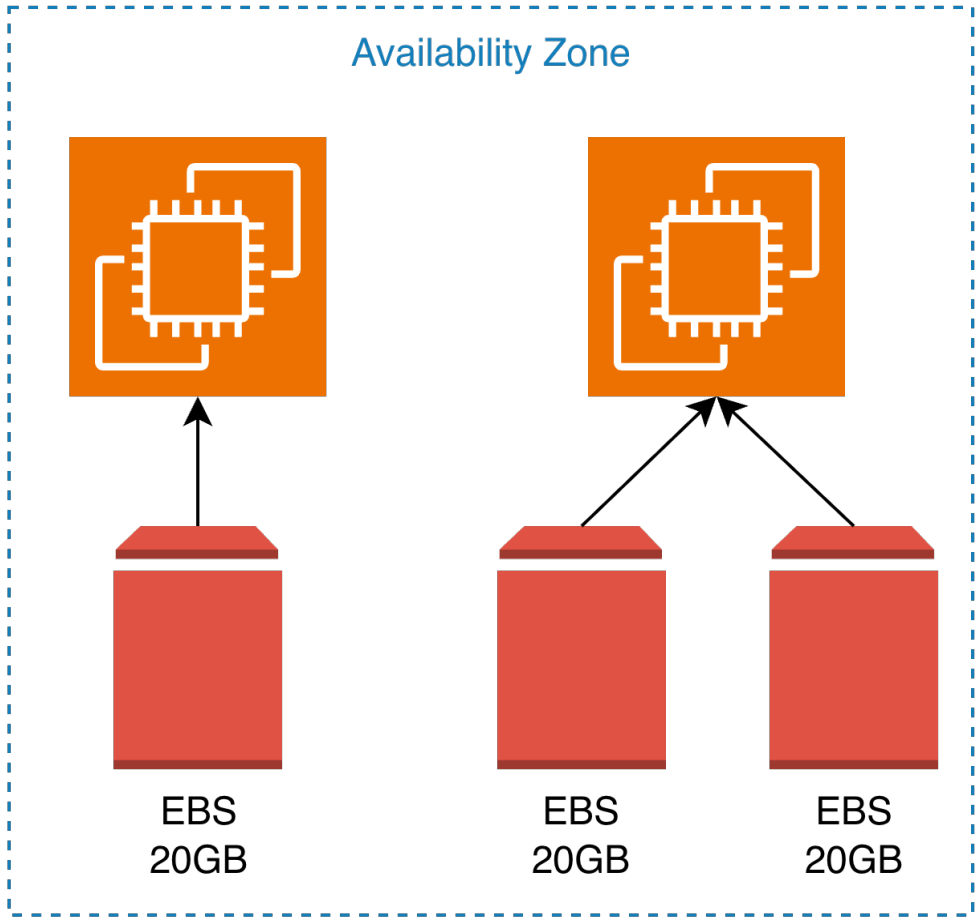
- Persistent storage – data remains even **if the EC2 instance stops or terminates**.
- Usually attached to ONE instance at a time (except io1/io2 Multi-Attach).
- Bound to a specific Availability Zone (AZ).

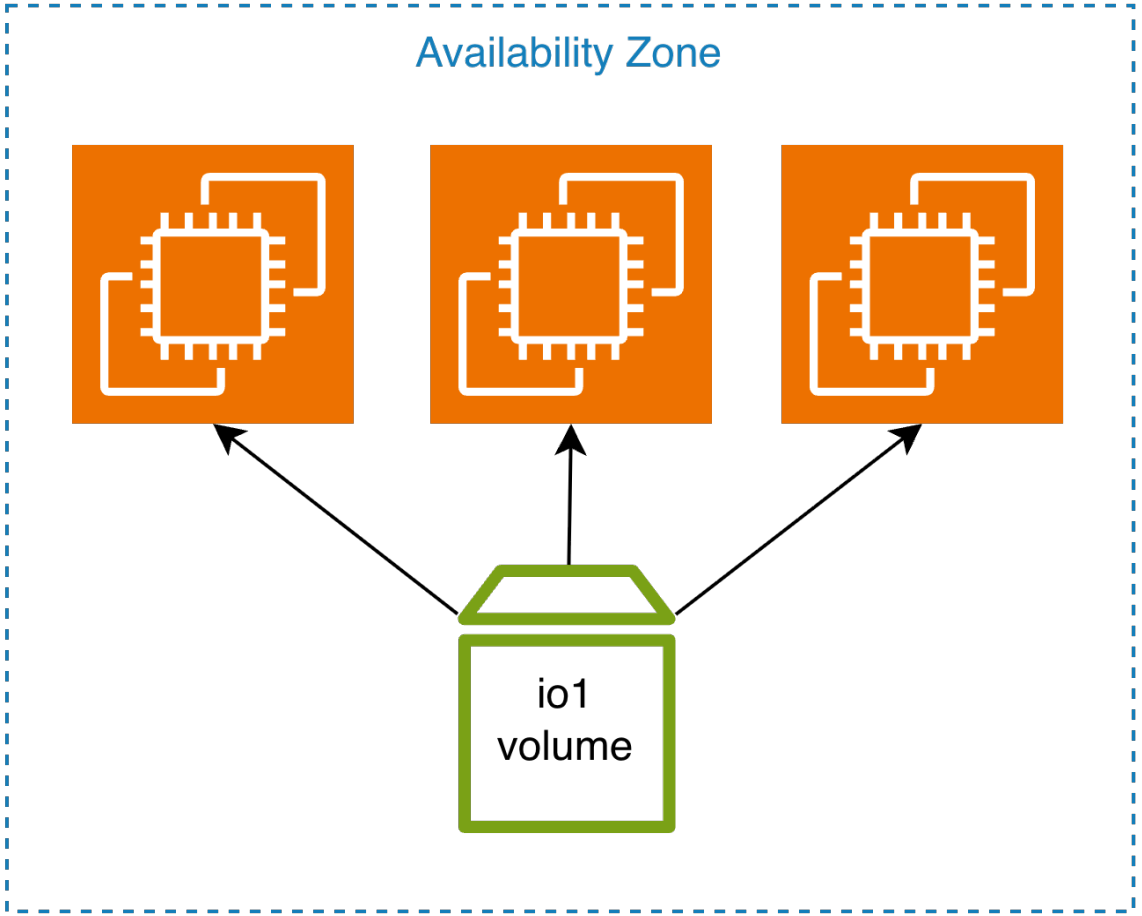
- AWS Free Tier includes **30 GB of EBS storage per month**, using **General Purpose SSD (gp2/gp3) or Magnetic volumes**.

Ref: https://docs.aws.amazon.com/ebs/latest/userguide/ebs-volumes.html

EC2 Storage

❖ Elastic Block Store (EBS)

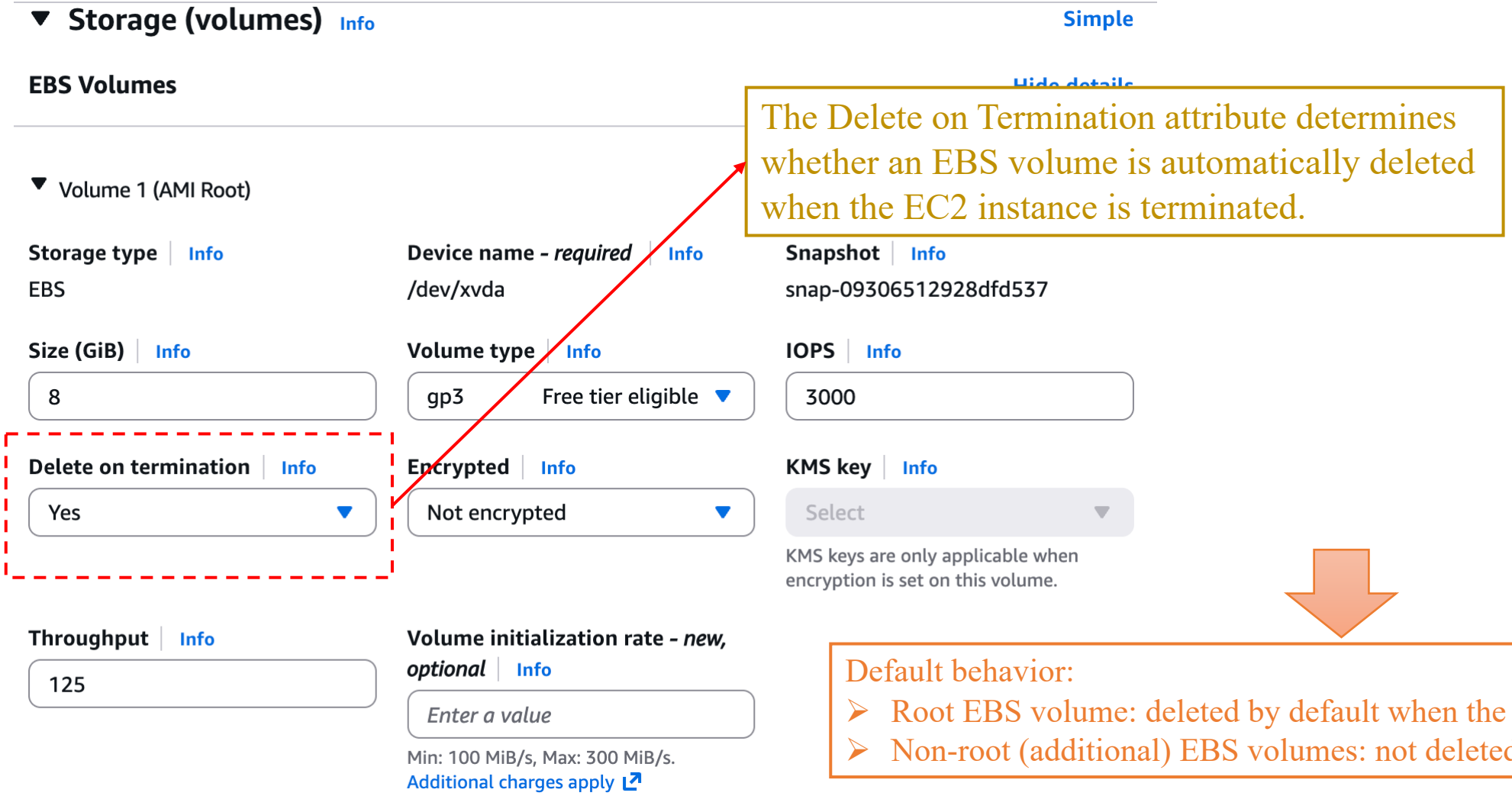




EBS Multi-Attach (io1/io2 family only) -
Up to 16 EC2 Instances at a time

EC2 Storage

❖ EBS Delete on Terminate



The Delete on Termination attribute determines whether an EBS volume is automatically deleted when the EC2 instance is terminated.

Default behavior:

- Root EBS volume: deleted by default when the instance is terminated.
- Non-root (additional) EBS volumes: not deleted by default.

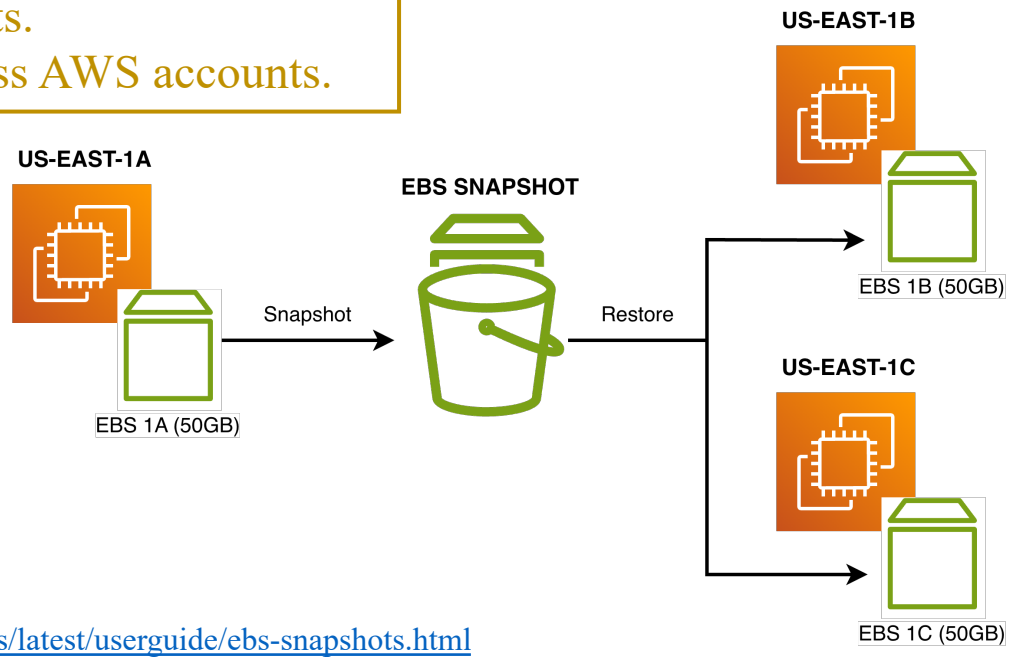
EC2 Storage

❖ EBS - Snapshots

- An EBS Snapshot is a **point-in-time** backup of an Amazon EBS volume.
- Snapshots allow you to restore data or create new EBS volumes from that backup.

- Incremental backups – only changed data blocks are saved.
- Stored in Amazon S3 (managed by AWS) with high durability.
- Can create new EBS volumes from snapshots.
- Can be copied across regions or shared across AWS accounts.

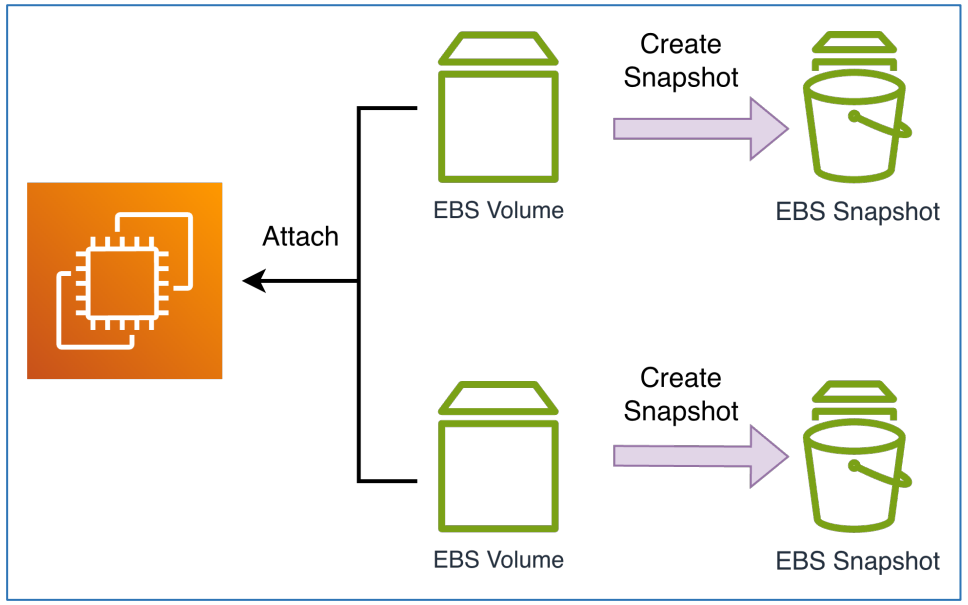
- Backup and disaster recovery
- Creating identical volumes for testing or scaling
- Migrating data across regions or accounts

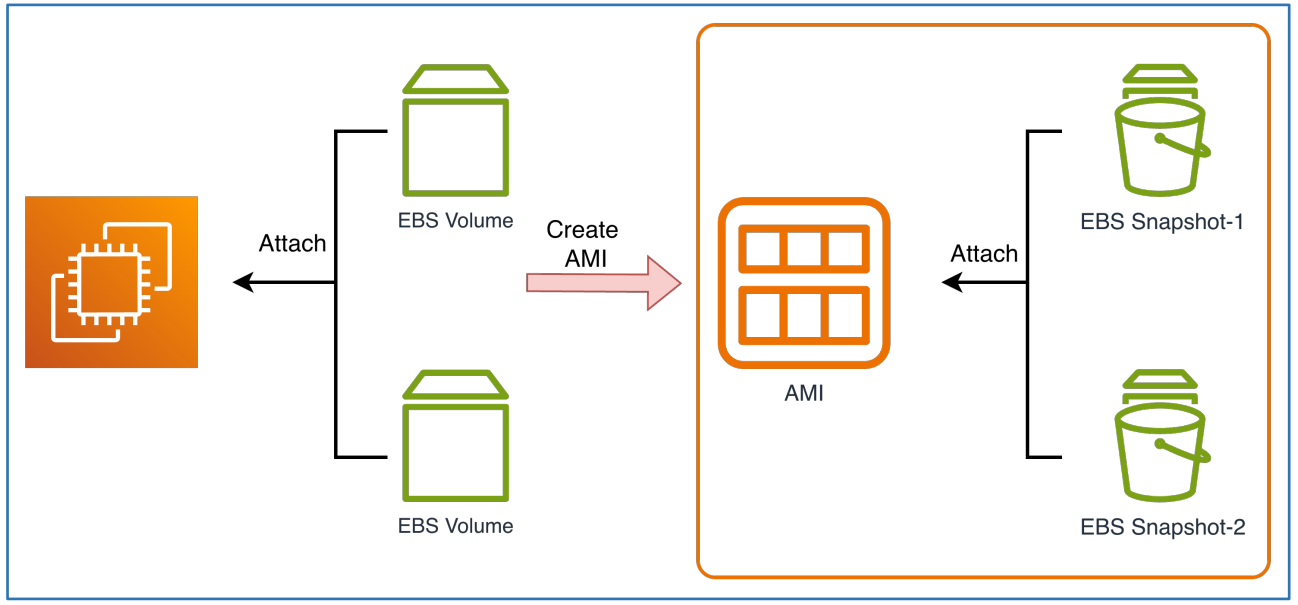


Ref: https://docs.aws.amazon.com/ebs/latest/userguide/ebs-snapshots.html

EC2 Storage

❖ EBS - Snapshots vs AMI





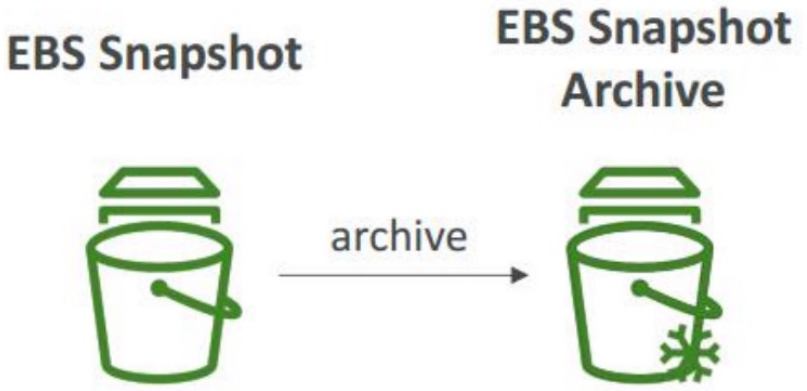
	EBS Snapshot	AMI
Purpose	Backup of a **single EBS volume**	Template used to **launch EC2 instances**
Content	Data from a **specific volume**	OS, configuration, software, and EBS snapshots
Launch EC2	Cannot launch EC2 directly	Can launch EC2 instances
Use case	Backup or restore **data volumes**	Create **preconfigured EC2 environments**

EC2 Storage

❖ EBS - Snapshots Features

Snapshot Archive

- Move snapshots to a low-cost archive tier (up to ~75% cheaper).
- Restoring an archived snapshot typically takes 24–72 hours.



Recycle Bin for Snapshots

- Protects against accidental deletion of snapshots.
- Deleted snapshots are retained according to defined retention rules.
- Retention period can be 1 day to 1 year.

Fast Snapshot Restore (FSR)

- Enables fully initialized volumes created from snapshots.
- Eliminates initial read latency when accessing the volume for the first time.
- Additional cost applies.

EC2 Storage

❖ EBS - Volume Types

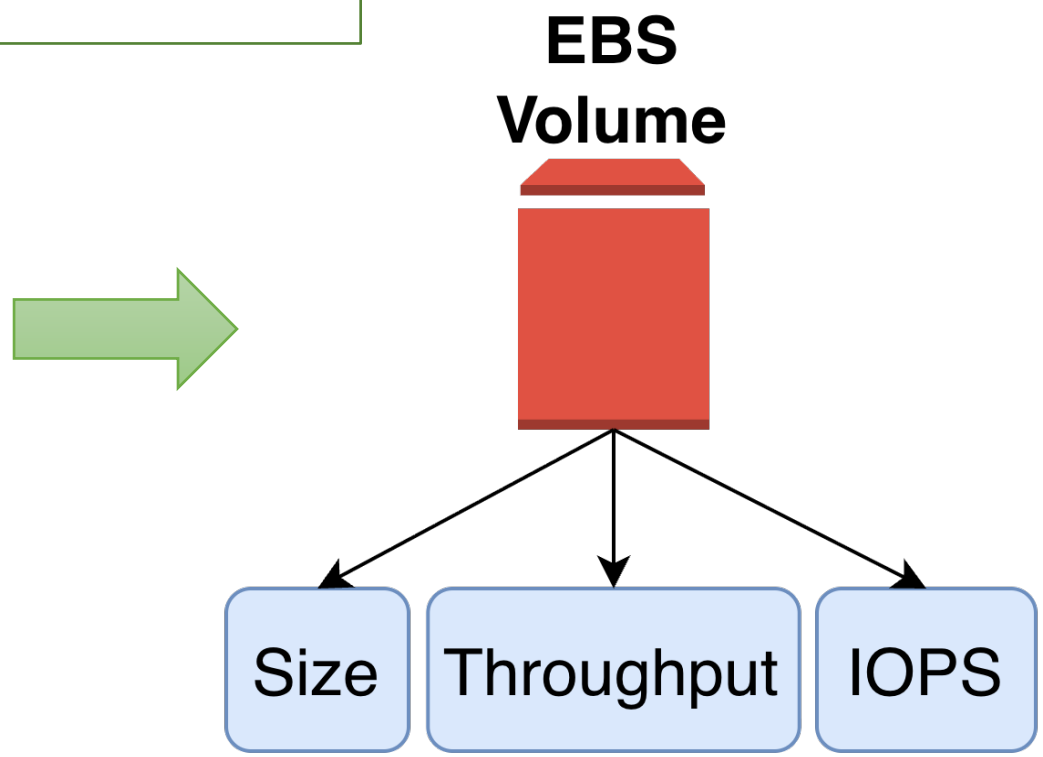
- ❑ Amazon EBS provides multiple volume types optimized for different workloads.
- ❑ EBS Volumes come in 6 types

SSD-based volumes (high performance)

- gp3 / gp2 (General Purpose SSD)
- io1 / io2 Block Express (Provisioned IOPS SSD)

HDD-based volumes (throughput-oriented)

- st1 (Throughput Optimized HDD)
- sc1 (Cold HDD)



Boot volume support

Only SSD-based volumes (gp2, gp3, io1, io2) can be used as EC2 boot volumes.

EC2 Storage

❖ General Purpose SSD

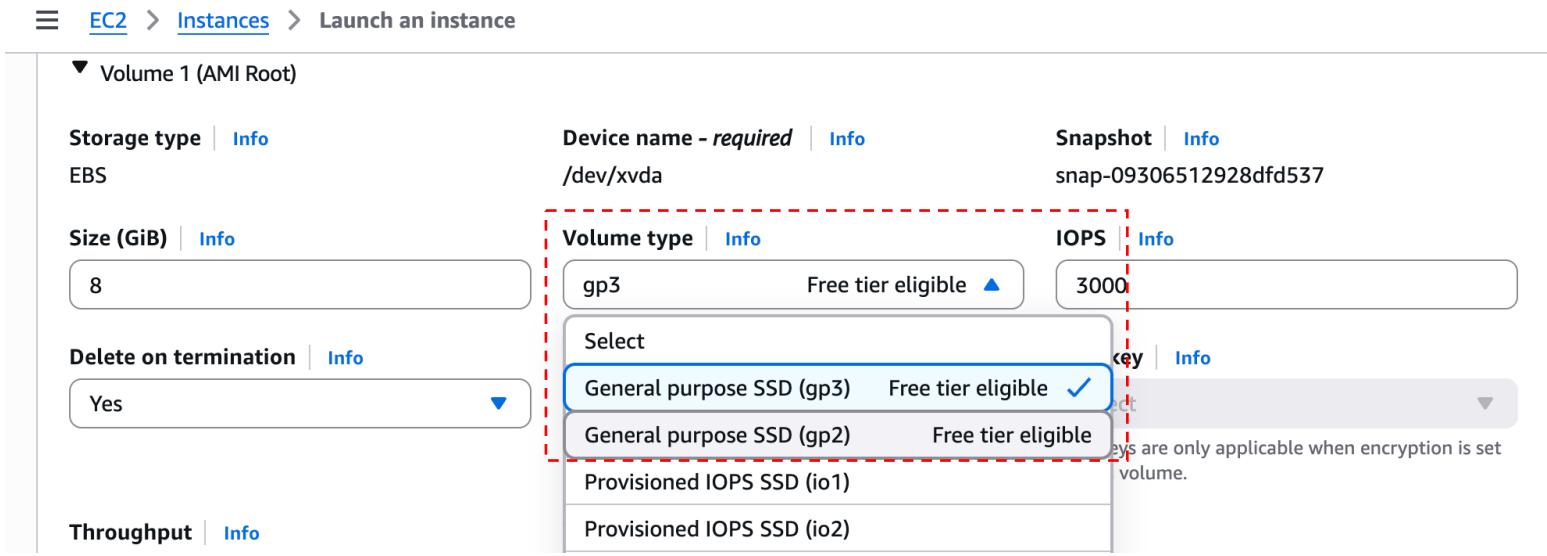
- ❑ General Purpose SSD volumes provide a balance of price, performance, and low latency.
- ❑ Suitable for **most common workloads** such as boot volumes, small databases, and development environments.

gp3 (latest generation)

- Baseline performance: 3,000 IOPS and 125 MiB/s throughput.
- Can scale up to 16,000 IOPS and 1,000 MiB/s throughput.
- IOPS and throughput can be configured independently from volume size.

gp2 (previous generation)

- Performance tied to volume size.
- Provides 3 IOPS per GB of storage.
- Maximum performance: 16,000 IOPS (reached at ~5,334 GB).
- Smaller volumes can burst up to 3,000 IOPS for short periods.



EC2 Storage

❖ Provisioned IOPS (PIOPS) SSD

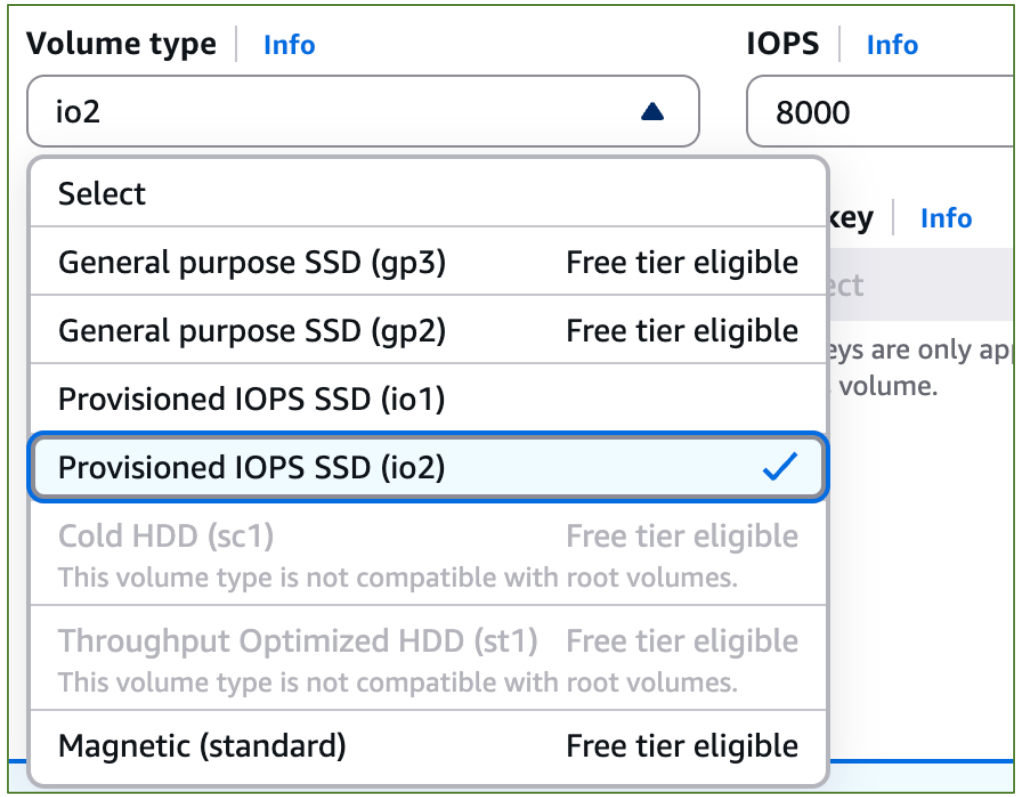
- ❑ Provisioned IOPS SSD volumes deliver the **highest performance EBS storage**.
- ❑ Designed for **mission-critical**, **low-latency**, and **high-throughput workloads**.
- ❑ Ideal for **database applications** requiring consistent and high I/O performance.

io1

- Volume size: 4 GiB – 16 TiB
- Maximum performance:
  - 64,000 IOPS on Nitro-based EC2 instances
  - 32,000 IOPS on other instances
- IOPS can be provisioned independently of storage size
- Supports EBS Multi-Attach

io2 Block Express

- Volume size: 4 GiB – 64 TiB
- Sub-millisecond latency
- Up to 256,000 IOPS
- IOPS-to-size ratio up to 1,000:1
- Supports EBS Multi-Attach



EC2 Storage

❖ Hard Disk Drives (HDD)

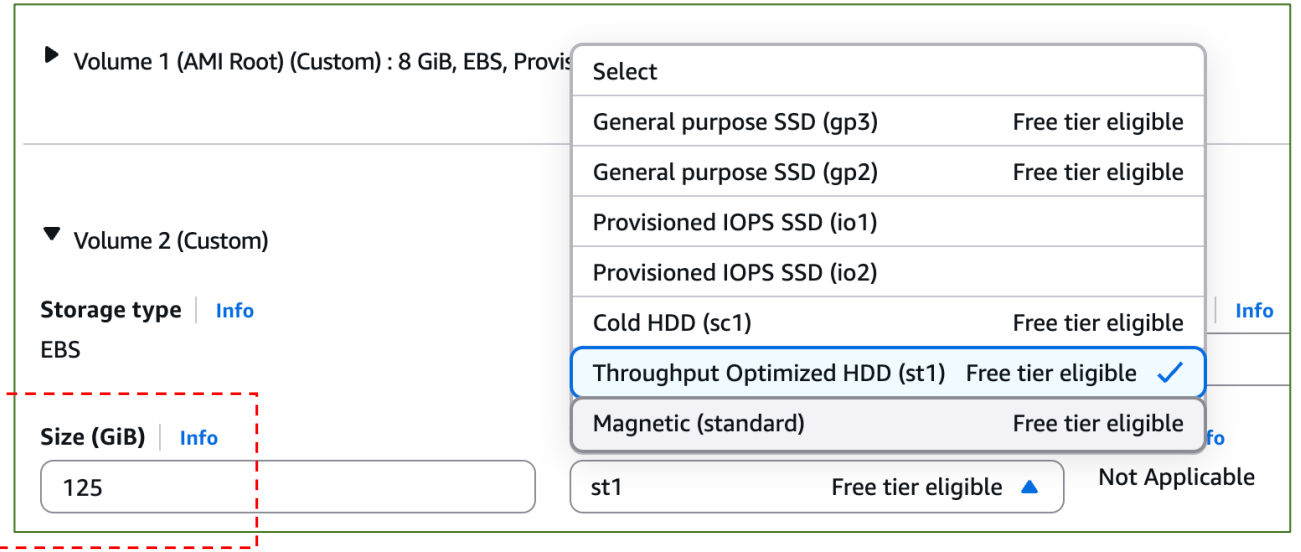
HDD-based EBS volumes provide low-cost storage optimized for **large, sequential throughput** workloads rather than high IOPS performance.

Throughput Optimized HDD (st1)

- Designed for large, sequential workloads requiring high throughput.
- Typical use cases:
  - Big data processing
  - Data warehouses
  - Log processing
- Performance:
  - Max throughput: ~500 MiB/s
  - Max IOPS: ~500

Cold HDD (sc1)

- Lowest-cost EBS storage option for infrequently accessed data.
- Suitable for workloads where cost is more important than performance.
- Performance:
  - Max throughput: ~250 MiB/s
  - Max IOPS: ~250



A minimum volume size of 125 GB

EC2 Storage

❖ Volume Types Summary - SSD

	Amazon EBS General Purpose SSD volumes		Amazon EBS Provisioned IOPS SSD volumes	
Volume type	gp3 [6]	gp2	io2 Block Express	io1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)		99.999% durability (0.001% annual failure rate)	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)
Use cases	• Transactional workloads • Virtual desktops • Medium-sized, single-instance databases • Low-latency interactive applications • Boot volumes • Development and test environments		Workloads that require: • Consistent sub-millisecond latency with average latency under 500 microseconds [5] • Sustained IOPS performance • More than 80,000 IOPS or 2,000 MiB/s of throughput	• Workloads that require sustained IOPS performance or more than 16,000 IOPS • I/O-intensive database workloads
Volume size	1 GiB - 64 TiB	1 GiB - 16 TiB	4 GiB - 64 TiB	4 GiB - 16 TiB
Max IOPS	80,000 [3] (64 KiB I/O [4])	16,000 (16 KiB I/O [4])	256,000 [3] (16 KiB I/O [4])	64,000 (16 KiB I/O [4])
Max throughput	2,000 MiB/s	250 MiB/s [1]	4,000 MiB/s	1,000 MiB/s [2]
Amazon EBS Multi-attach	Not supported		Supported	
NVMe reservations	Not supported		Supported	Not supported
Boot volume	Supported			

Ref: https://docs.aws.amazon.com/ebs/latest/userguide/ebs-volume-types.html

EC2 Storage

❖ Volume Types Summary - HDD

	Throughput Optimized HDD volumes	Cold HDD volumes
Volume type	st1	sc1
Durability	99.8% - 99.9% durability (0.1% - 0.2% annual failure rate)	
Use cases	• Big data • Data warehouses • Log processing	• Throughput-oriented storage for data that is infrequently accessed • Scenarios where the lowest storage cost is important
Volume size	125 GiB - 16 TiB	
Max IOPS per volume (1 MiB I/O)	500	250
Max throughput per volume	500 MiB/s	250 MiB/s
Amazon EBS Multi-attach	Not supported	
Boot volume	Not supported	

Ref: https://docs.aws.amazon.com/ebs/latest/userguide/ebs-volume-types.html

EC2 Storage

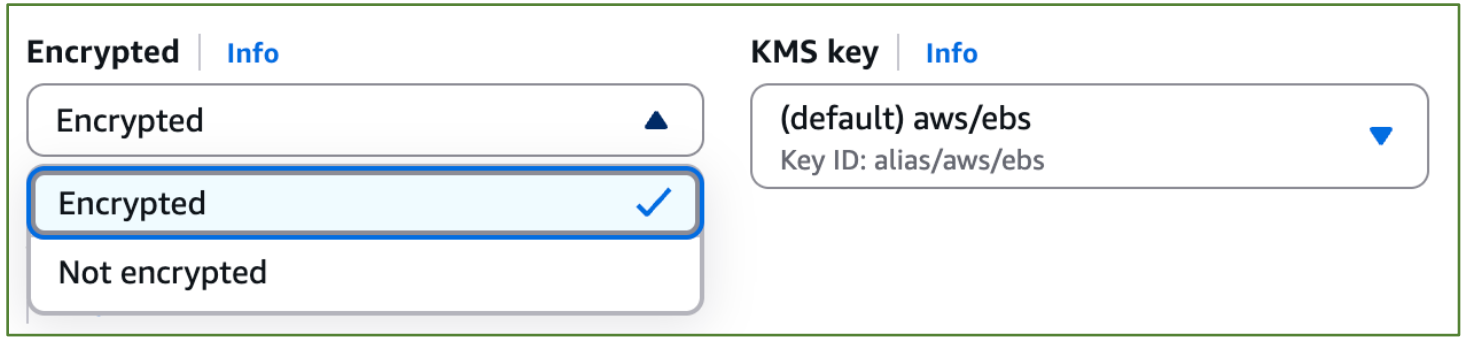
❖ EBS - Encryption

- ❑ Amazon EBS encryption protects data stored in EBS volumes and snapshots using **AWS Key Management Service (KMS).**
- ❑ It uses **AES-256** encryption and works automatically without changes to applications.

What gets encrypted ?

- Data at rest on the EBS volume
- Data in transit between the EC2 instance and the EBS volume
- All snapshots created from encrypted volumes
- New volumes created from encrypted snapshots

- You cannot change encryption on an existing volume, but you can copy a snapshot and encrypt it.
- Snapshots created from encrypted EBS volumes are automatically encrypted

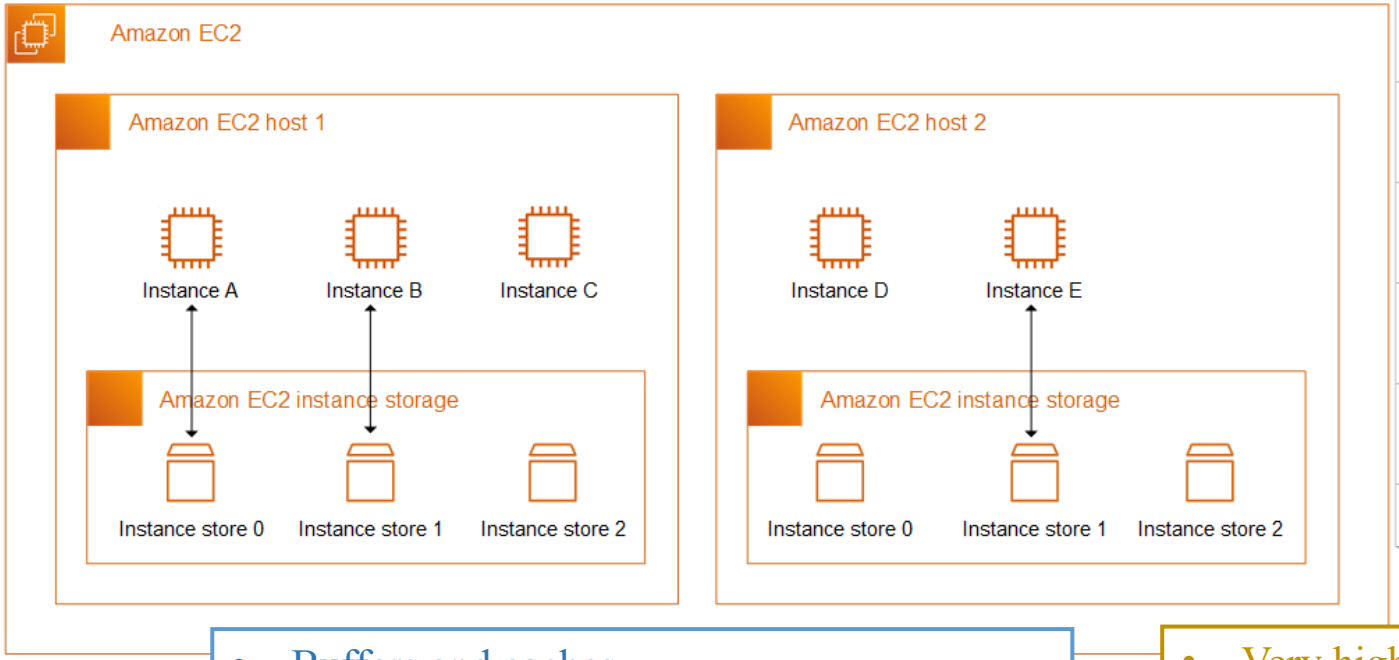


Ref: https://docs.aws.amazon.com/ebs/latest/userguide/ebs-encryption.html

EC2 Storage

❖ EC2 Instance Store

❑ EC2 Instance Store provides **temporary** block storage located on the **physical host machine** attached to the EC2 instance.



Instance type	Instance store volumes	Instance store type	100% random read IOPS / Write IOPS
		M5ad	
m5ad.large	1 x 75 GB	NVMe SSD	30,000 / 15,000
m5ad.xlarge	1 x 150 GB	NVMe SSD	59,000 / 29,000
m5ad.2xlarge	1 x 300 GB	NVMe SSD	117,000 / 57,000
m5ad.4xlarge	2 x 300 GB	NVMe SSD	234,000 / 114,000
m5ad.8xlarge	2 x 600 GB	NVMe SSD	466,666 / 233,334
m5ad.12xlarge	2 x 900	NVMe	700,000 /

- Buffers and caches
- Scratch data or temporary files
- High-performance temporary storage for compute workloads.

- Very high I/O performance
- Ephemeral storage - data is lost if the instance is stopped, terminated, or the underlying host fails.
- Available only on certain EC2 instance types

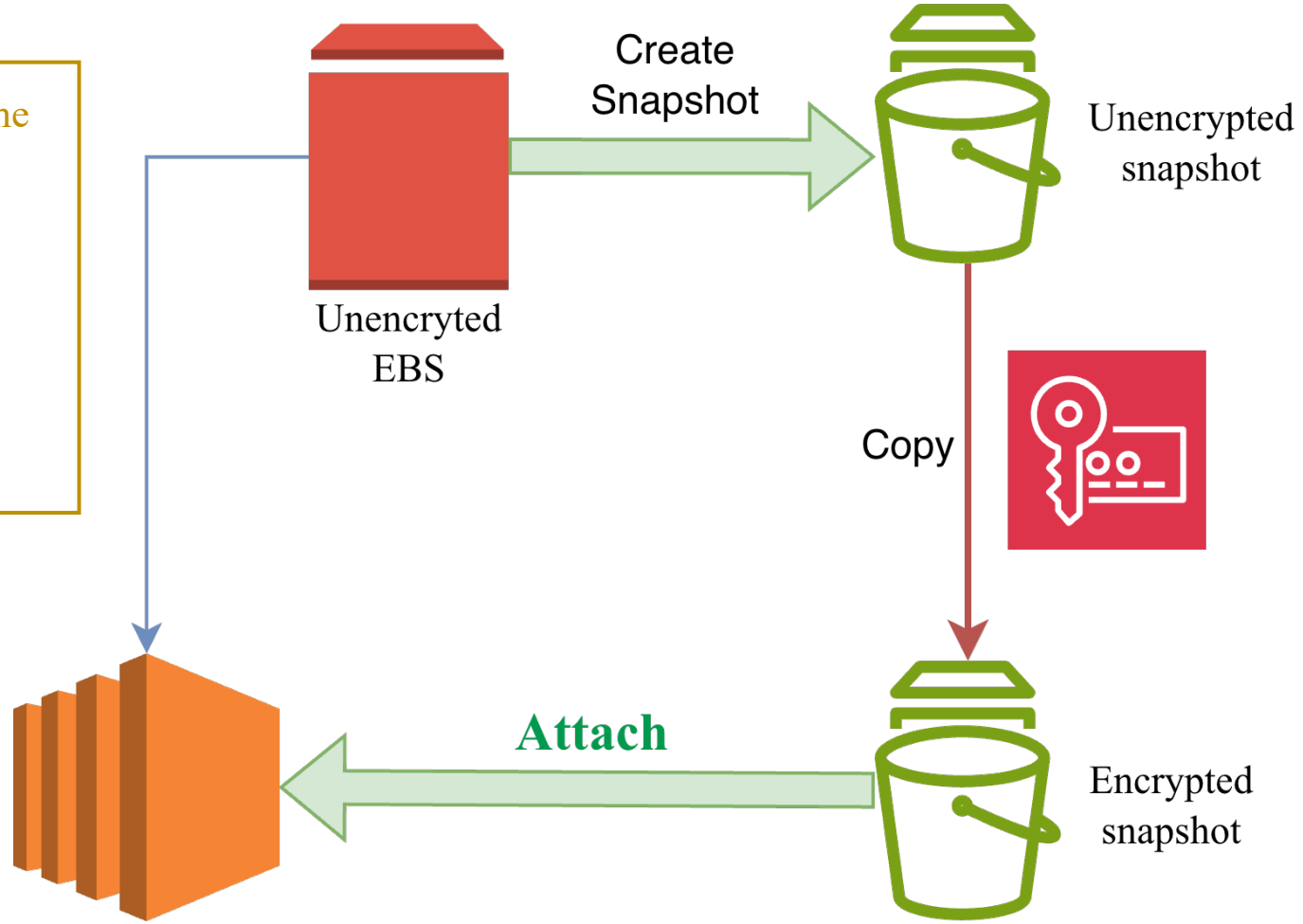
Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/InstanceStorage.html

Lab Session

Lab Session

❖ Encrypt an unencrypted EBS

1) Create a snapshot of the unencrypted EBS volume
2) Copy the snapshot with encryption enabled
3) Create a new encrypted EBS volume
4) Detach the old volume
5) Attach the encrypted volume
6) Start the instance



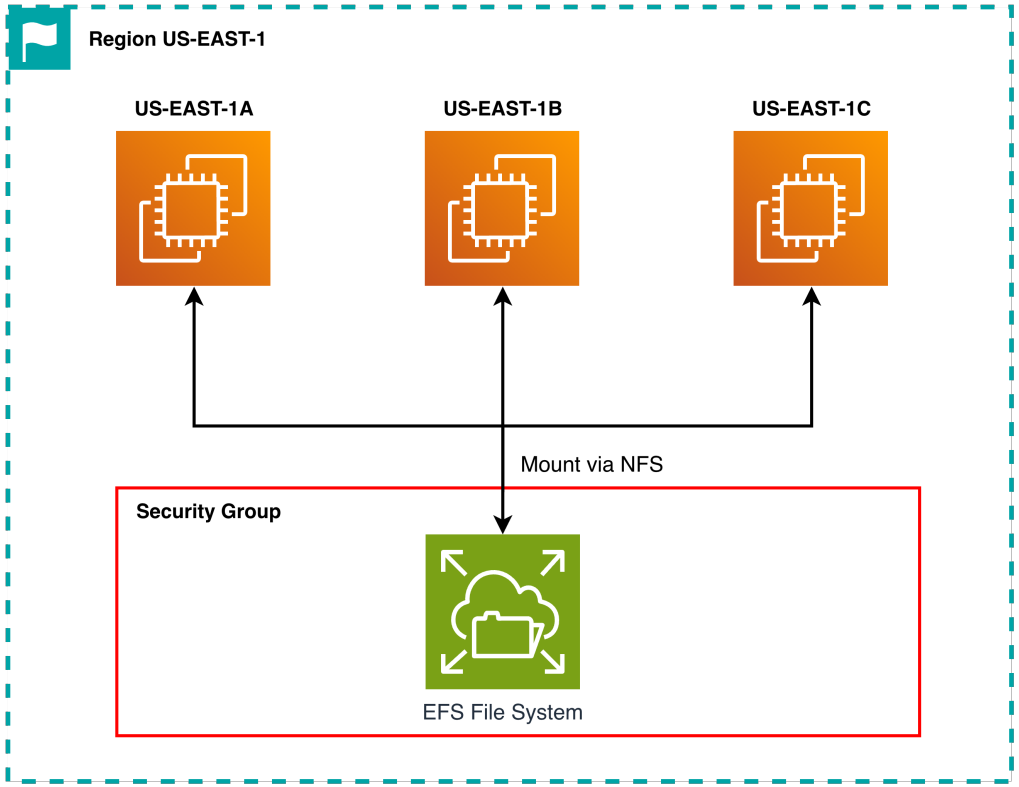
Amazon EFS

Amazon EFS

❖ Getting Started

❑ Amazon Elastic File System (EFS) is a serverless, fully managed **Network File System (NFS)**, elastic file storage service that allows **multiple compute resources** to share files concurrently.

- Can be mounted by multiple EC2 instances simultaneously.
- Works across multiple Availability Zones (Multi-AZ) for high availability.
- Automatically scales as storage grows.
- Typically more expensive than EBS (≈3× gp2/gp3). Uses NFSv4.1 protocol.
- Compatible with Linux-based AMIs and supports POSIX file system permissions.



Use case

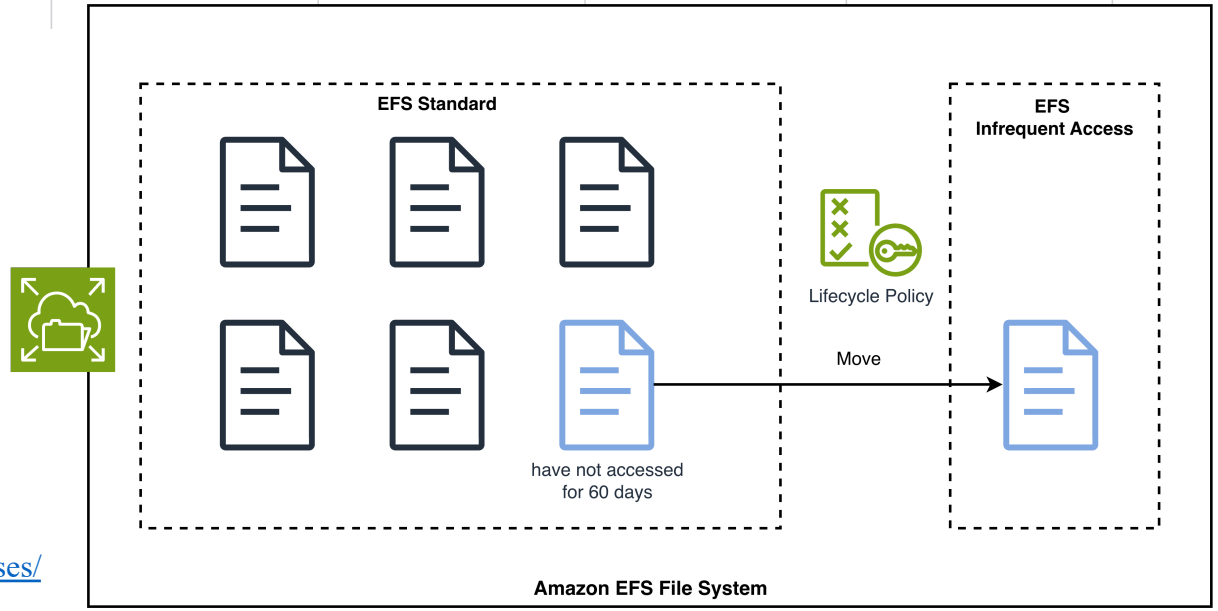
- Content management systems.
- Web serving and shared application data.
- NFSv4.1 protocol
- File sharing across multiple EC2 instances.

Ref: https://docs.aws.amazon.com/efs/latest/ug/whatisefs.html

Amazon EFS

❖ EFS - Storage Classes

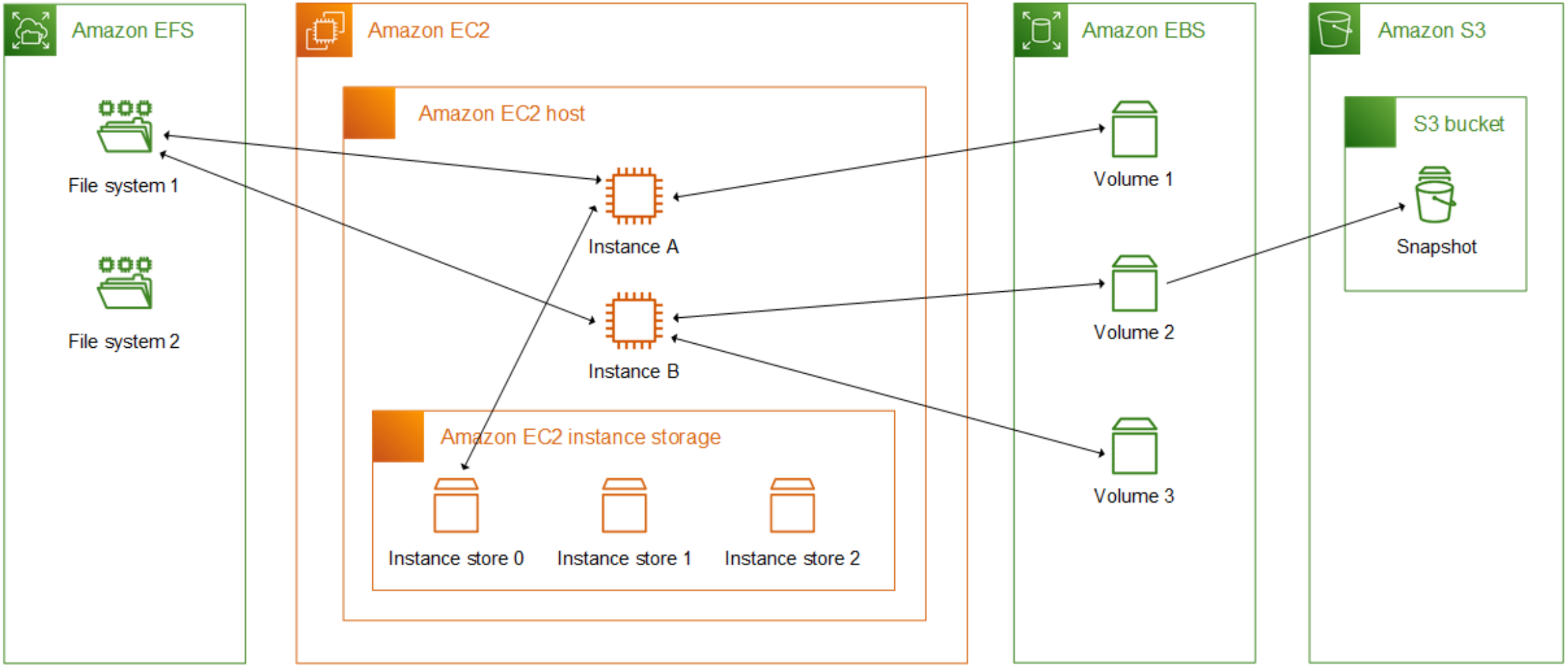
Storage class	Designed for	First byte read latency	Durability (designed for) (1)	Availability SLA	Availability zones	Minimum billing charge per file (2)	Minimum storage duration
EFS Standard	Active data requiring fast sub-millisecond latency performance	Sub-millisecond	99.999999999%(11 9's)	99.99% (Regional)99.9% (One Zone)	=>3 (Regional)1 (One Zone)	Not applicable	Not applicable
EFS Infrequent Access	Inactive data that is accessed only a few times each quarter	Tens of milliseconds	99.999999999%(11 9's)	99.99% (Regional) 99.9% (One Zone)	=>3 (Regional)1 (One Zone)	128 KiB	Not applicable
EFS Archive	Inactive data that is accessed a few times each year or less	Tens of milliseconds	99.999999999%(11 9's)	99.9% (Regional)	=>3 (Regional)	128 KiB	90 days



Ref: https://aws.amazon.com/efs/storage-classes/

Amazon EFS

❖ EBS vs EFS vs Instance store

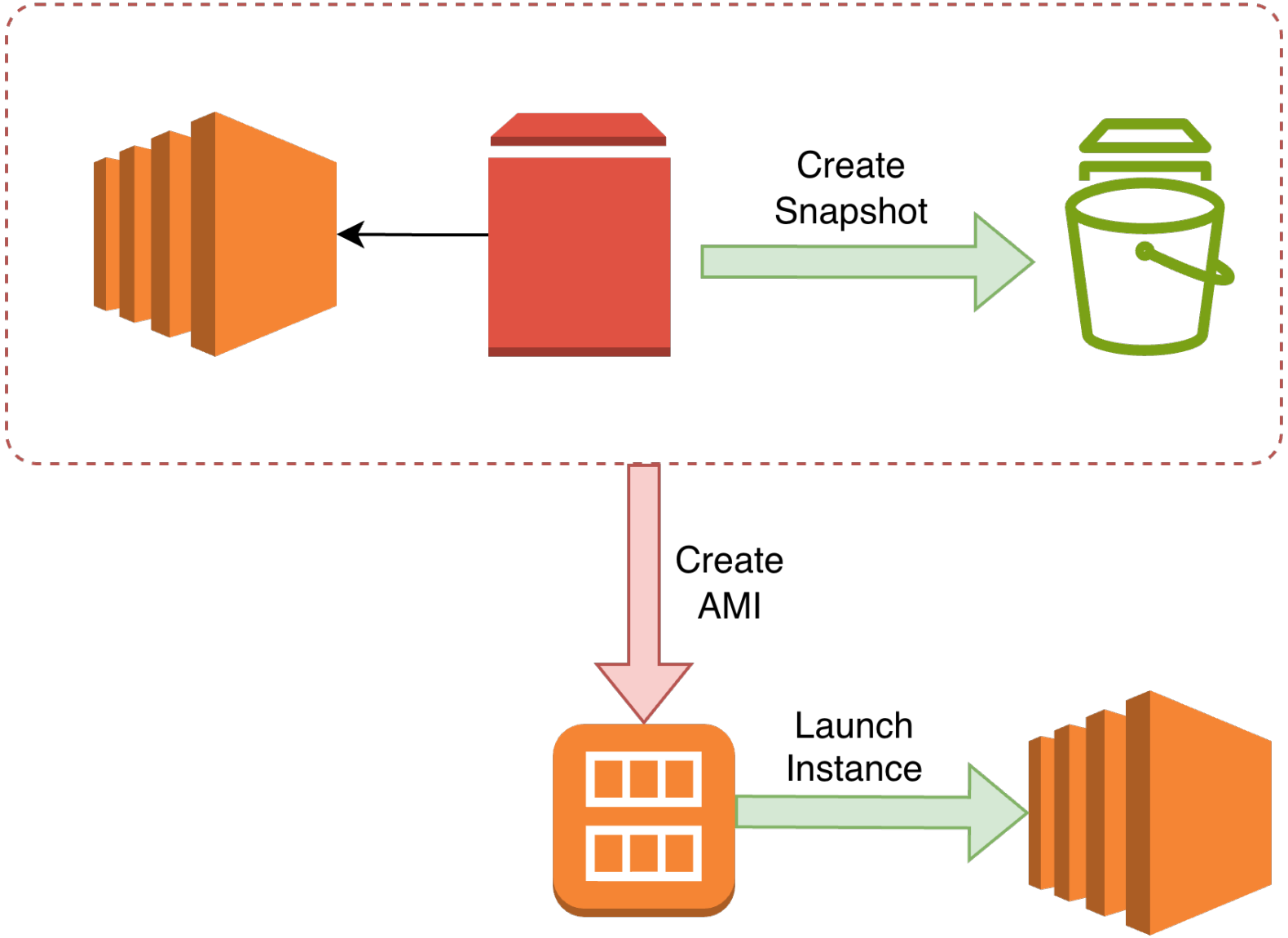


Ref: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/Storage.html

Lab Session

Lab Session

❖ Creat AMI and Launch Instance



Scalability & High Availability

Scalability & High Availability

❖ Scalability

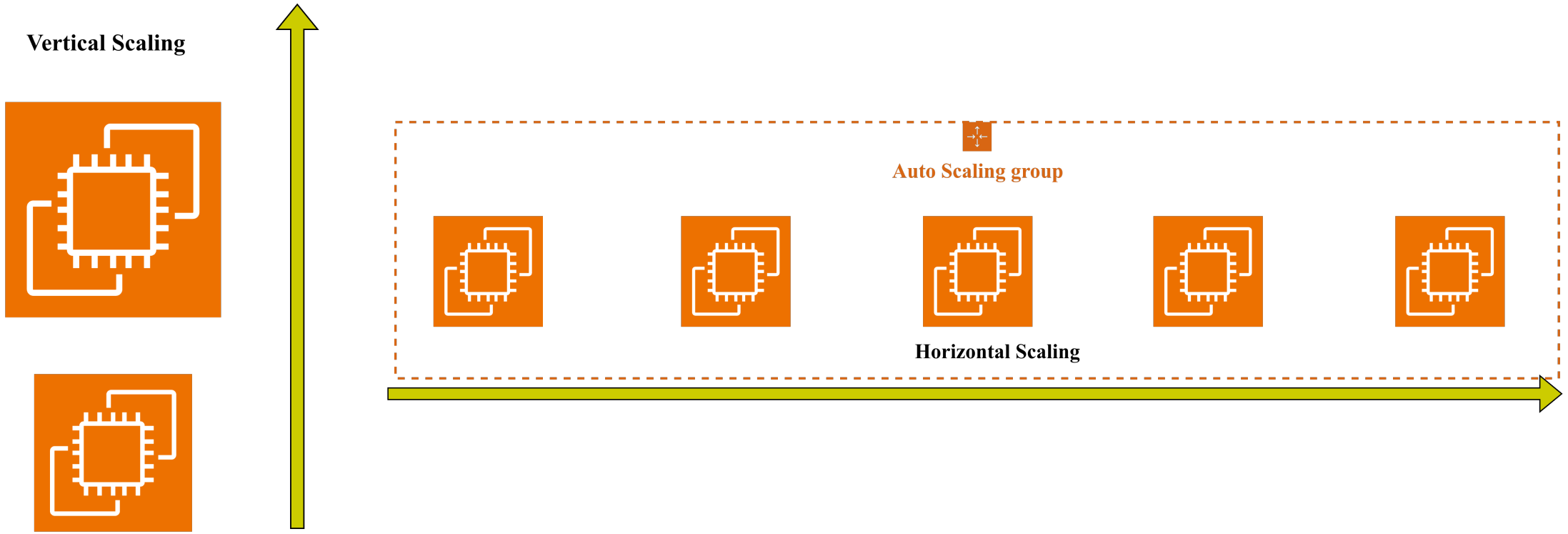
❑ Scalability is the ability of a system to handle increasing workloads by adding or adjusting resources.

Vertical Scalability (Scale Up)

Increase the capacity of a single instance (e.g., more CPU, RAM).

Horizontal Scalability (Scale Out / Elasticity)

Add more instances or servers to distribute the workload.

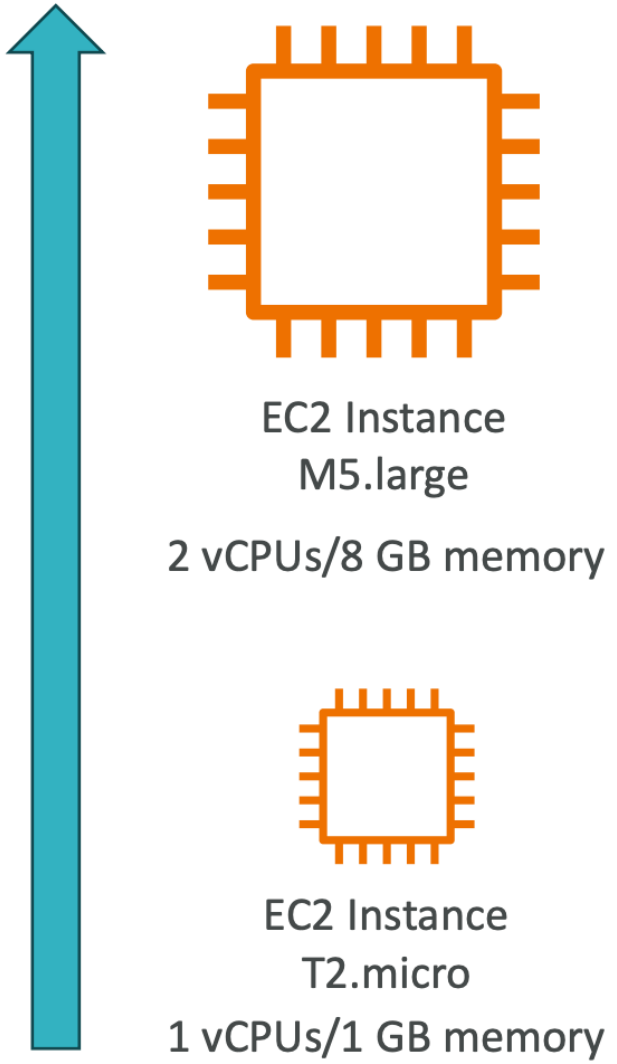


Scalability & High Availability

❖ Vertical Scalability

❑ **Vertical scalability (scale up / scale down)** means **increasing or decreasing the resources of a single instance**, such as CPU, RAM, or storage

- Typically used for non-distributed systems, such as databases.
- Amazon RDS and ElastiCache are services that support vertical scaling.
- Improves performance by upgrading the capacity of one machine instead of adding more machines.



Scalability & High Availability

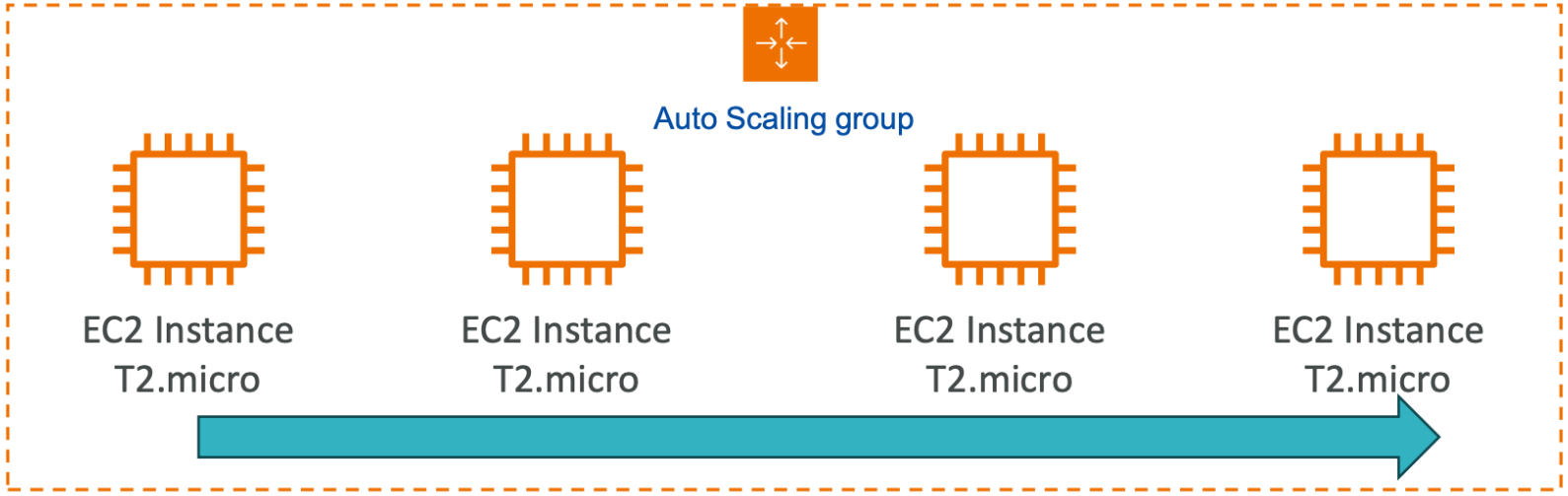
❖ Horizontal Scalability

❑ Horizontal scalability (scale out / scale in) means adding or removing instances to handle workload changes.

- Requires a distributed system architecture.
- Workload is distributed across multiple servers or instances.
- Improves performance, scalability, and fault tolerance.

Use case

- Web applications
- Microservices architectures
- Cloud-native and modern distributed systems.



Scalability & High Availability

❖ High Availability

❑ High Availability (HA) ensures a system remains operational even when some components fail.

Key idea

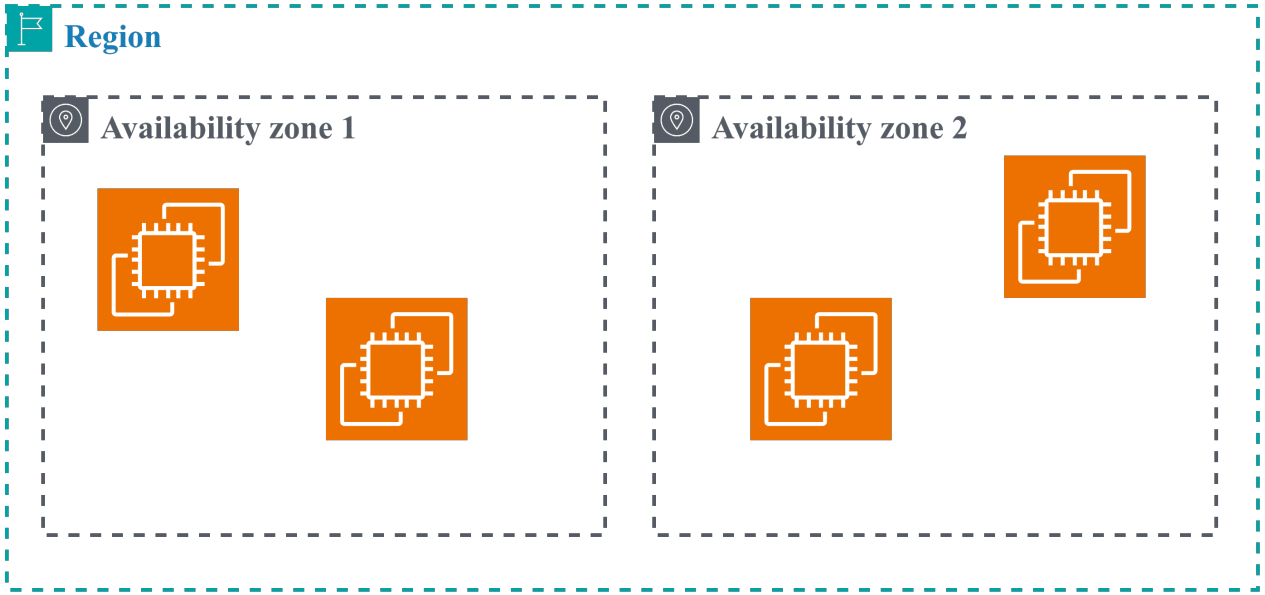
- Scalability focuses on handling growth in demand.
- High Availability focuses on maintaining service reliability and uptime.

- Typically implemented by running systems in multiple Availability Zones (AZs) – at least 2 data centers.
- The goal is to continue operating even if one data center fails.

Types of High Availability

Passive HA
- A standby instance is kept ready for failover.
- Example: Amazon RDS Multi-AZ.

Active HA
- Multiple active instances handle traffic simultaneously.
- Common in horizontally scaled architectures (e.g., with load balancers).



Scalability & High Availability

❖ High Availability & Scalability For EC2

Vertical Scalability (Scale Up)

Increase or decrease the size of an EC2 instance.

- FROM: t2.nano – 0.5 GB RAM, 1 vCPU
- TO: u-12tb1.metal – up to 12.3 TB RAM, 448 vCPUs

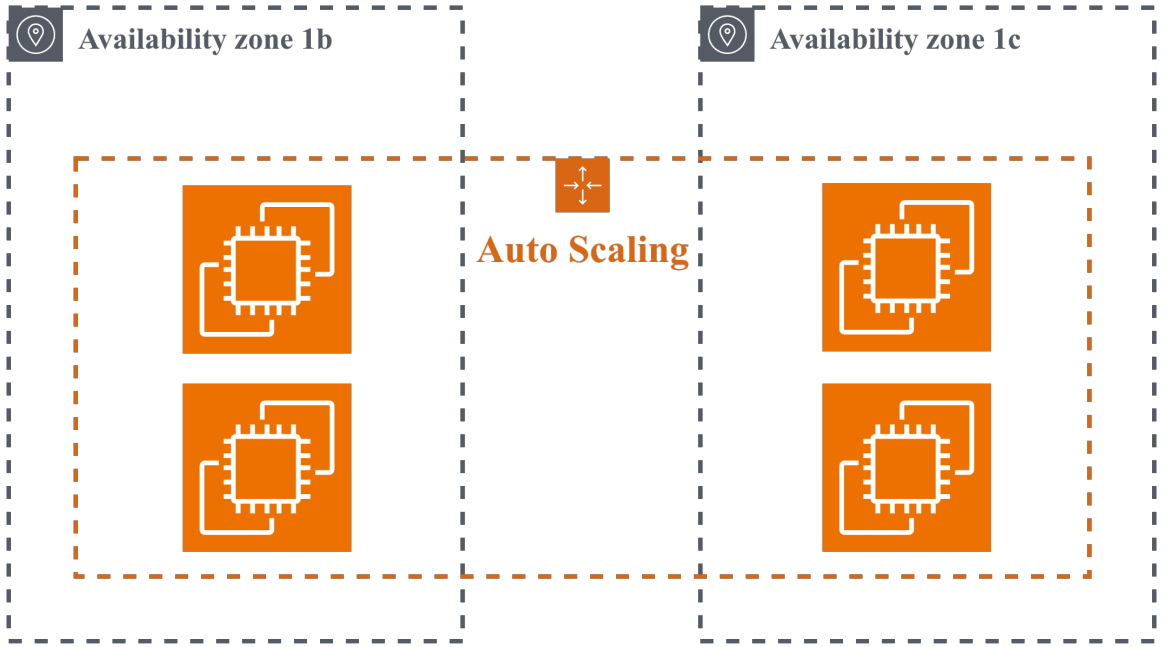
Horizontal Scalability (Scale Out / In)

Increase or decrease the size of an EC2 instance.

- Auto Scaling Groups (ASG)
- Elastic Load Balancers (ELB)

High Availability

- Deploy instances across multiple Availability Zones (AZs).
- Ensures the application continues running if one AZ fails.
  - Auto Scaling Group across multiple AZs
  - Load Balancer distributing traffic across AZs

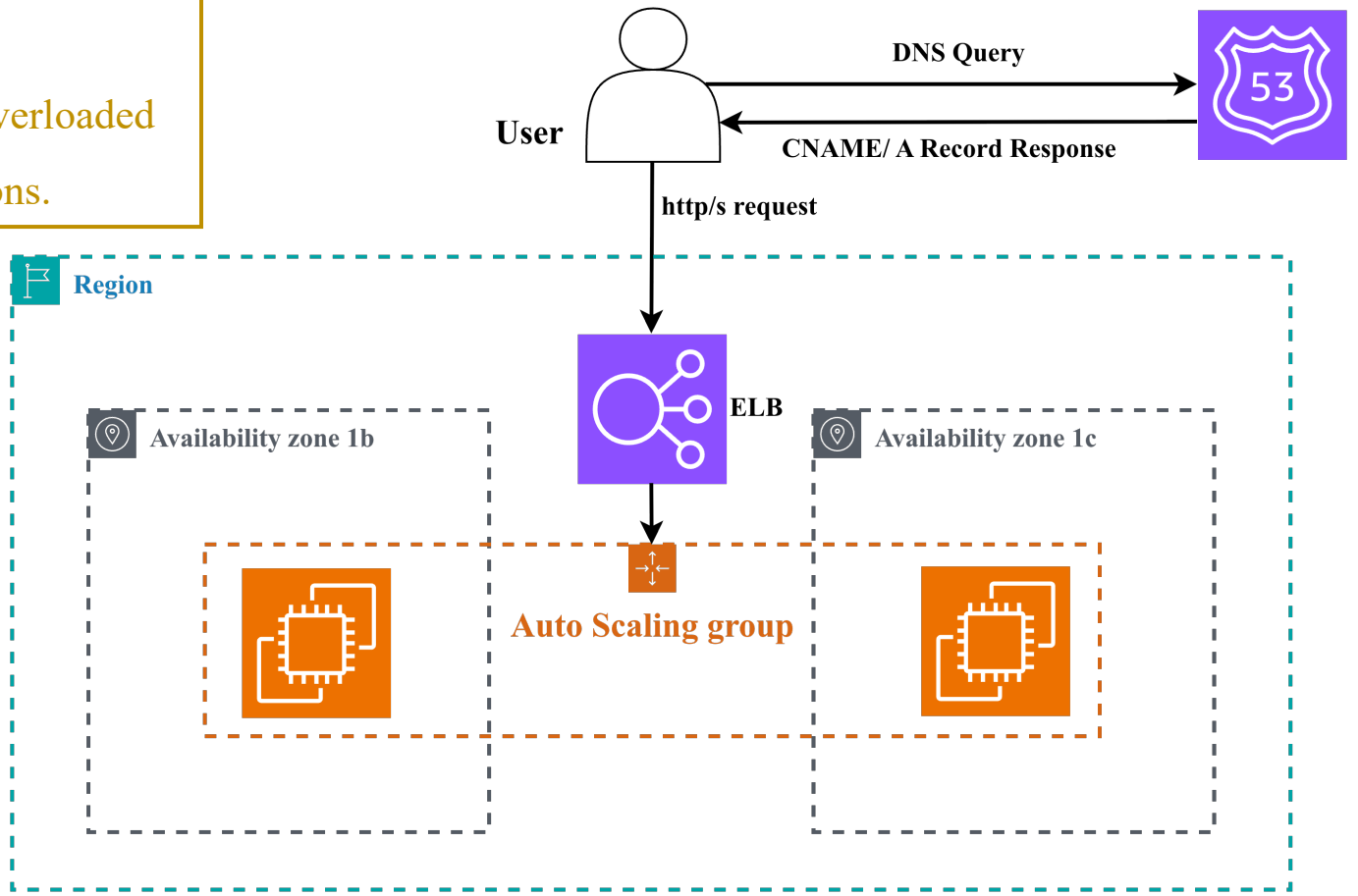


Scalability & High Availability

❖ Load Balancing

❑ Load balancing is the process of **distributing** incoming network traffic across **multiple servers**.

- Improves availability and reliability
- Prevents any single server from being overloaded
- Supports horizontal scaling of applications.



Scalability & High Availability

❖ Load Balancing

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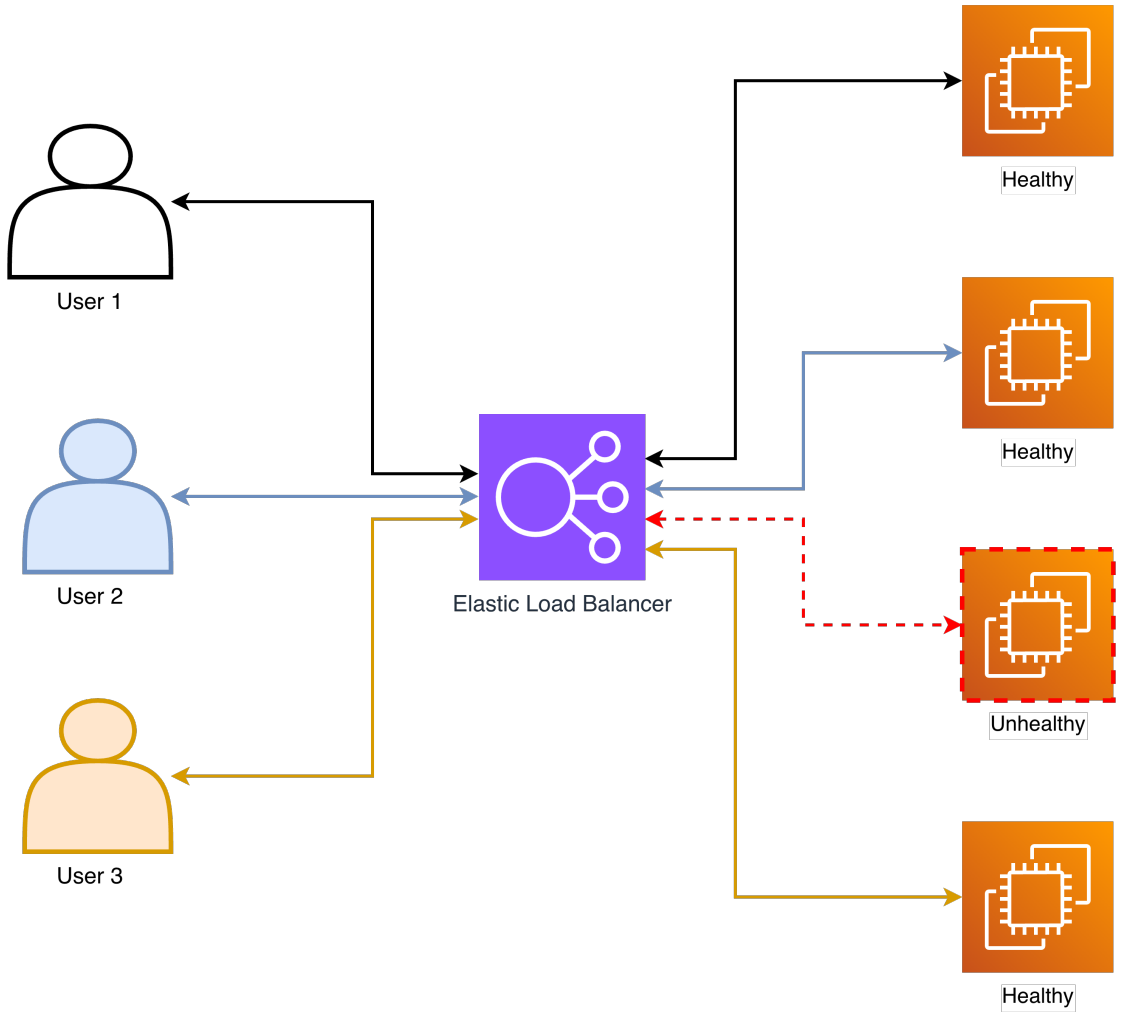


Elastic Load Balancer

Elastic Load Balancer

❖ Elastic Load Balancer

❑ Elastic Load Balancer (ELB) is a fully managed load balancing service provided by AWS.

Key benefits

- Highly reliable – AWS manages infrastructure to ensure availability.
- Automatic maintenance and updates – AWS handles upgrades, patching, and scaling.
- Built-in high availability across multiple Availability Zones.
- Simple configuration with only a few required settings.

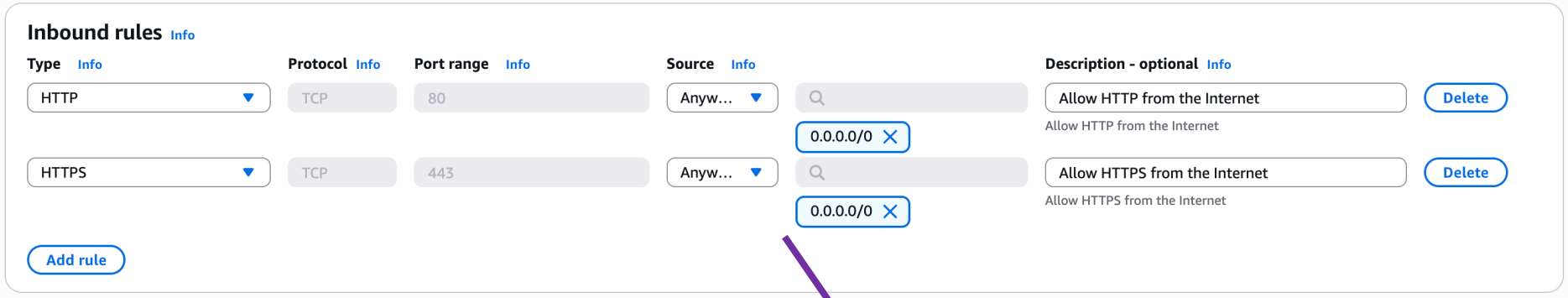
Integration with AWS services

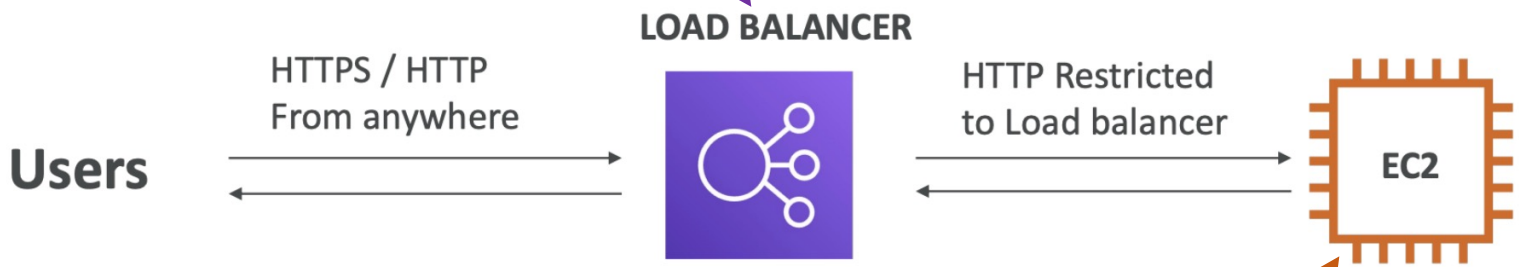
- Amazon EC2 and Auto Scaling Groups
- Amazon ECS and container services
- AWS Certificate Manager (ACM) for SSL/TLS Amazon
- CloudWatch for monitoring
- Amazon Route 53, AWS WAF, and AWS Global Accelerator.

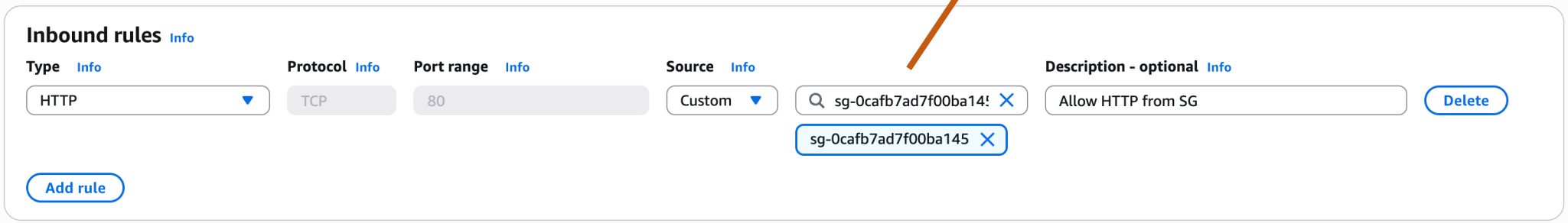
Ref: https://docs.aws.amazon.com/elasticloadbalancing/latest/userguide/what-is-load-balancing.html

Elastic Load Balancer

❖ Security Groups







Elastic Load Balancer

❖ Types of Load Balancer

Feature	Application Load Balancer	Network Load Balancer	Gateway Load Balancer	Classic Load Balancer
Load Balancer type	Layer 7	Layer 4	Layer 3 Gateway + Layer 4 Load Balancing	Layer 4/7
Target type	IP, Instance, Lambda	IP, Instance, Application Load Balancer	IP, Instance	-
Terminates flow/proxy behavior	Yes	Yes	No	Yes
Protocol listeners	HTTP, HTTPS, gRPC	TCP, UDP, TLS	IP	TCP, SSL/TLS, HTTP, HTTPS
Reachable via	VIP	VIP	Route table entry	-

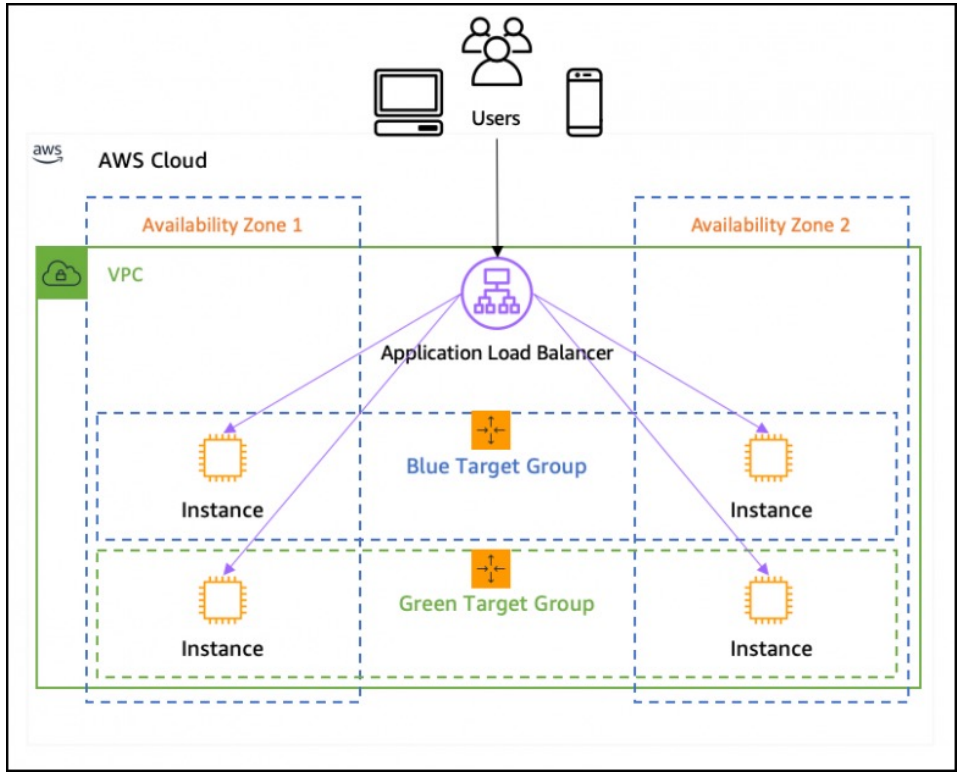
Ref: https://aws.amazon.com/elasticloadbalancing/features/#Product_comparisons

Elastic Load Balancer

❖ Application Load Balancer

- ❑ Application Load Balancer operates at Layer 7 (Application layer) of the OSI model.
- ❑ Designed for HTTP and HTTPS traffic and advanced request routing

- Routes traffic to multiple target groups (e.g., EC2 instances, containers, IP addresses).
- Supports multiple applications on the same server, commonly used with containers (ECS, Kubernetes).
- Supports modern protocols such as HTTP/2 and WebSocket.
- Can perform HTTP → HTTPS redirects and other routing rules.



Use case

- Microservices architectures
- Container-based applications
- Modern web applications requiring advanced routing rules.

Ref: https://docs.aws.amazon.com/elasticloadbalancing/latest/application/introduction.html

Elastic Load Balancer

❖ Routing capabilities

Path-based routing

- example.com/users → Service A
- example.com/posts → Service B

Host-based routing

- one.example.com → Service A
- other.example.com → Service B

Request-based routing

Based on query strings or HTTP headers

- example.com/users?id=123&order=false

- ❑ Ideal for microservices architectures.
- ❑ Works well with container platforms such as Docker and Amazon ECS.
- ❑ Supports dynamic port mapping, allowing traffic to be routed to containers running on different ports.

Elastic Load Balancer

❖ Target Groups

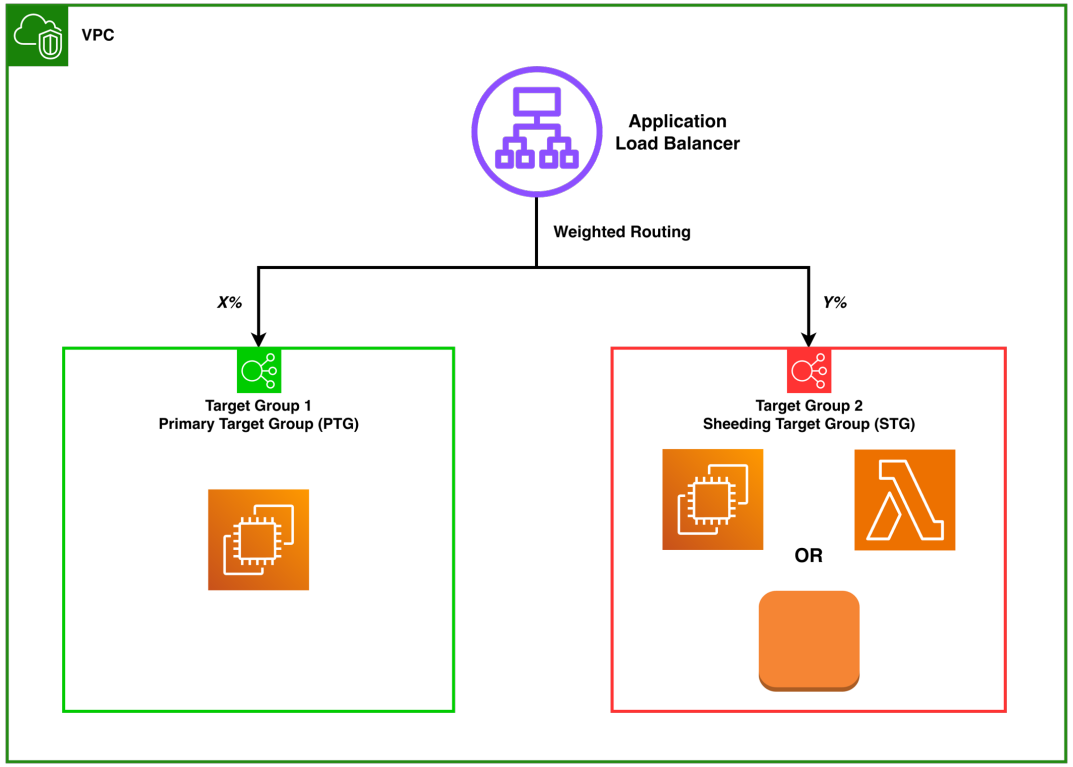
- ❑ Target groups define where the ALB forwards incoming requests.
- ❑ An ALB can route traffic to **multiple target groups**.

Supported targets

- EC2 instances (often managed by Auto Scaling Groups) – HTTP/HTTPS
- ECS tasks (managed by Amazon ECS) – HTTP/HTTPS
- AWS Lambda functions – HTTP request is converted into a JSON event
- IP addresses – must be private IPs

Health checks

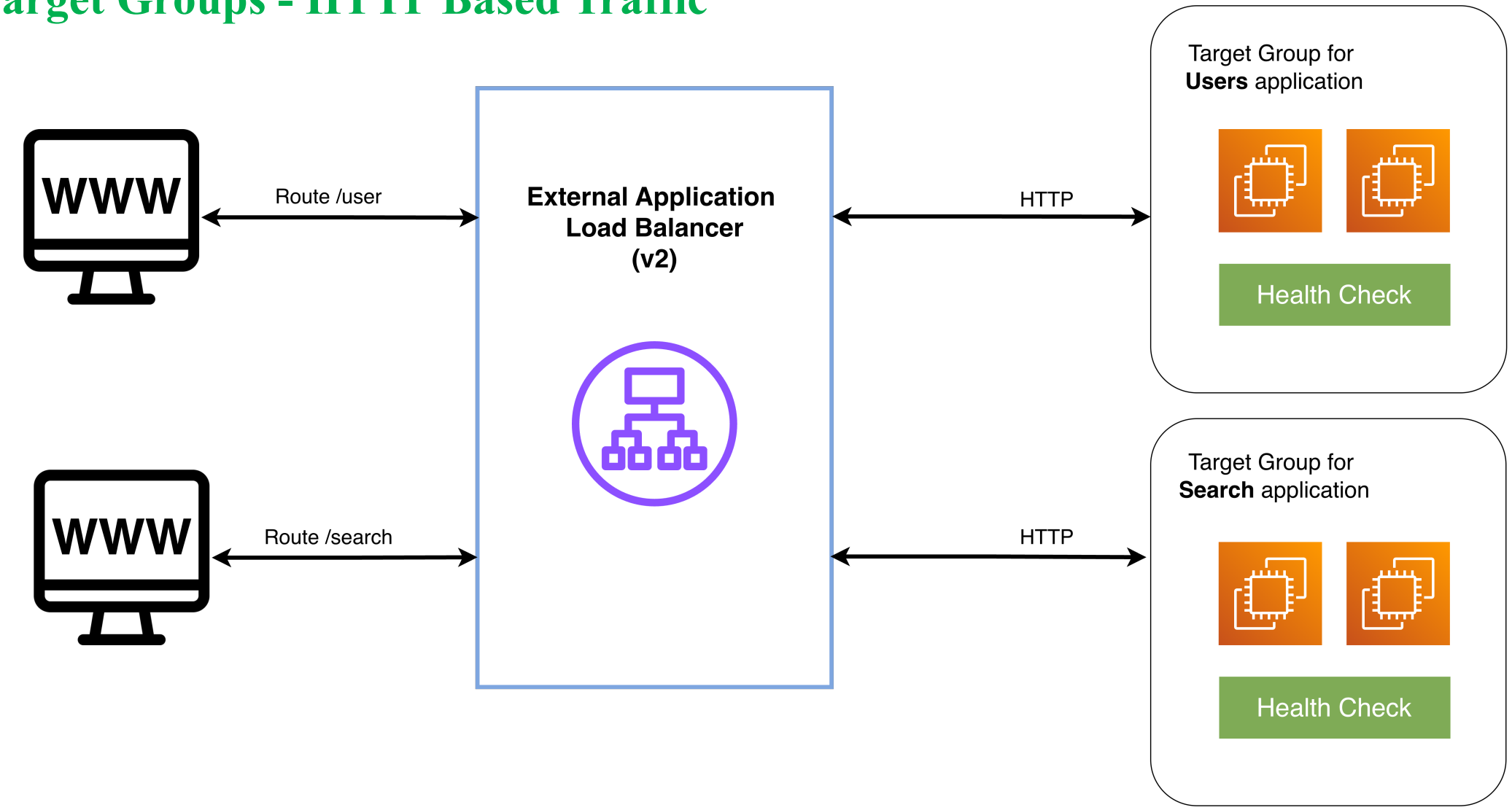
- Health checks are configured at the target group level.
- Only healthy targets receive traffic from the load balancer.



Ref: https://docs.aws.amazon.com/elasticloadbalancing/latest/application/load-balancer-target-groups.html

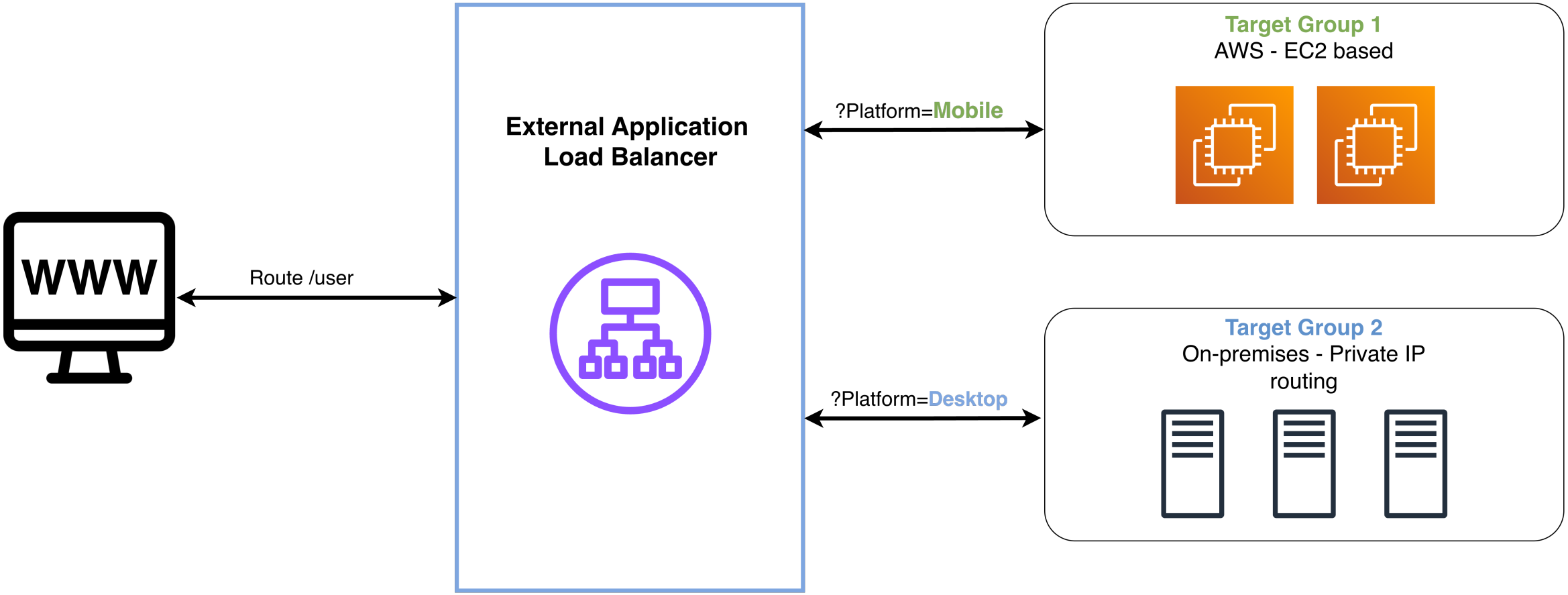
Elastic Load Balancer

❖ Target Groups - HTTP Based Traffic



Elastic Load Balancer

❖ Target Groups - Query Strings/Parameters Routing



Elastic Load Balancer

❖ Network Load Balancer

- ❑ Network Load Balancer operates at Layer 4 (Transport layer) of the OSI model.
- ❑ Designed for **ultra-high performance** and low-latency traffic handling.

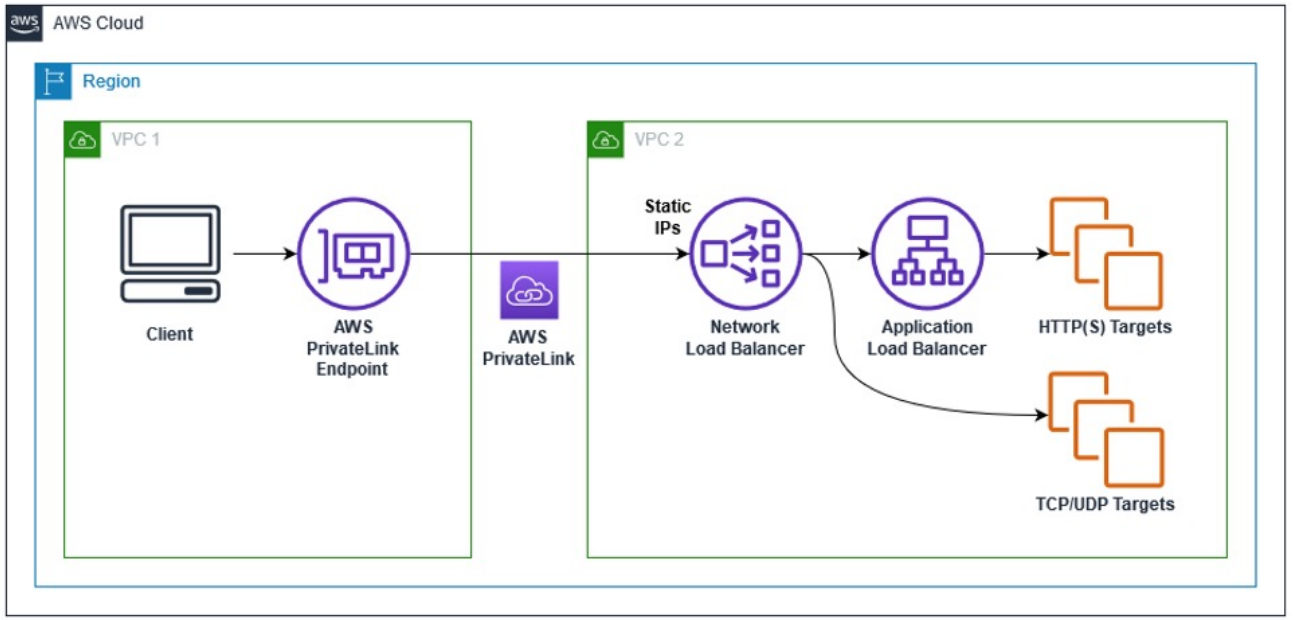
- Supports TCP, UDP, and TLS traffic forwarding to targets.
- Can handle millions of requests per second with very low latency.
- Provides static IP addresses per Availability Zone and supports Elastic IP assignment (useful for IP whitelisting).

Noticed

Network Load Balancer is not included in the AWS Free Tier.

Use case

- Applications requiring extreme performance and low latency.
- Workloads using TCP or UDP protocols.
- Systems needing static IP addresses for networking or security policies.



Ref: https://docs.aws.amazon.com/elasticloadbalancing/latest/network/introduction.html

Elastic Load Balancer

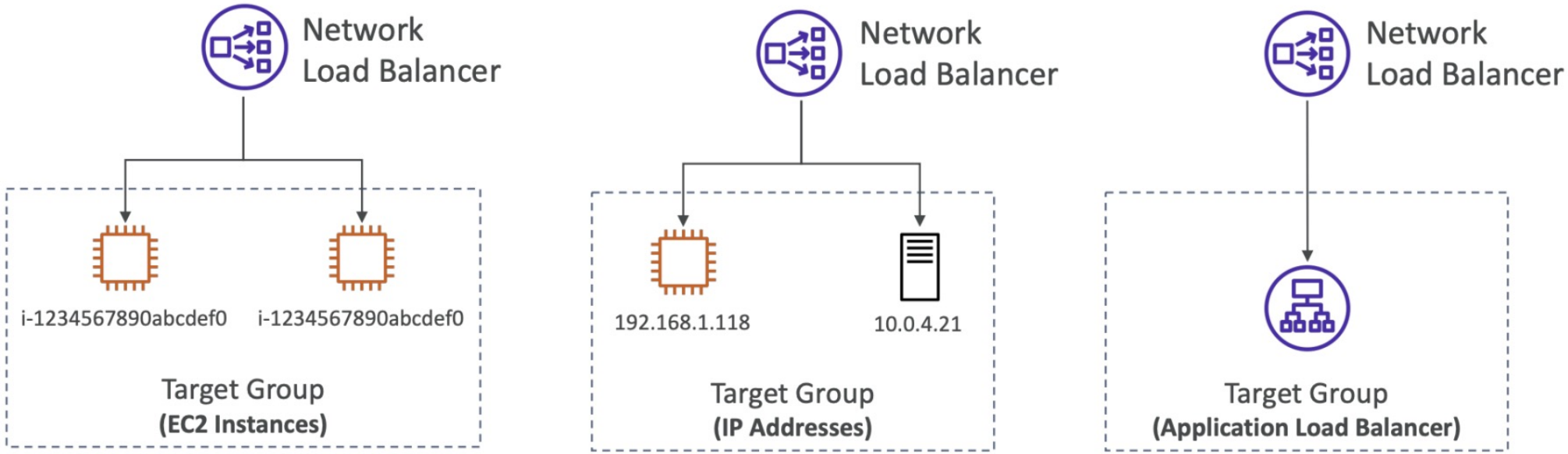
❖ NLB - Target Groups

Supported targets

- EC2 instances
- IP addresses (must be private IPs)
- Application Load Balancer (ALB)

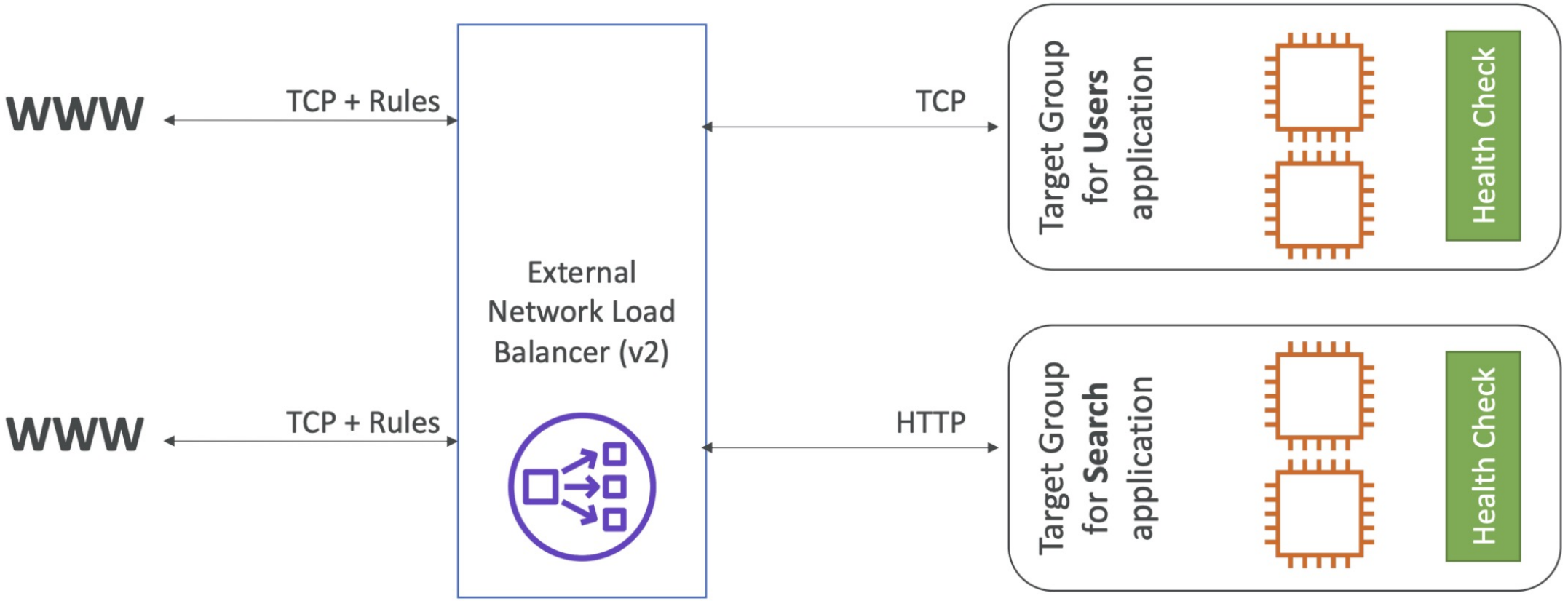
Health checks

- NLB supports health checks using:
  - TCP
  - HTTP
  - HTTPS
- Only healthy targets receive traffic from the load balancer.



Elastic Load Balancer

❖ NLB – TCP (Layer 4) Based Traffic



Elastic Load Balancer

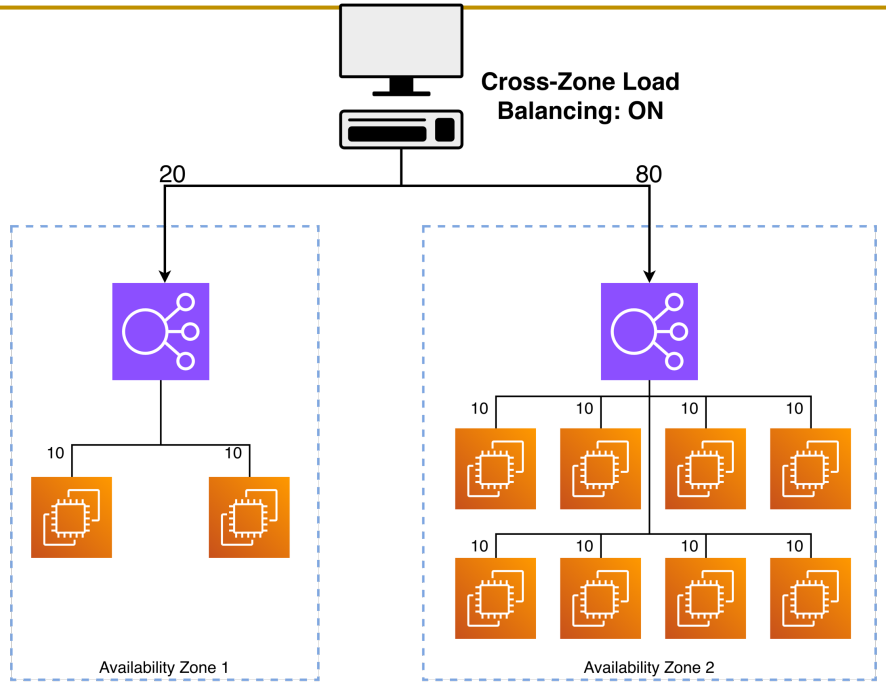
❖ Cross-Zone Load Balancing

Ref: https://docs.aws.amazon.com/elasticloadbalancing/latest/classic/enable-disable-crosszone-lb.html

❑ Cross-Zone Load Balancing allows a load balancer to distribute traffic evenly across all registered targets in all Availability Zones (AZs).

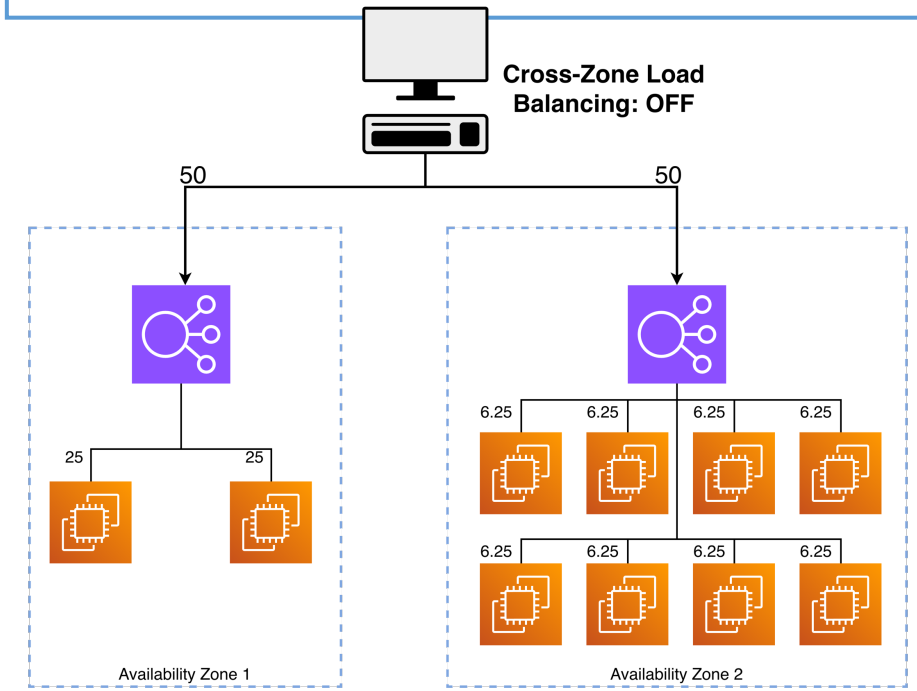
With Cross-Zone Load Balancing

- Each load balancer node distributes requests across targets in all AZs.
- Results in more balanced traffic distribution.



Without Cross-Zone Load Balancing

- Each load balancer node only distributes traffic to targets in its own AZ.
- Traffic distribution may become uneven if AZs have different numbers of instances.

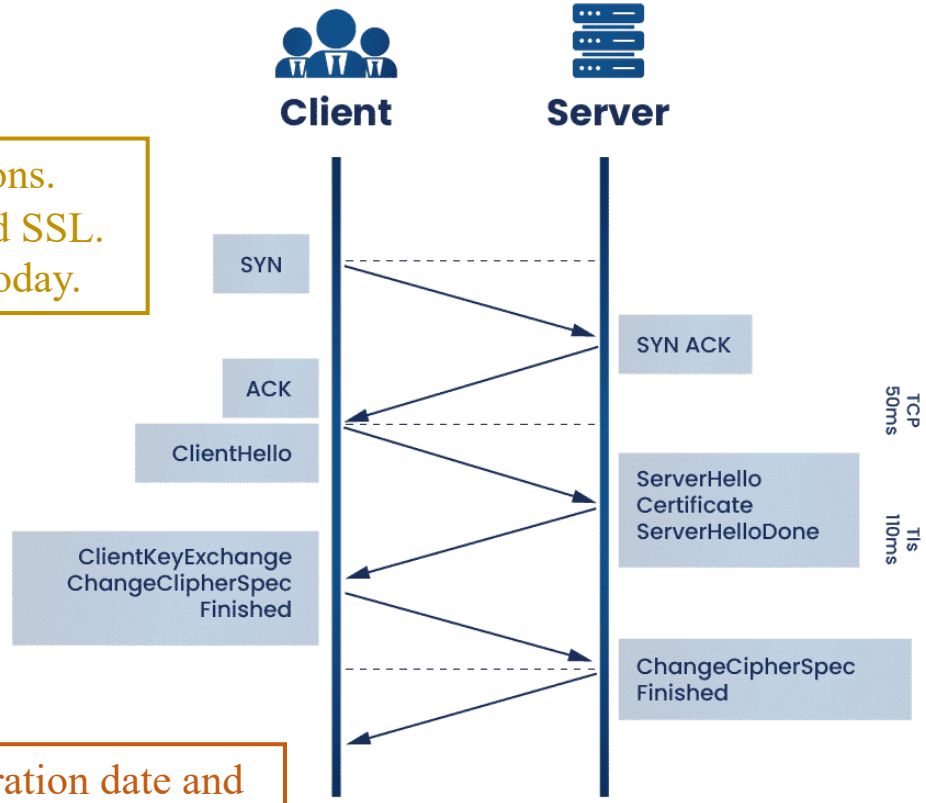


Elastic Load Balancer

❖ SSL/TLS

❑ SSL/TLS certificates encrypt traffic between clients and servers (or load balancers) to protect data in transit.

- SSL (Secure Sockets Layer) – older protocol used to secure network connections.
- TLS (Transport Layer Security) – newer and more secure version that replaced SSL.
- In practice, people often still say "SSL certificate", even though TLS is used today.



Certificate Authorities (CA)

Public certificates are issued by trusted Certificate Authorities, such as:

- DigiCert
- GlobalSign
- Let's Encrypt
- GoDaddy Sectigo (Comodo)

Important note

SSL/TLS certificates have an expiration date and must be renewed periodically.

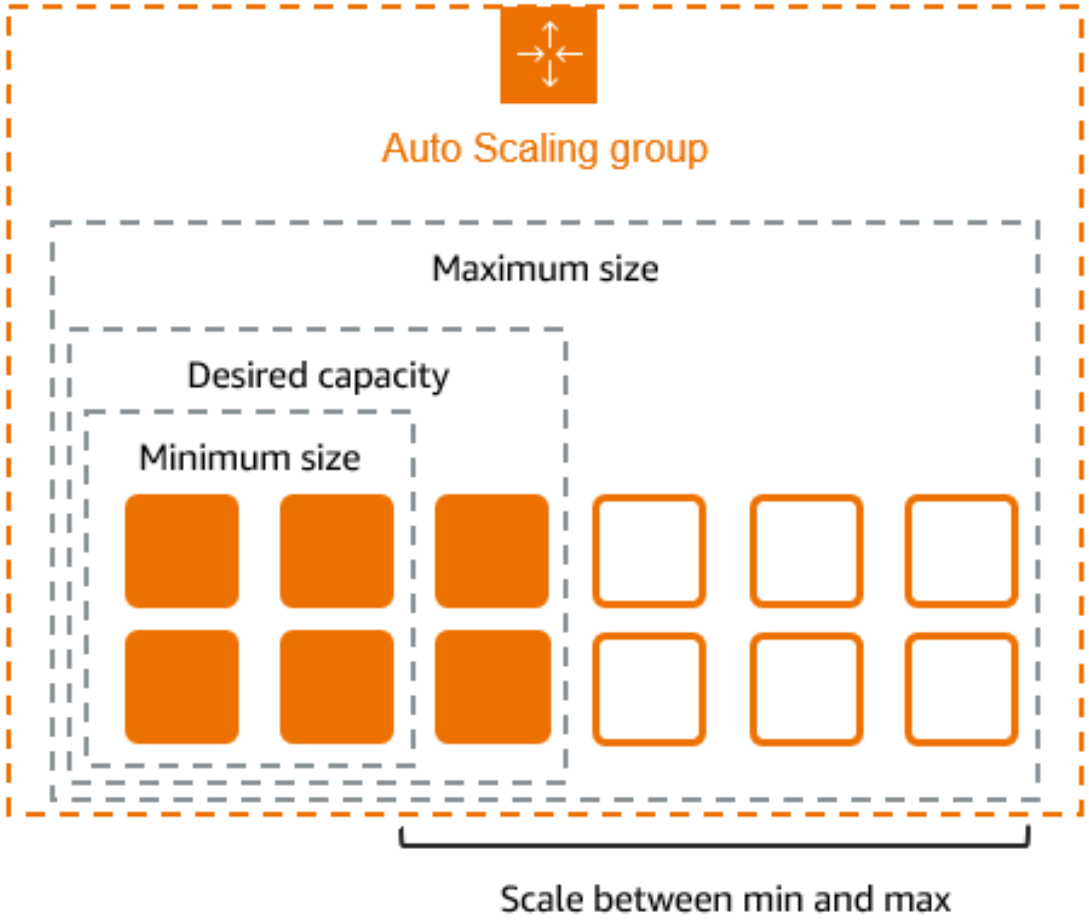
Auto Scaling Group

Auto Scaling Group

❖Getting Started

❑ An Auto Scaling Group (ASG) automatically **manages a group** of EC2 instances to handle changing application demand.

- Scale out – add EC2 instances when demand increases.
- Scale in – remove EC2 instances when demand decreases.
- Maintain a defined minimum, desired, and maximum number of instances.
- Automatically register new instances with a load balancer.
- Replace unhealthy instances automatically.



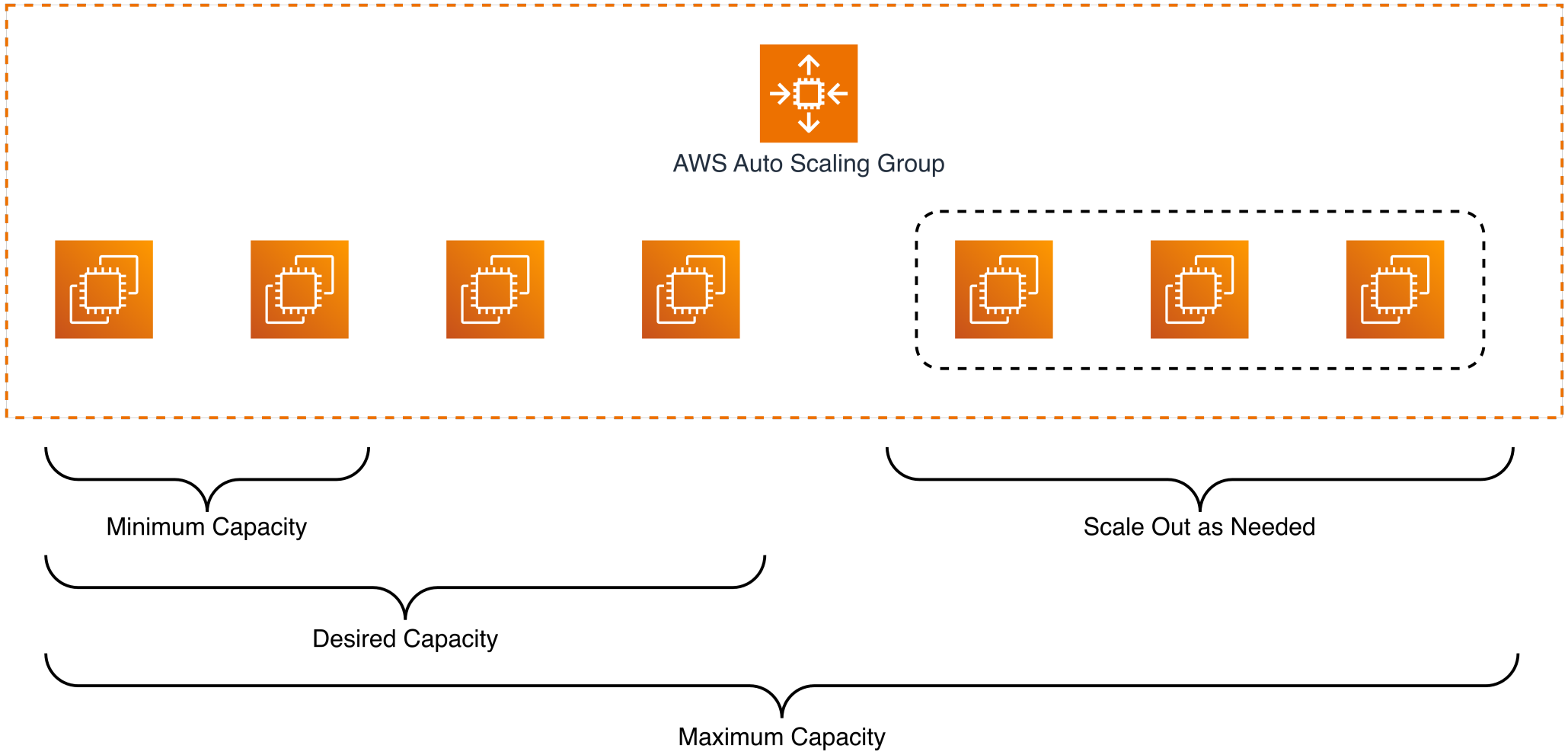
Cost

Auto Scaling itself is free – you only pay for the EC2 instances and resources used.

Ref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/auto-scaling-groups.html

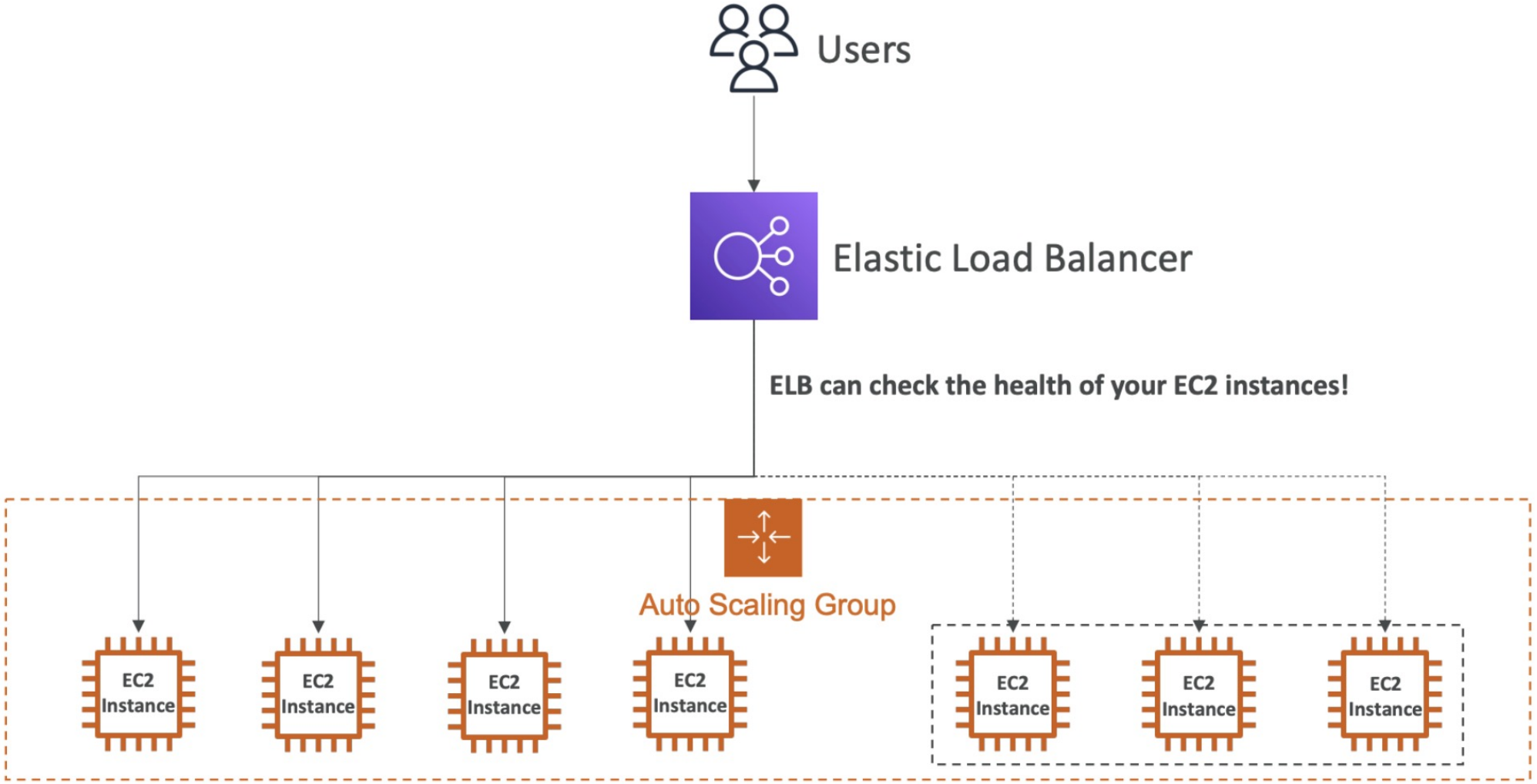
Auto Scaling Group

❖ AGS in AWS With Load Balancer



Auto Scaling Group

❖ ASG in AWS With Load Balancer



Auto Scaling Group

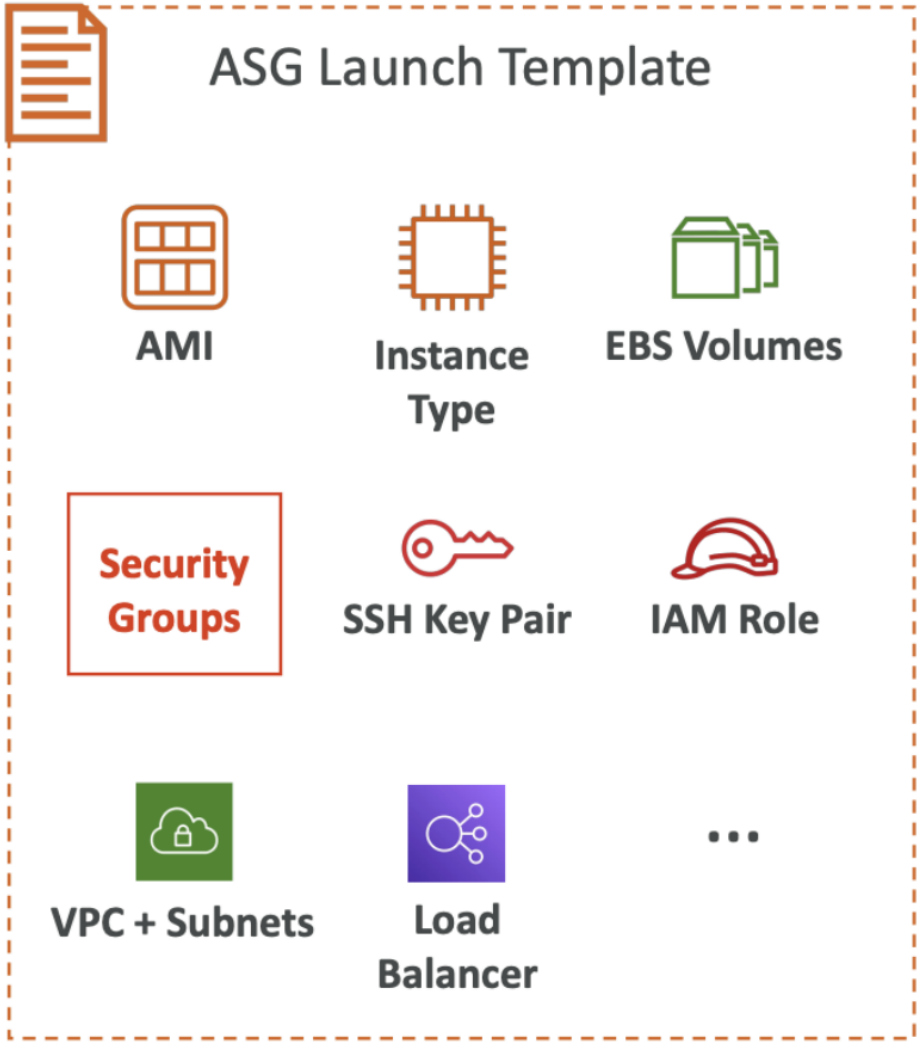
❖ Attributes

❑ **Launch Template d**efines how EC2 instances are created.

- AMI and instance type
- EC2 User Data (startup scripts)
- EBS volume configuration
- Security groups
- SSH key pair
- IAM role for the EC2 instances
- Network configuration and subnets
- Load Balancer information

Additional ASG settings

- Load balancer configuration
- Minimum, maximum, and desired capacity
- Scaling policies (rules for scaling in or out)



Ref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/launch-templates.html

Auto Scaling Group

❖ AGS with CloudWatch Alarms & Scaling

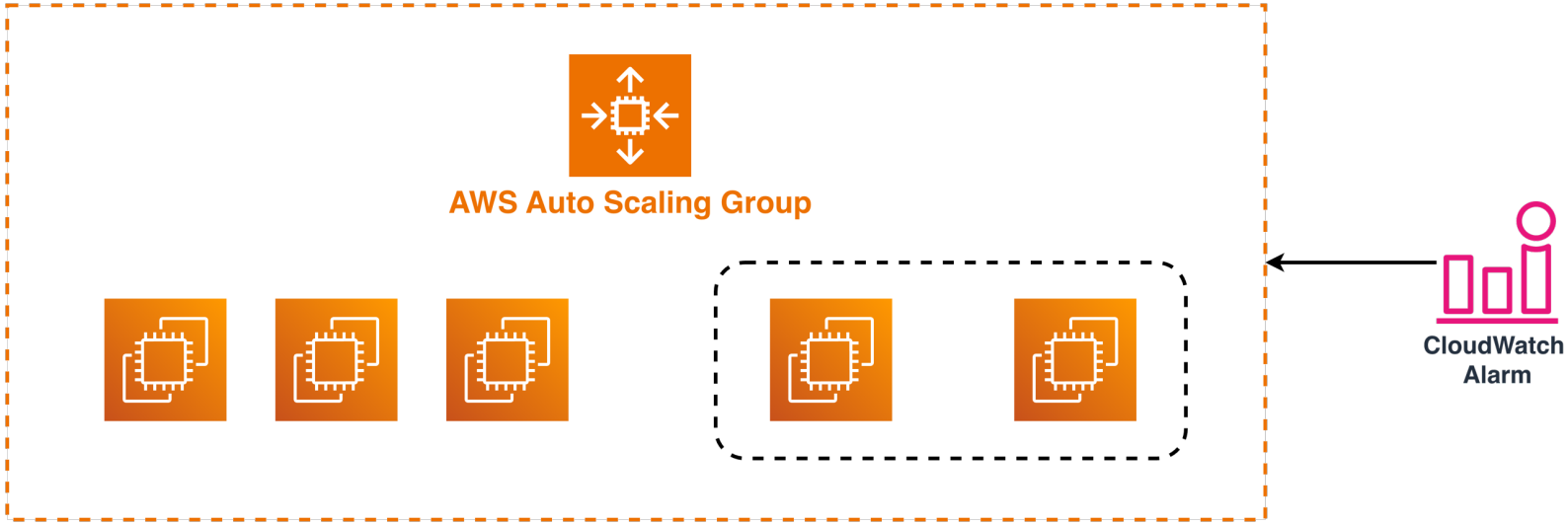
❑ Auto Scaling Groups (ASG) can automatically scale based on **Amazon CloudWatch alarms**.

How it works?

- A CloudWatch alarm monitors a metric, such as:
  - Average CPU utilization
  - Network traffic
  - Custom application metrics
- Metrics like Average CPU are calculated across all instances in the Auto Scaling Group.

Scaling actions

- Scale-out policy – add EC2 instances when demand increases.
- Scale-in policy – remove EC2 instances when demand decreases.



Ref: https://docs.aws.amazon.com/AmazonCloudWatch/latest/monitoring/CloudWatch_Alarms.html

Auto Scaling Group

❖ Scaling Policies

Dynamic Scaling

Automatically adjusts capacity based on real-time metrics.

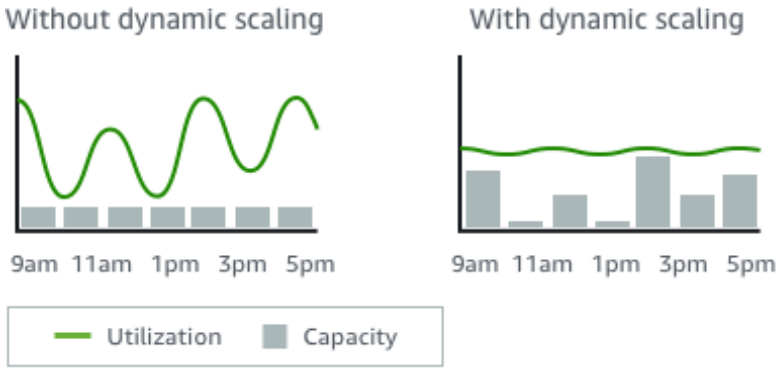
- Target Tracking Scaling (recommended)
  - Maintains a target value for a CloudWatch metric.
  - Example: keep average CPU utilization around 40%.
- Simple / Step Scaling
  - Scaling actions are triggered by CloudWatch alarms.
  - Example:
    - ✓ CPU > 70% → add 2 instances
    - ✓ CPU < 30% → remove 1 instance

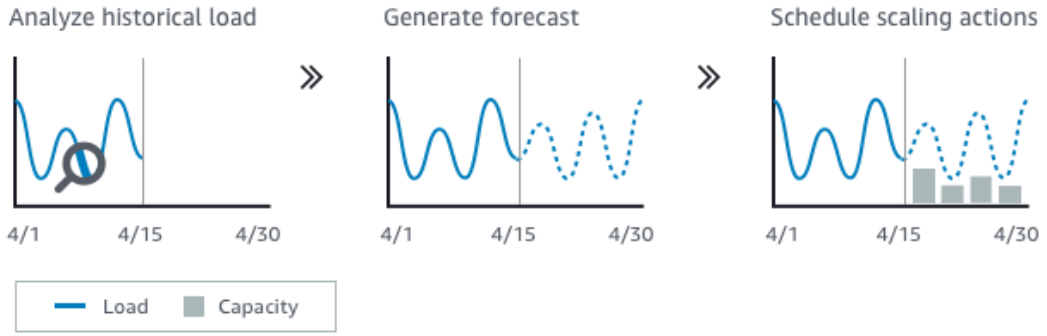
Scheduled Scaling

- Adjust capacity based on predictable usage patterns.
- Example: increase minimum capacity to 10 at 5 PM on Fridays.

Predictive scaling

- Uses machine learning with historical CloudWatch data to forecast demand and automatically scale capacity.





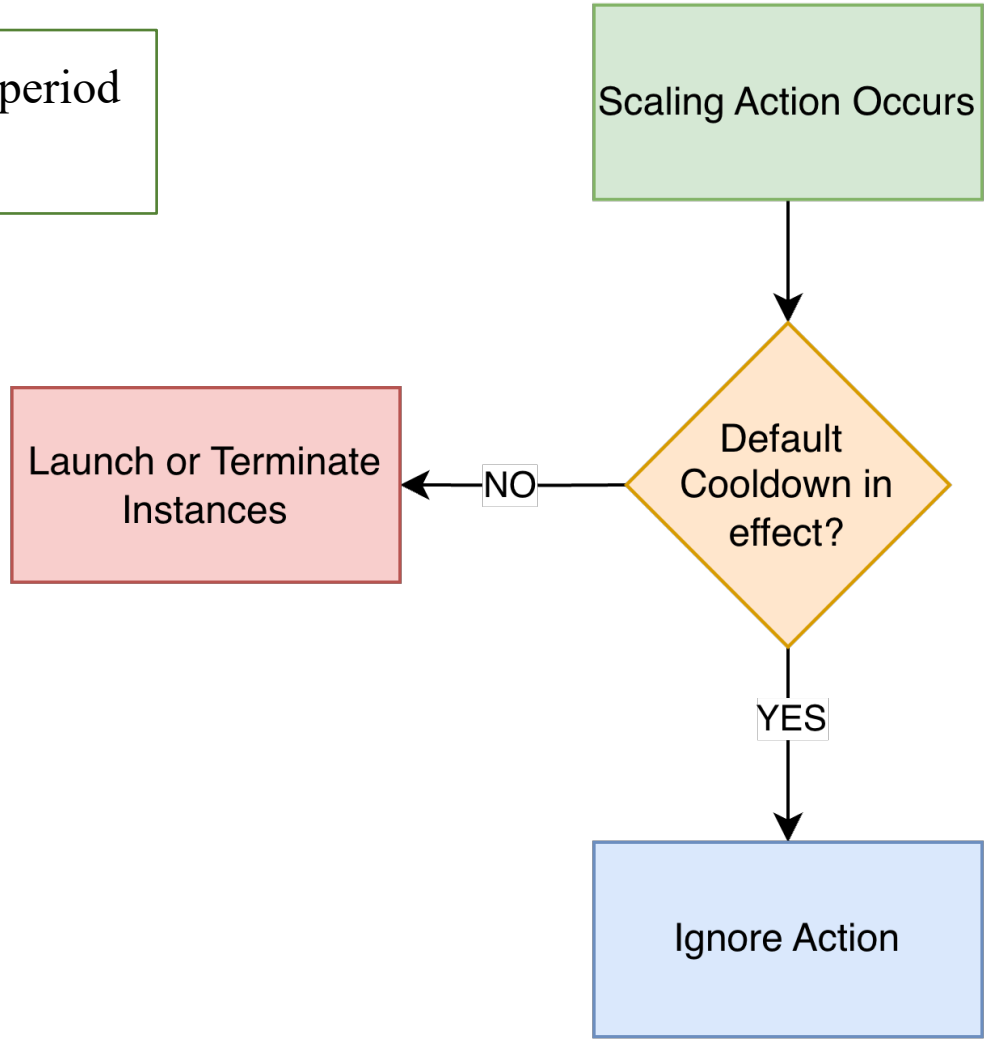
Ref: https://docs.aws.amazon.com/autoscaling/plans/userguide/what-is-a-scaling-plan.html

Auto Scaling Group

❖ Scaling Cooldowns

❑ After a scaling activity, the Auto Scaling Group enters a cooldown period **(default: 300 seconds).**

- During this time, the ASG temporarily pauses additional scaling actions.
- This allows metrics to stabilize before making further scaling decisions.

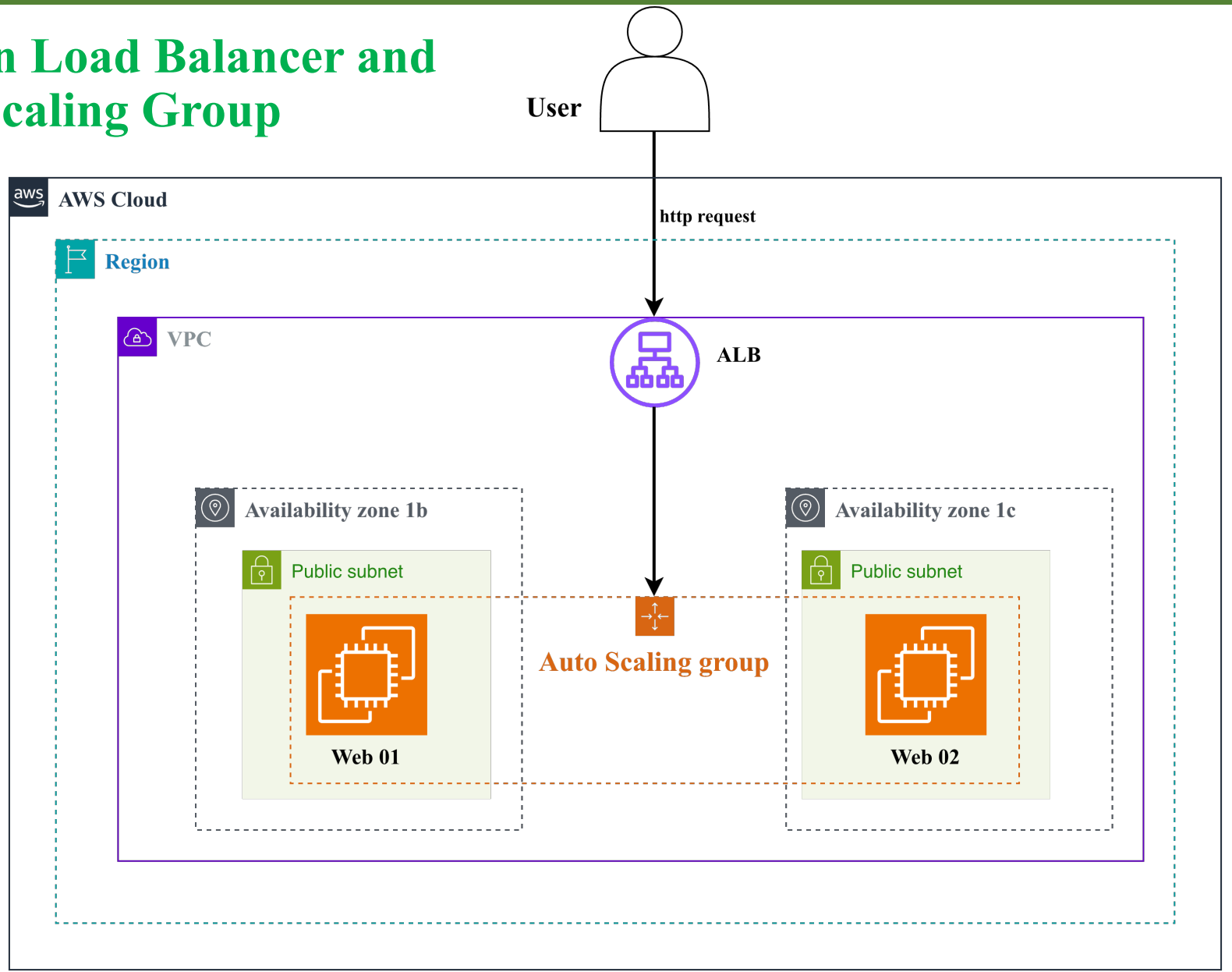


Ref: https://docs.aws.amazon.com/autoscaling/ec2/userguide/ec2-auto-scaling-scaling-cooldowns.html

Lab Session

Lab Session

❖ Application Load Balancer and EC2 Auto Scaling Group



Serverless

Serverless

❖ Getting Started

- ❑ Serverless is a **cloud computing paradigm** where developers run applications without managing servers or infrastructure.
- ❑ The cloud provider automatically provisions, scales, and manages the underlying compute resources.
- ❑ Developers primarily focus on **writing and deploying code**, while the cloud platform handles infrastructure operations.

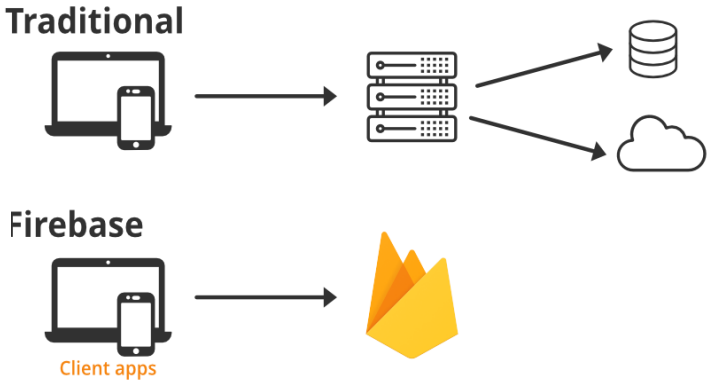
- No server management
- Automatic scaling
- Event-driven execution
- Pay-per-use pricing
- High availability by default

FaaS (Function as a Service)

Serverless was initially associated with Function as a Service (FaaS).
- Developers deploy small functions
- Functions execute when triggered by an event
- The platform manages scaling and infrastructure

▲Vercel

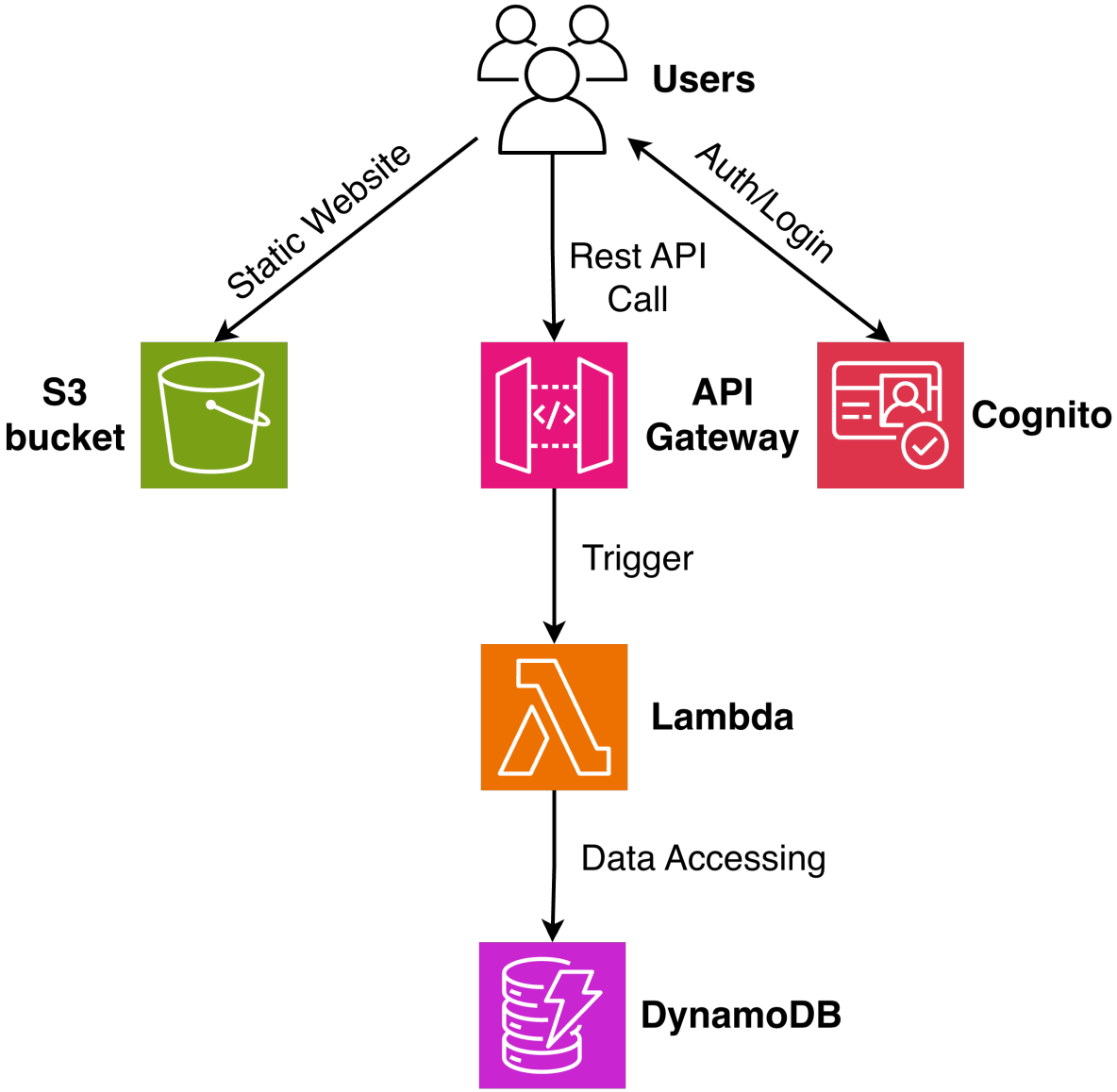
serverless



Serverless

❖ Serverless in AWS

- ❑ AWS Lambda
- ❑ DynamoDB
- ❑ AWS Cognito
- ❑ AWS API Gateway
- ❑ Amazon S3
- ❑ AWS SNS & SQS
- ❑ AWS Kinesis Data Firehose
- ❑ Aurora Serverless
- ❑ Step Functions
- ❑ Fargate

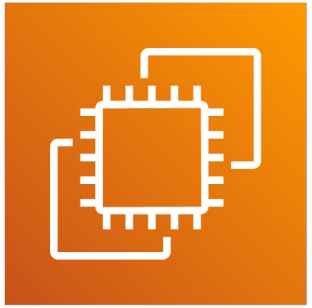


Lambda

Lambda

❖ Getting Started

- ❑ **AWS Lambda** is a serverless compute service that allows developers to run code without provisioning or managing servers.
- ❑ Instead of deploying applications on virtual machines, developers deploy **functions** that execute in response to events.
- ❑ Lambda automatically handles **infrastructure management, scaling, and execution**, allowing teams to focus only on application logic.



- Applications run on virtual machines in the cloud
- Limited by server CPU and RAM capacity
- Servers are continuously running
- Scaling requires adding or removing servers

- Run functions instead of managing servers
- No infrastructure management
- Functions execute for short, limited durations
- On-demand execution (runs only when triggered)
- Automatic scaling handled by AWS

Ref: https://docs.aws.amazon.com/lambda/latest/dg/welcome.html

Lambda

❖ Getting Started

Cost-Effective Pricing

- Pay only for number of requests and compute execution time
- Free tier: 1,000,000 requests and 400,000 GB-seconds of compute per month

Deep AWS Integration

Easily integrates with many AWS services (storage, databases, messaging, APIs)

Multi-Language Support

Supports multiple programming languages (e.g., Python, Node.js, Java, Go)

Built-in Monitoring

Integrated with AWS CloudWatch for logs, metrics, and monitoring

Flexible Resource Allocation

Functions can be configured with up to **10 GB RAM**

Performance Scaling

Increasing memory also **improves CPU power and network throughput**

Lambda

❖ Integrations

API Gateway

Kinesis

DynamoDB

S3

CloudFront

CloudWatch Events
EventBridge

CloudWatch Logs

SNS

SQS

Cognito

DynamoDB

DynamoDB

❖ Getting Started

- ❑ Fully managed NoSQL database service.
- ❑ Designed for high performance and massive scale.
- ❑ Provides high availability through replication across **multiple Availability Zones.**

- NoSQL database (non-relational) with transaction support
- Distributed architecture for large-scale workloads
- Handles millions of requests per second
- Supports trillions of items and hundreds of TB of storage
- **Single-digit** millisecond latency

Scalability & Cost

- Automatic scaling for read/write capacity
- Low operational cost
- Supports Standard and Infrequent Access (IA) table classes for cost optimization.

Security & Management

- Integrated with AWS IAM for authentication and authorization
- Fully managed: no server management, maintenance, or patching required
- Built-in high availability

Ref: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Introduction.html

DynamoDB

❖ Getting Started

Data Structure

- DynamoDB stores data in tables
- Each table has a Primary Key (defined when the table is created)
- A table can contain an unlimited number of items (rows)

Items and Attributes

- Each item consists of attributes (columns)
- Attributes can be added or modified over time
- Attributes can also be null or absent

Item Size Limit

Maximum size of a single item: 400 KB

Supported Data Types

- Scalar Types
  - String
  - Number
  - Binary
  - Boolean
  - Null
- Document Types
  - List
  - Map
- Set Types
  - String Set
  - Number Set
  - Binary Set

Ref: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Introduction.html

DynamoDB

❖ Read/Write Capacity Modes

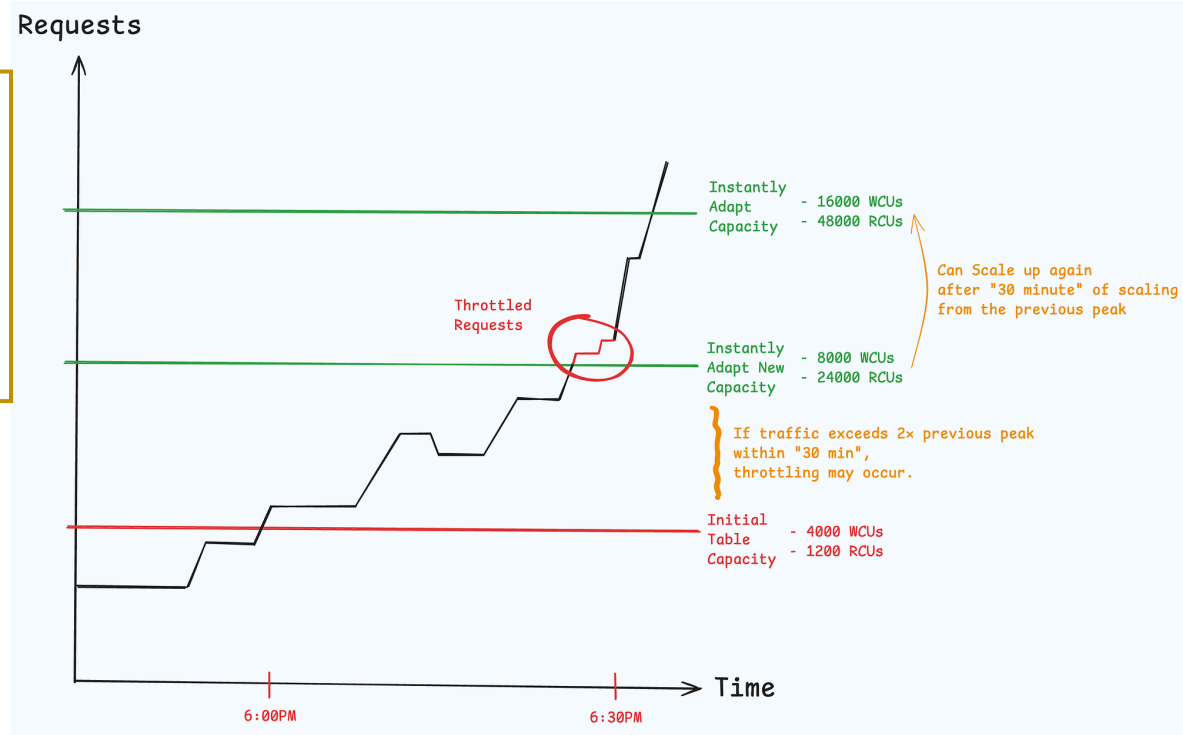
❑ DynamoDB provides two capacity modes to control how read and write throughput is managed.

Provisioned Capacity Mode (Default)

- You define the number of reads and writes per second
- Requires capacity planning in advance
- Pay for provisioned throughput
  - RCU (Read Capacity Units)
  - WCU (Write Capacity Units)
- Supports Auto Scaling to automatically adjust RCU and WCU

On-Demand Capacity Mode

- Automatic scaling based on traffic
- No need for capacity planning
- Pay only for actual read and write requests
- Typically more expensive



DynamoDB

❖ DynamoDB Accelerator (DAX)

- ❑ DAX is a fully managed, highly available in-memory cache for DynamoDB.
- ❑ Designed to improve read performance and reduce load on DynamoDB tables.

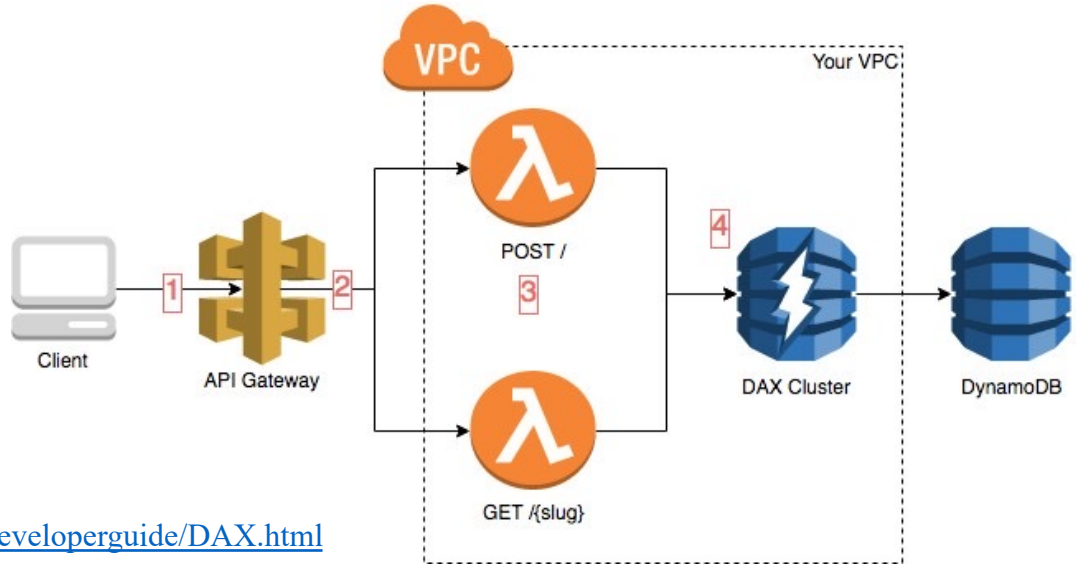
- Provides microsecond latency for cached data
- Helps reduce read congestion by caching frequently accessed items
- Fully managed by AWS

Integration

- Compatible with existing DynamoDB APIs
- Applications usually do not require code changes

Cache Behavior

- Uses TTL (Time-To-Live) to control cache lifetime
- Default cache TTL: 5 minutes



Ref: https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/DAX.html

DynamoDB

❖ Stream Processing

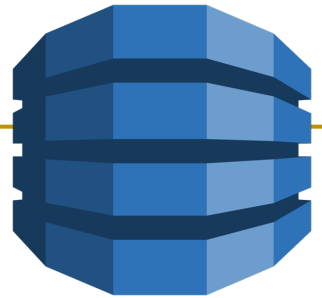
- ❑ DynamoDB Streams capture an ordered sequence of item-level changes in a table.
- ❑ Tracks create, update, and delete operations.

Use case

- Real-time reactions to data changes (e.g., send welcome emails)
- Real-time analytics and monitoring
- Update derivative tables or materialized views
- Cross-region data replication
- Trigger serverless workflows with AWS Lambda

DynamoDB Streams

- 24-hour data retention
- Limited number of concurrent consumers
- Processing options:
  - AWS Lambda triggers
  - Kinesis adapter

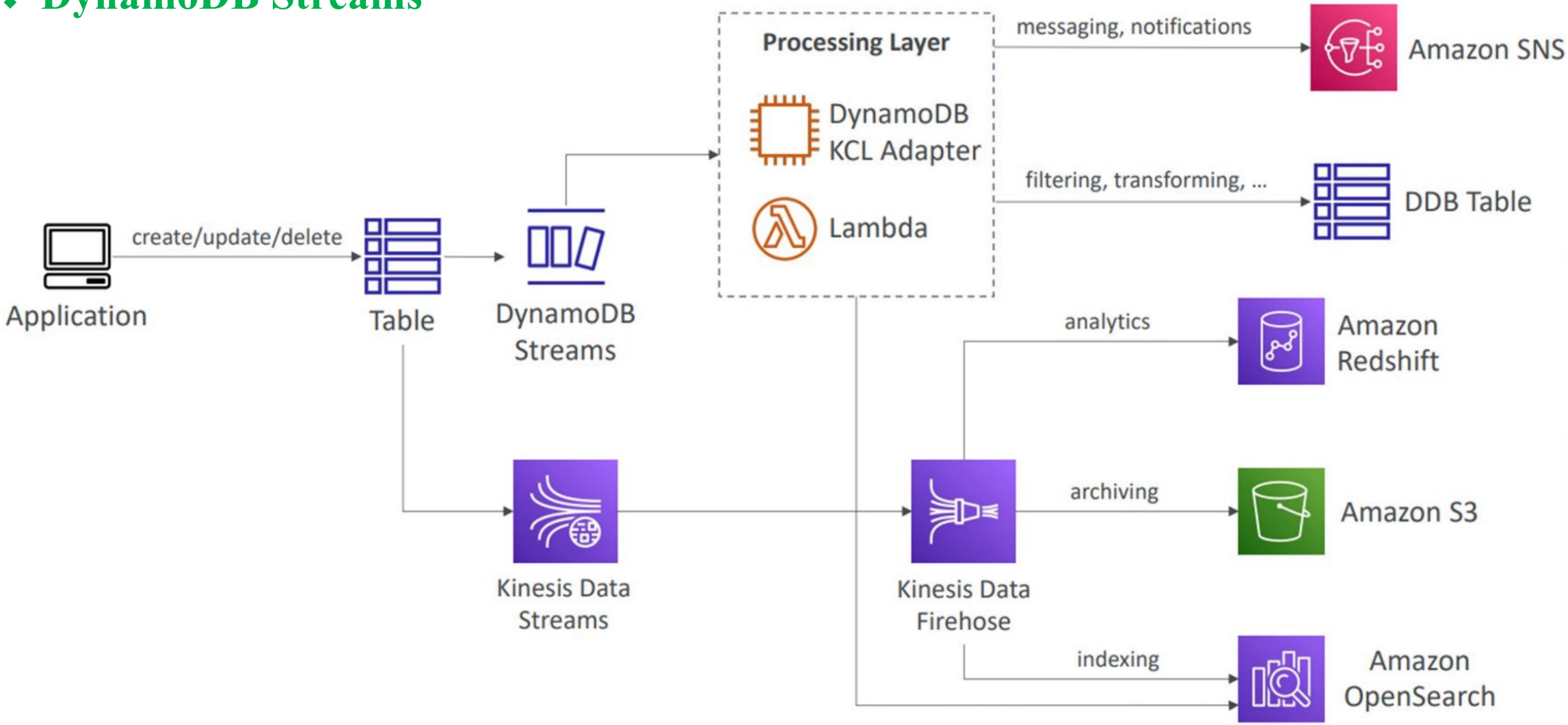


Kinesis Data Streams Integration

- Alternative streaming solution with longer retention
- Up to 1 year data retention
- Supports many consumers
- Can integrate with:
  - AWS Lambda Kinesis
  - Data Analytics Kinesis
  - Data Firehose
  - AWS Glue Streaming ETL

DynamoDB

❖ DynamoDB Streams



DynamoDB

❖ Global Tables

- ❑ Enables a DynamoDB table to be replicated across multiple AWS Regions
- ❑ Provides low-latency data access for global applications

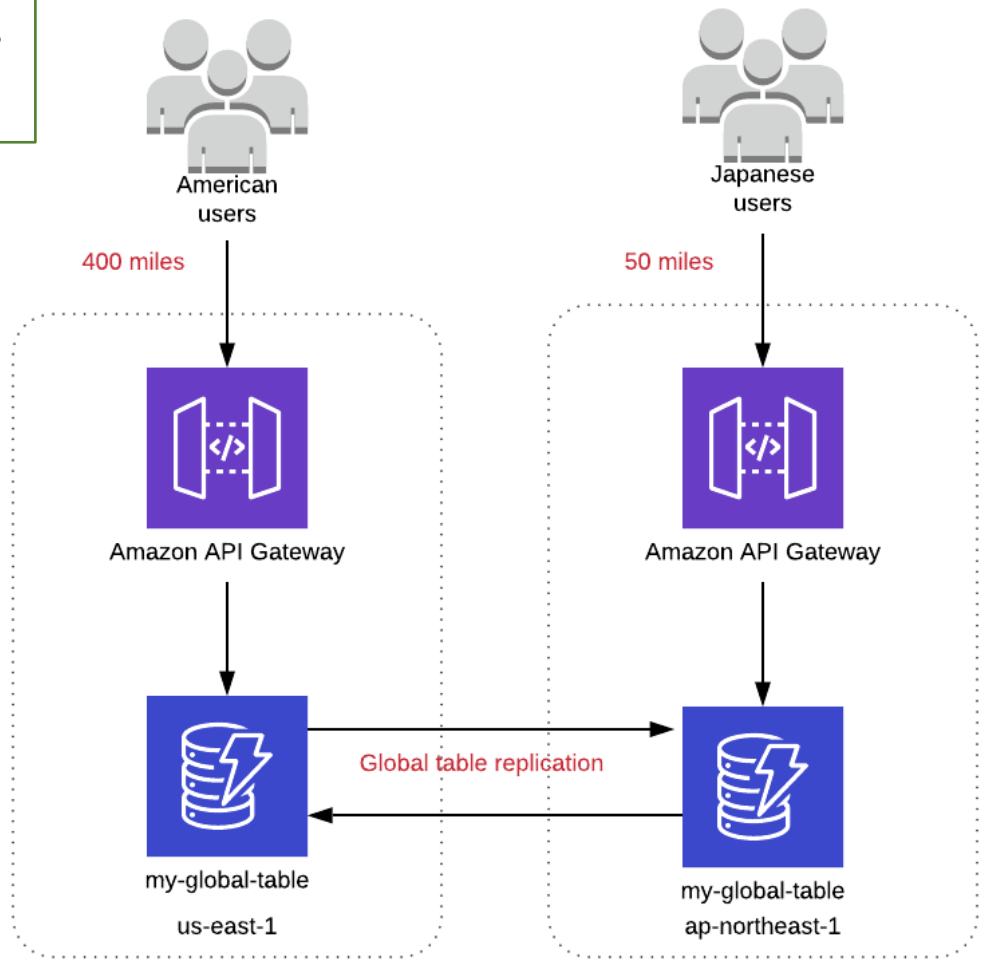
- Active–Active replication between regions
- Applications can read and write to the table in any region
- Data changes are automatically synchronized across regions

Use case

- Global applications with users distributed worldwide
- Disaster recovery and high availability
- Multi-region architectures

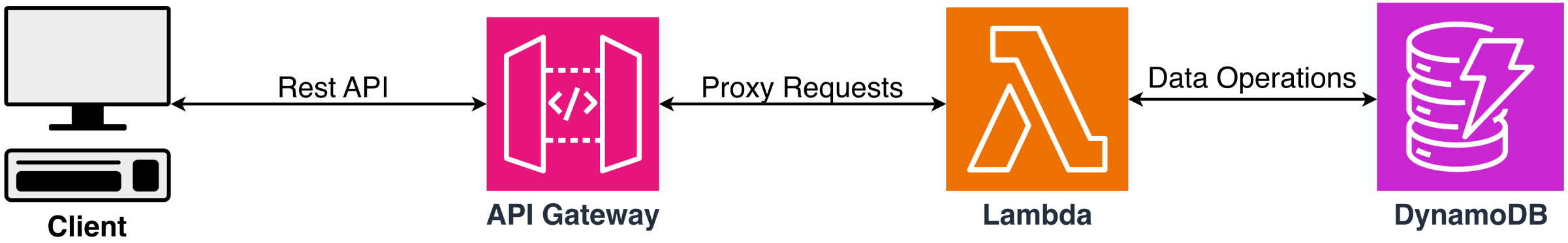
Note

- DynamoDB Streams must be enabled as a prerequisite to support replication between regions.



DynamoDB

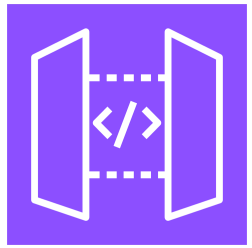
❖ Serverless API



API Gateway

API Gateway

❖ Getting Started



- ❑ Fully managed service to create, publish, and manage APIs
- ❑ Commonly used with AWS Lambda for serverless APIs
- ❑ No infrastructure management required

- Supports REST APIs and WebSocket APIs
- API versioning support (v1, v2, etc.)
- Manage multiple environments (dev, test, prod)

Security & Access Control

- Built-in authentication and authorization
- Supports API keys
- Request throttling and rate limiting

API Management

- Import APIs using Swagger / OpenAPI
- Request and response transformation
- Request validation

API Gateway

❖Integrations

❑ API Gateway can integrate with different backend services to process API requests.

Lambda Integration

- API Gateway invokes an AWS Lambda function
- Common approach for serverless REST APIs
- No servers or infrastructure to manage

HTTP Integration

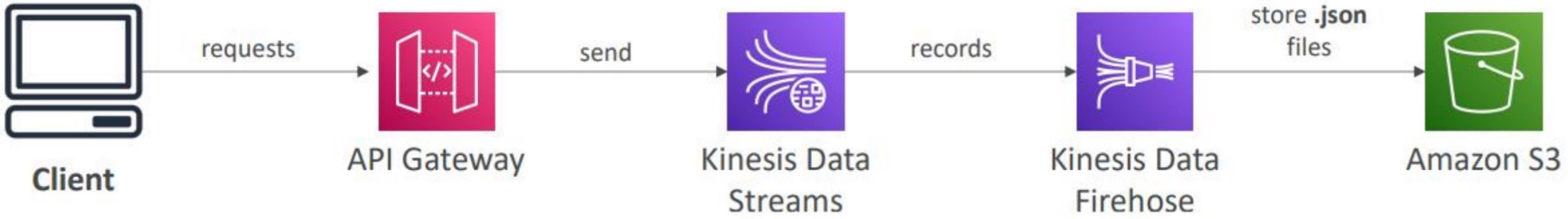
- API Gateway forwards requests to HTTP endpoints
- Examples:
  - Internal HTTP APIs
  - On-premise services
  - Application Load Balancer (ALB)

AWS Service Integration

- API Gateway can directly call AWS services
- Examples:
  - Start an AWS Step Functions workflow
  - Send messages to Amazon SQS

API Gateway

❖ AWS Service Integration



API Gateway

❖ Endpoint Types

Edge-Optimized (Default)

- Designed for global clients
- Requests are routed through CloudFront Edge Locations
- Helps reduce latency for worldwide users
- API Gateway itself remains deployed in a single region

Regional

- Intended for clients within the same region
- Requests go directly to the regional API Gateway endpoint
- Can be combined with CloudFront manually for more control over caching and distribution

Private

- Accessible only within a VPC
- Uses an Interface VPC Endpoint (ENI)
- Access controlled using API Gateway resource policies

Ref: https://docs.aws.amazon.com/apigateway/latest/developerguide/api-gateway-api-endpoint-types.html

Step Functions

Step Functions

❖ Getting Started

❑ Service to build serverless workflows that coordinate multiple AWS services

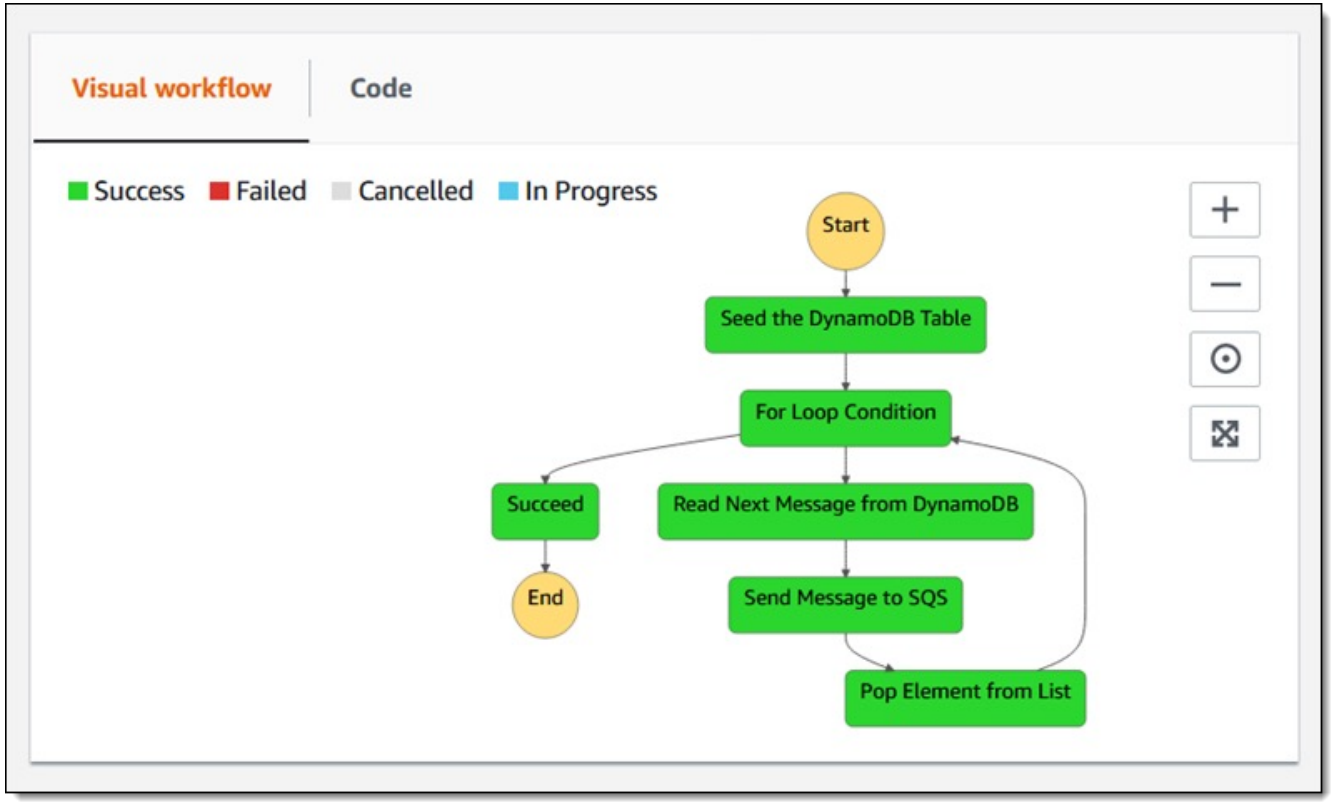
Often used to orchestrate AWS Lambda functions.

Features

- Visual workflow design
- Supports workflow logic:
  - Sequence
  - Parallel execution
  - Conditional branching
  - Timeouts
  - Error handling

Integrations

- AWS Lambda
- EC2
- ECS
- API Gateway
- SQS
- On-premises systems



Ref: https://docs.aws.amazon.com/step-functions/latest/dg/welcome.html

Amazon Cognito

Amazon Cognito

❖ Getting Started

- ❑ Service for user authentication, authorization, and user management.
- ❑ Allows users to sign in and interact with web or mobile applications.

Cognito User Pools

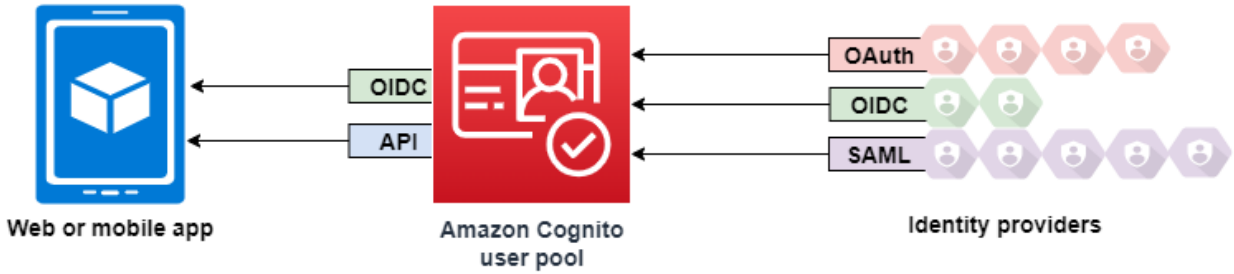
- Provides sign-up and sign-in functionality
- Manages user authentication
- Supports integration with:
  - API Gateway
  - Application Load Balancer
- Used for application user identity management

Cognito vs IAM

- IAM: designed for AWS service users and administrators
- Cognito: designed for application users
  - Large number of users (hundreds or millions)
  - Mobile or web users
  - Authentication using federated identity providers (e.g., SAML)

Cognito Identity Pools (Federated Identity)

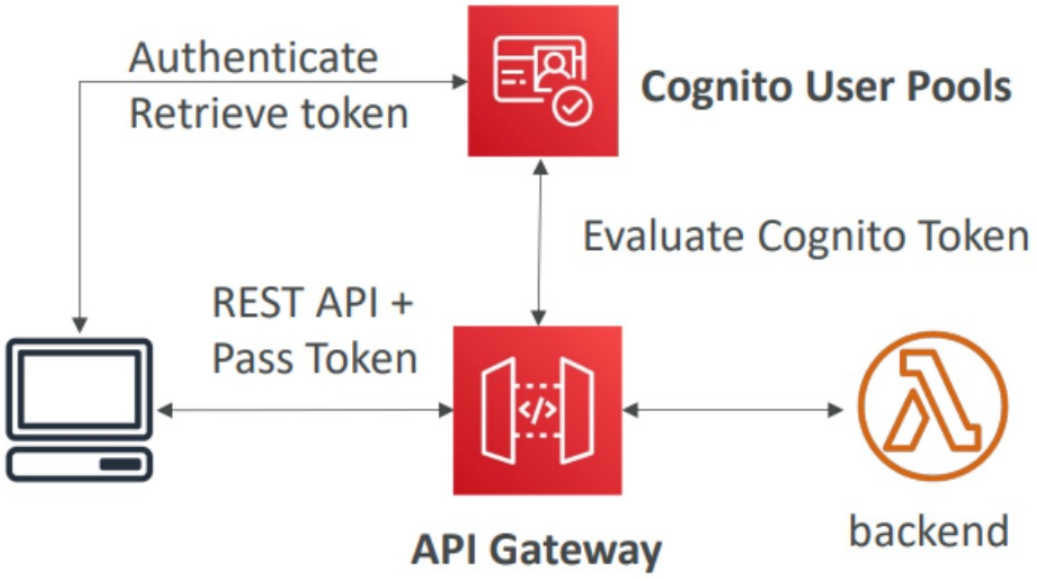
- Provide temporary AWS credentials to authenticated users
- Allows users to access AWS services directly
- Can use Cognito User Pools as an identity provider

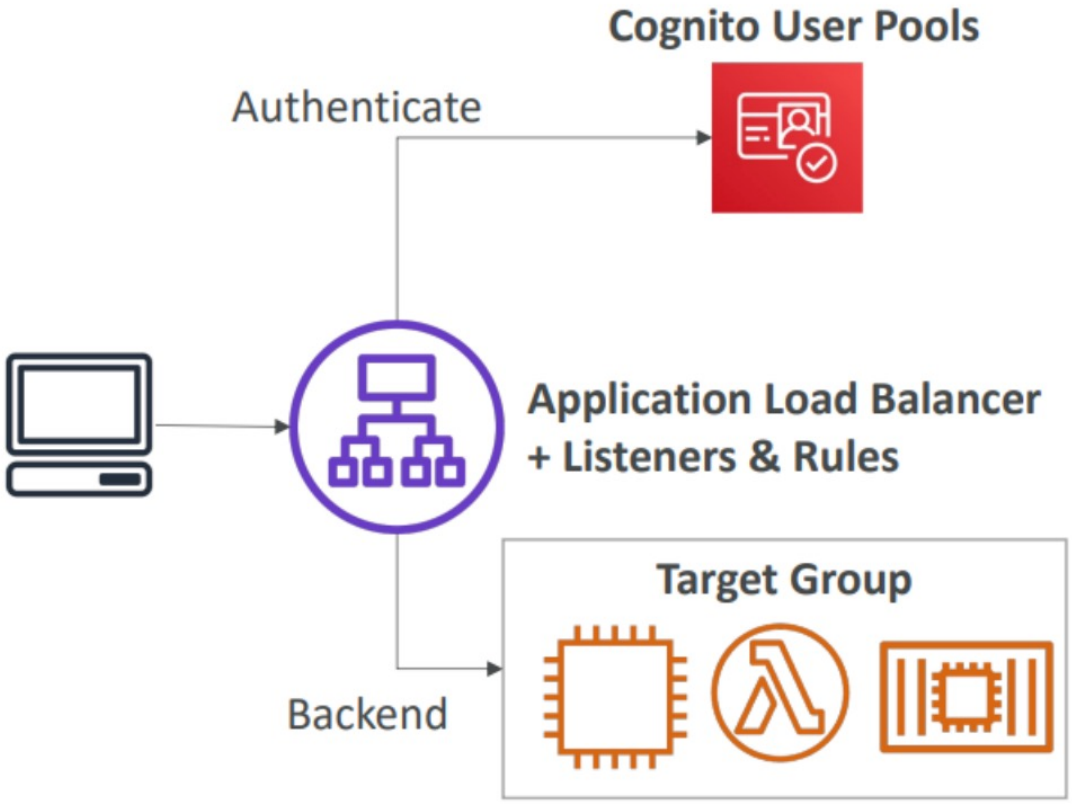


Ref: https://docs.aws.amazon.com/cognito/latest/developerguide/cognito-user-pools.html

Amazon Cognito

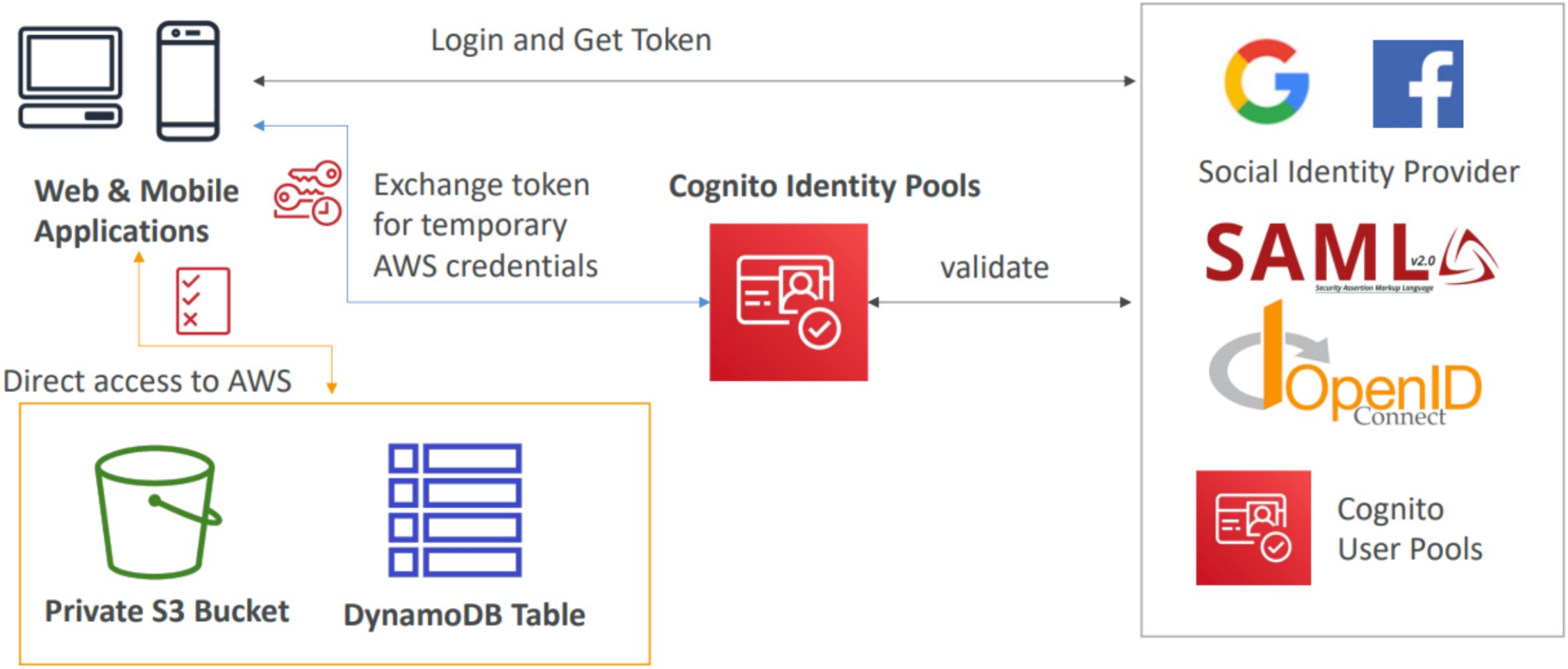
❖ Cognito User Pools (CUP) - Integrations





Amazon Cognito

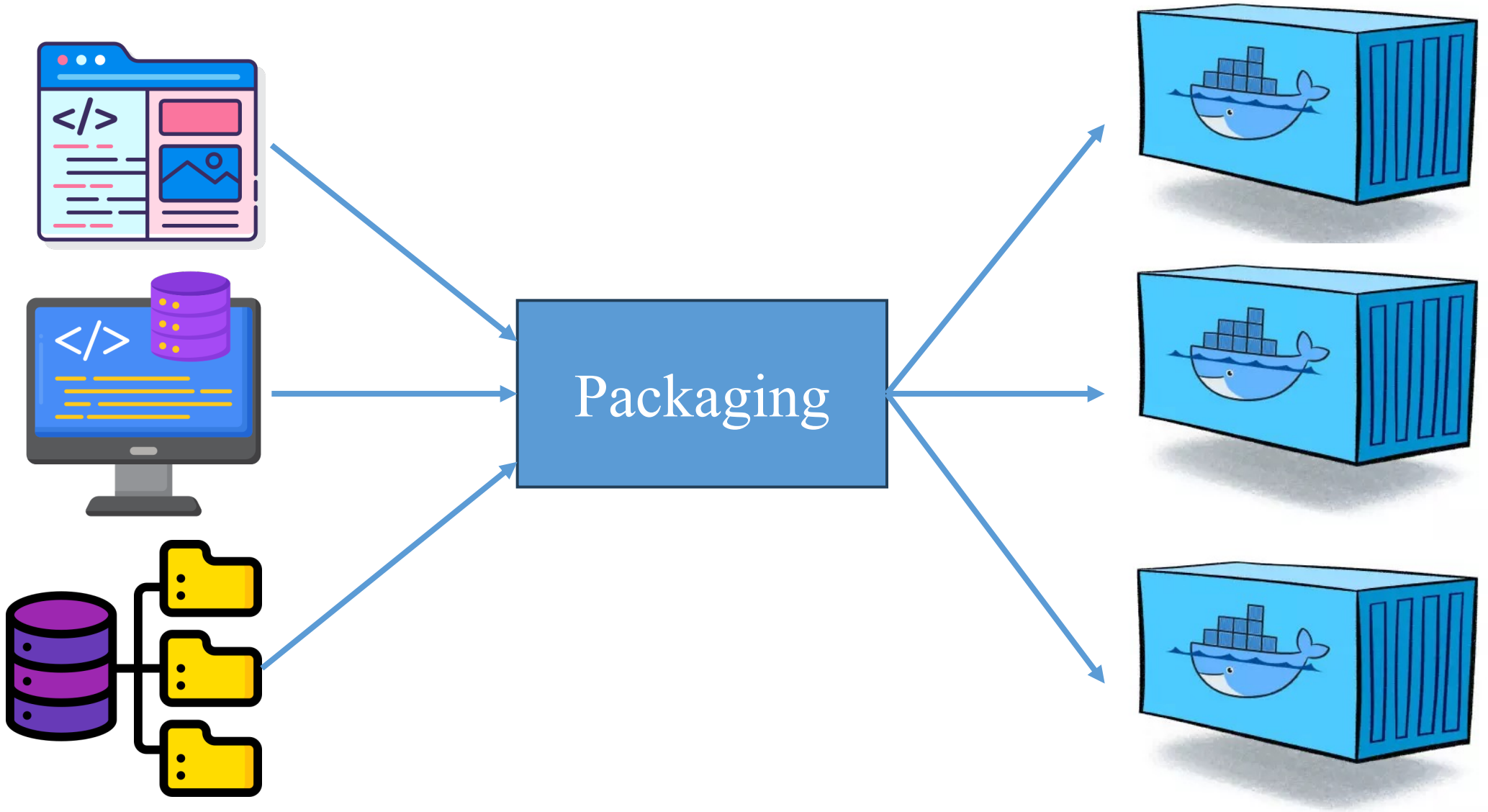
❖ Cognito Identity Pools – Diagram



Containerize

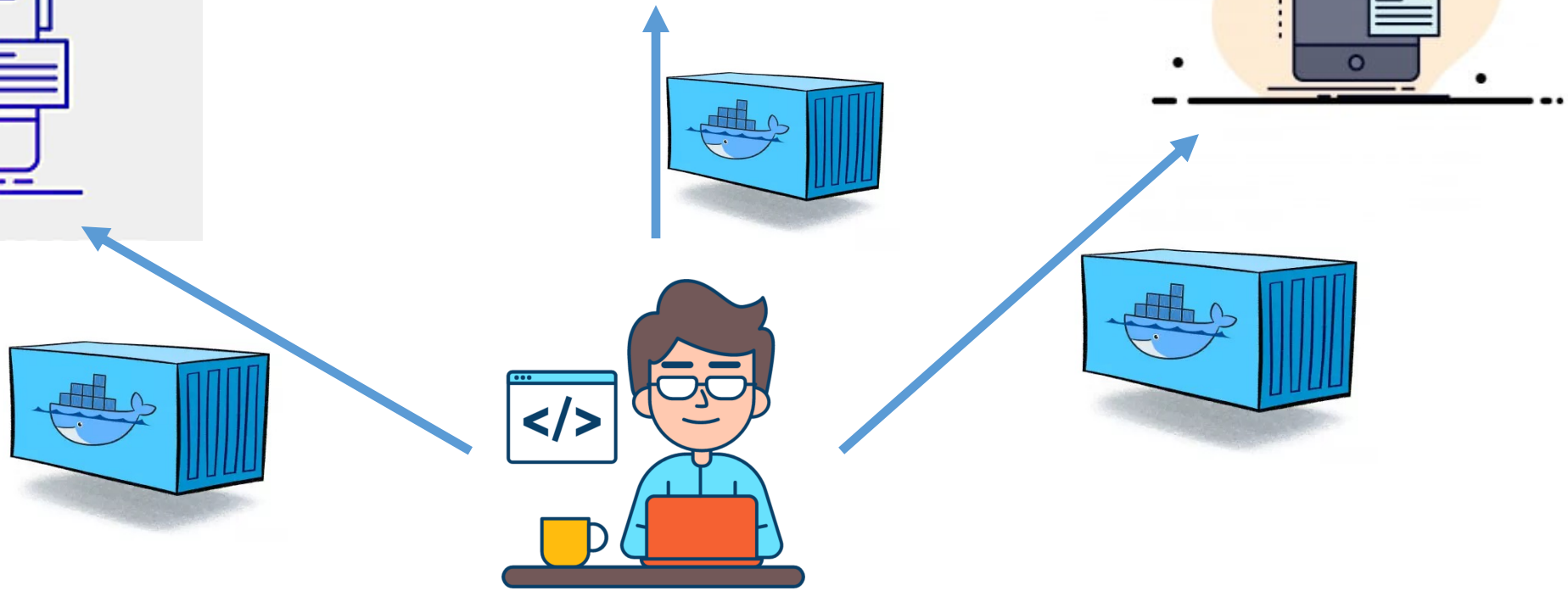
Docker in AWS

❖ Getting Started



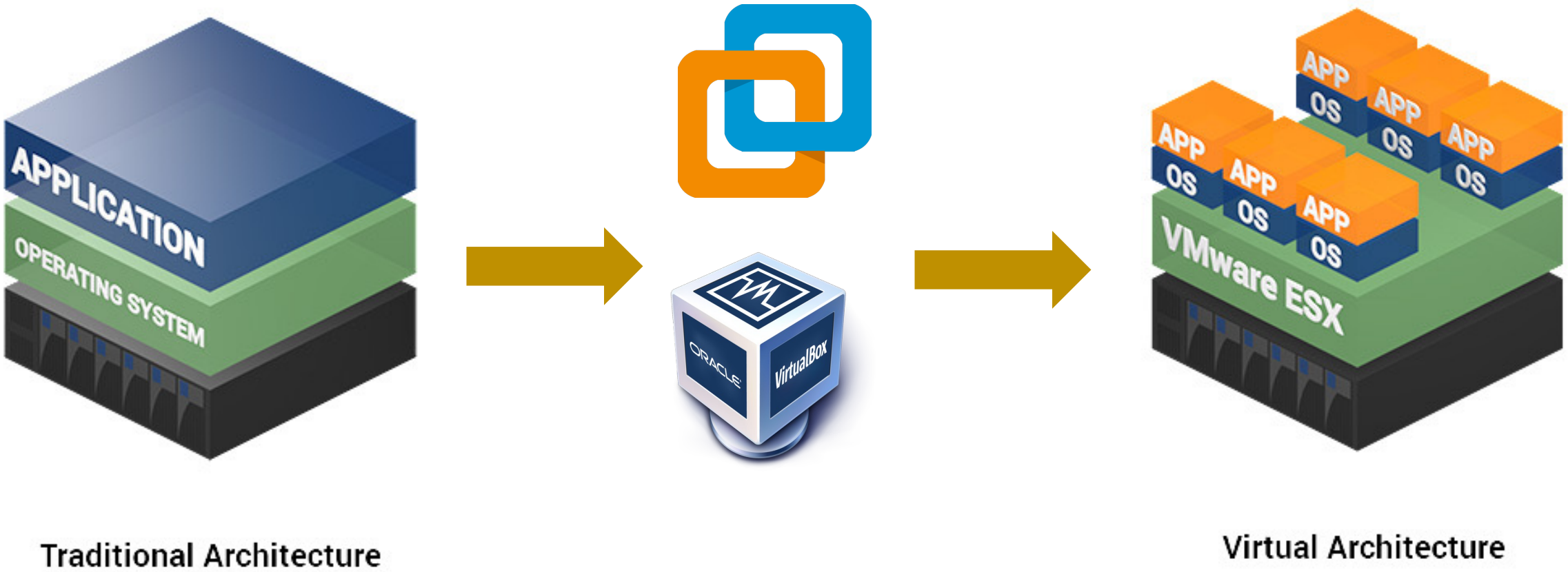
Docker in AWS

❖ Deloyment



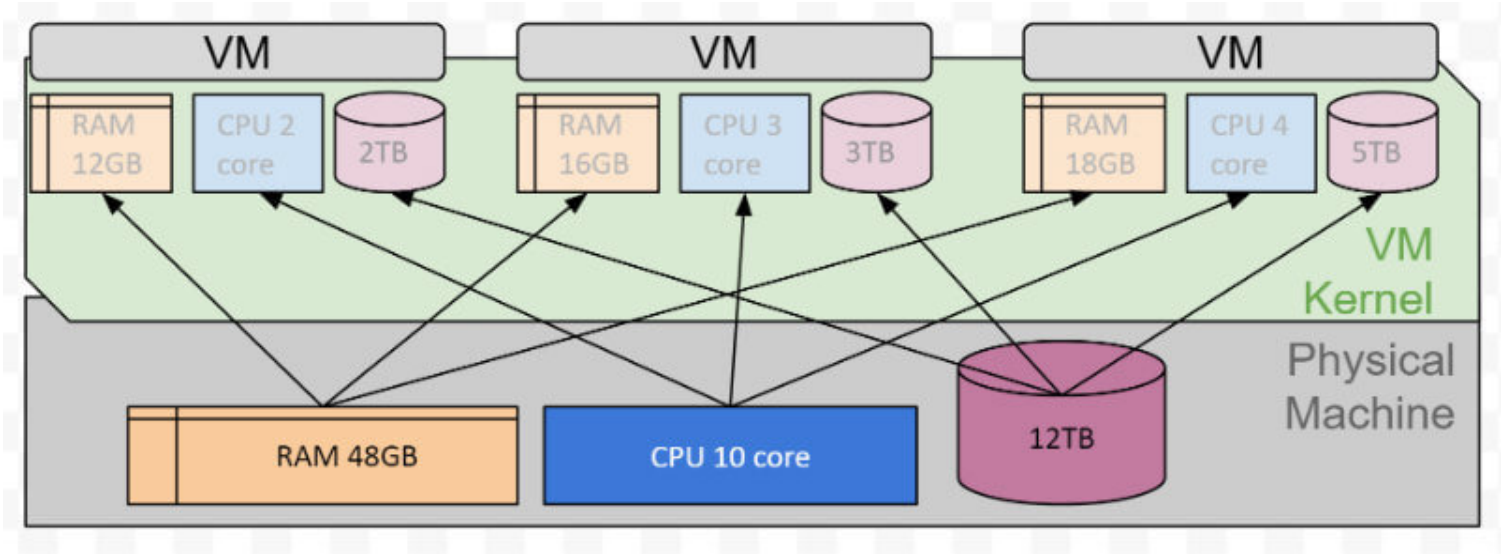
Docker in AWS

❖ Previous Solution



Docker in AWS

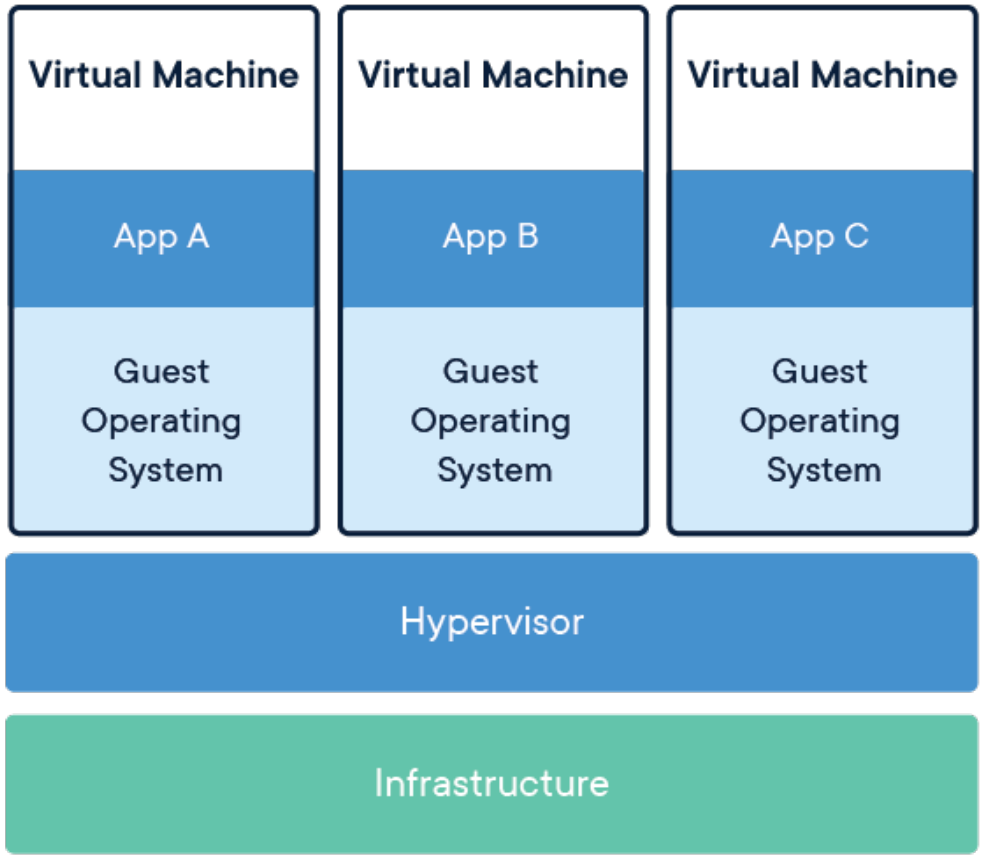
❖ Previous Solution

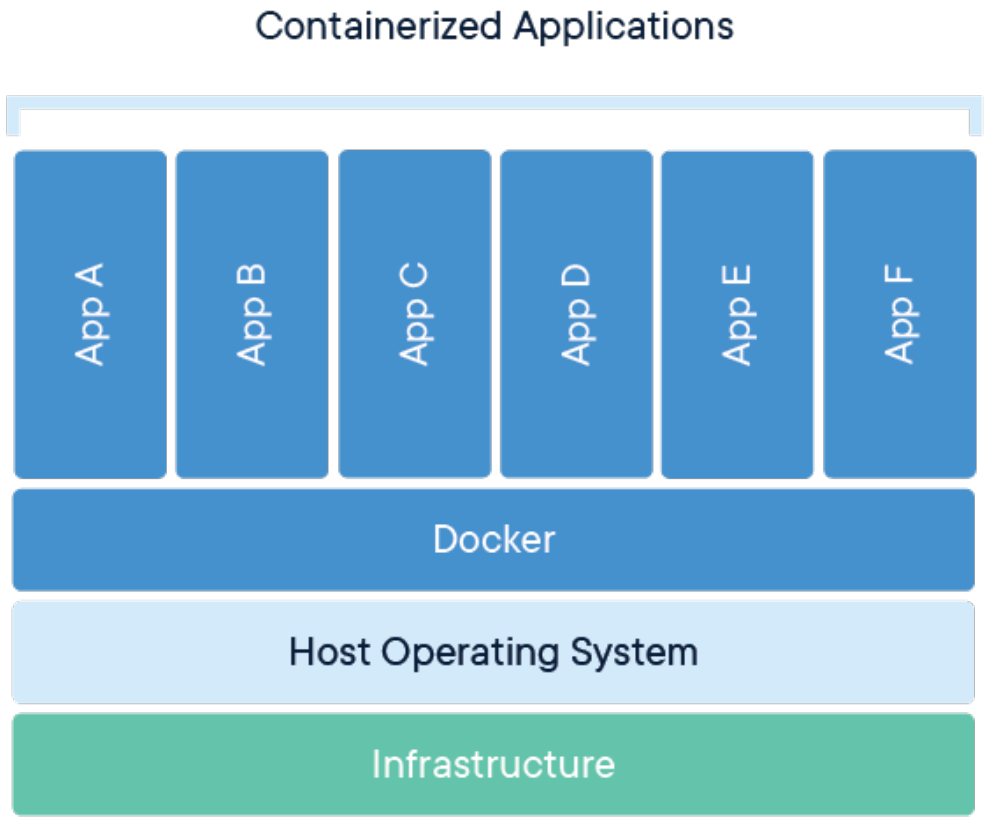


Allocate resources when initializing VM

Docker in AWS

❖ Container Method





Docker in AWS

❖ Containerization platforms

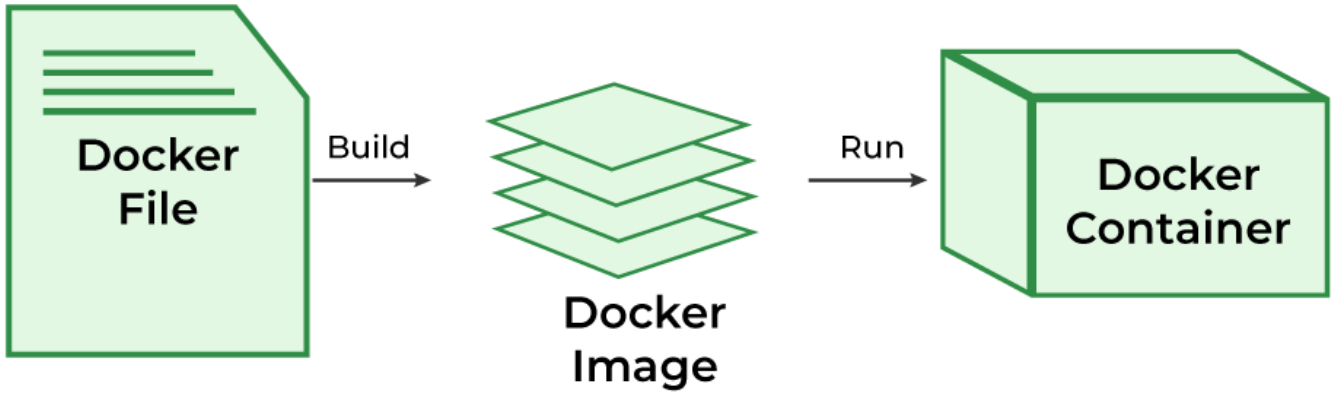
OrbStack

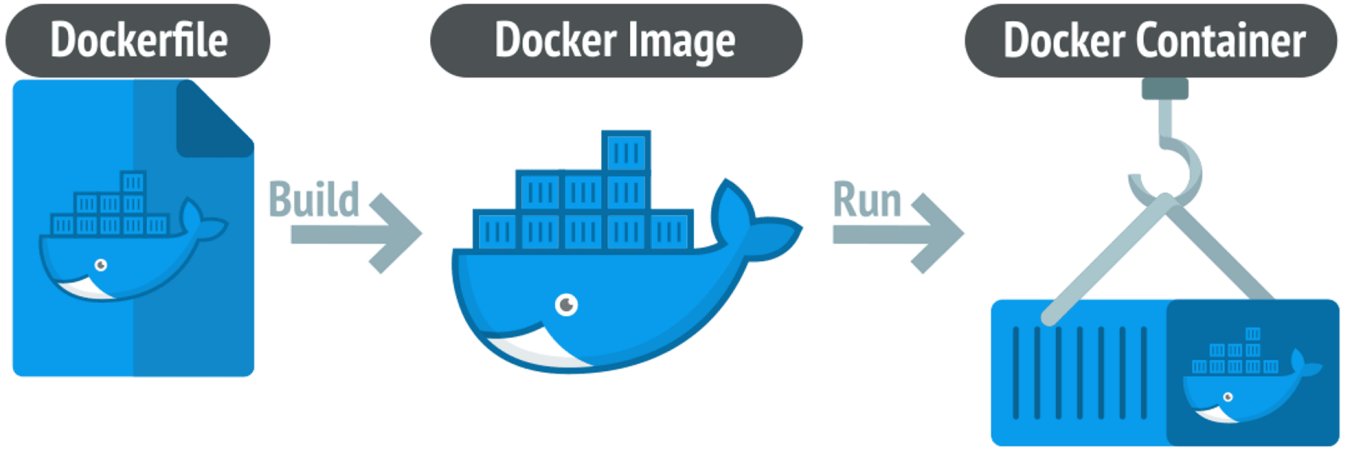
podman

Docker in AWS

❖ Docker Image

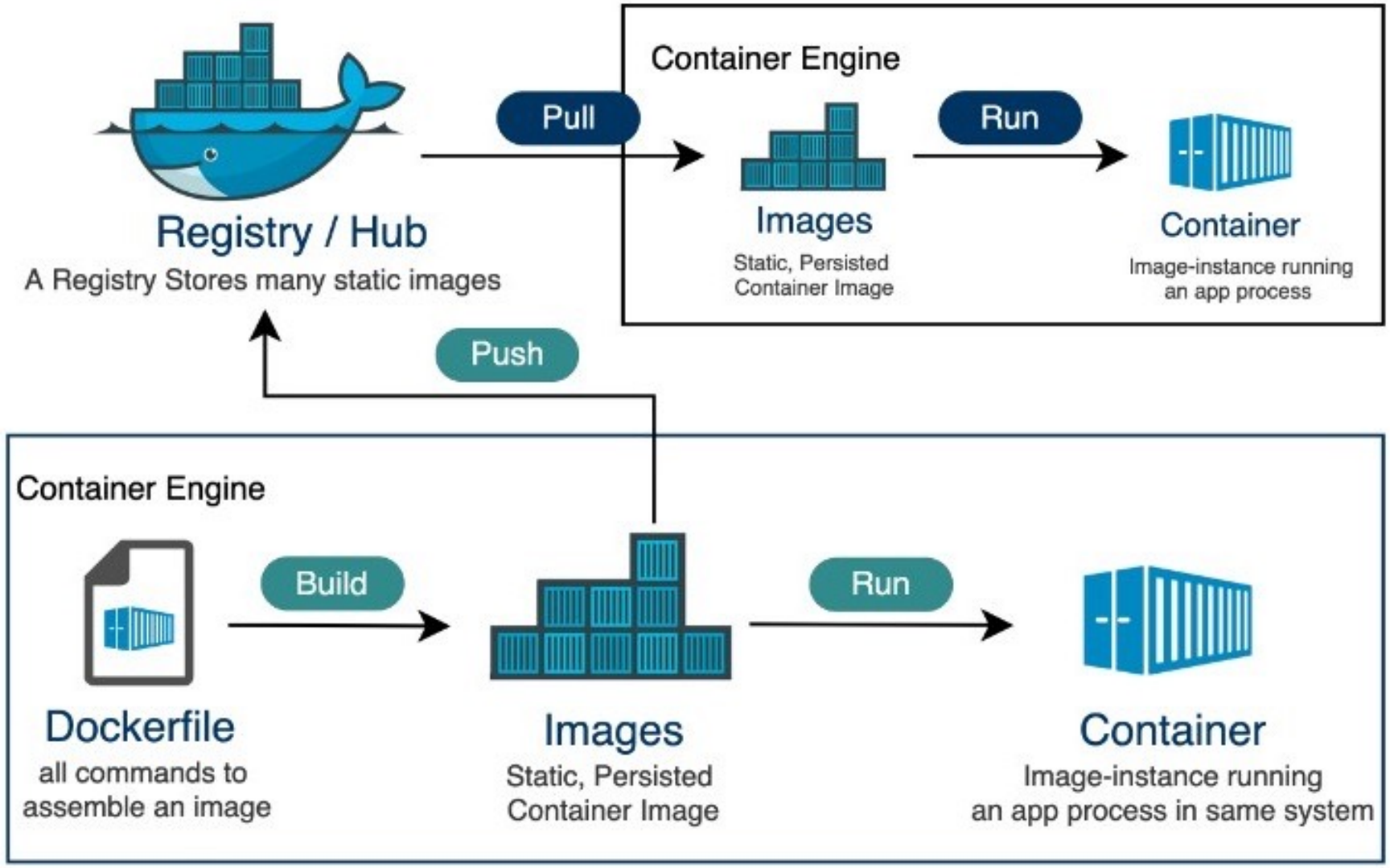
- ❑ An image is a template with instructions for creating a Docker container.
- ❑ Each instruction in Dockerfile creates a layer in image.





Docker in AWS

❖ Registry



Docker Containers on AWS

Docker Containers on AWS

❖ Getting Started

Amazon Elastic Container Service (Amazon ECS)

- Amazon's own container platform

Amazon ECS

Amazon Elastic Kubernetes Service (Amazon EKS)

- Amazon's managed Kubernetes (open source)

Amazon EKS

AWS Fargate

- Amazon's own Serverless container platform
- Works with ECS and with EKS

AWS Fargate

Amazon ECR

- Store container images

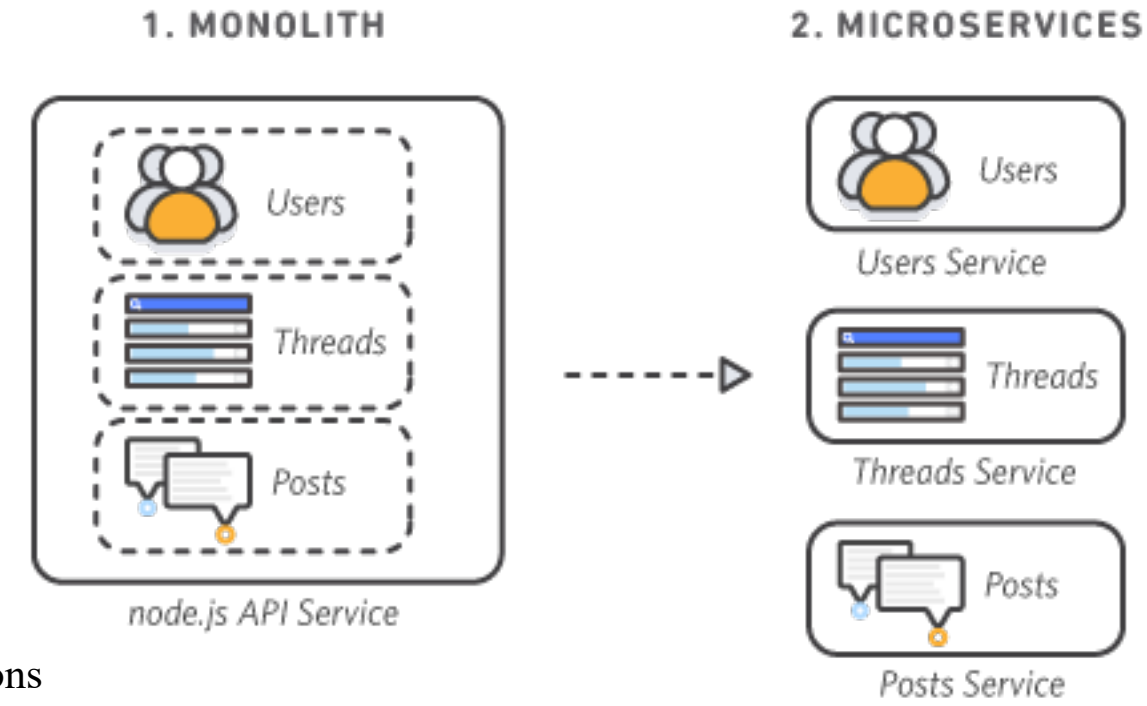
Amazon ECR

Amazon Elastic Container Service

Amazon ECS

❖ Use Cases

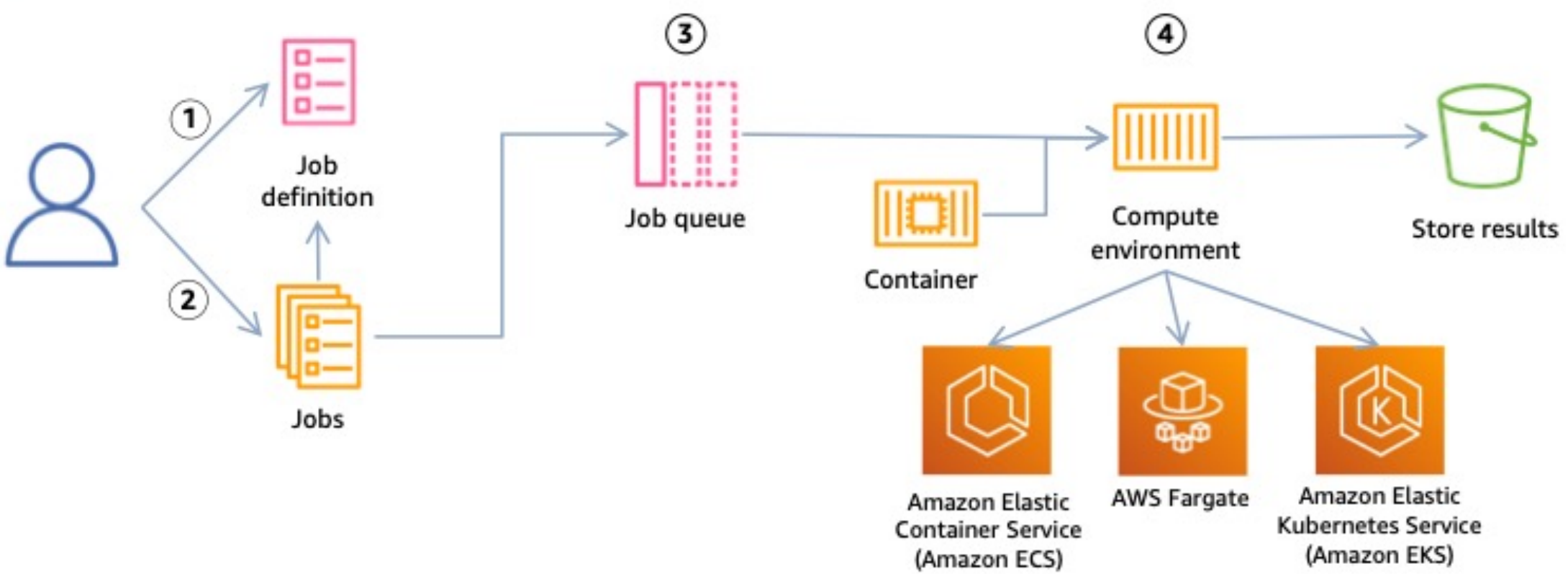
- ❑ Crash Isolation
  - ➢ Each microservice runs in a separate container
  - ➢ Failure in one service does not affect other services
- ❑ Increased Security
  - ➢ Services are isolated at the container level
  - ➢ Easier to apply fine-grained security policies
- ❑ Independent Scaling
  - ➢ Each service can scale independently
  - ➢ Resources are allocated based on the service workload
- ❑ Easy Rollback
  - ➢ Container images allow quick rollback to previous versions
  - ➢ Supports safe and controlled deployments



Ref: https://aws.amazon.com/ecs/

Amazon ECS

❖ Use Cases

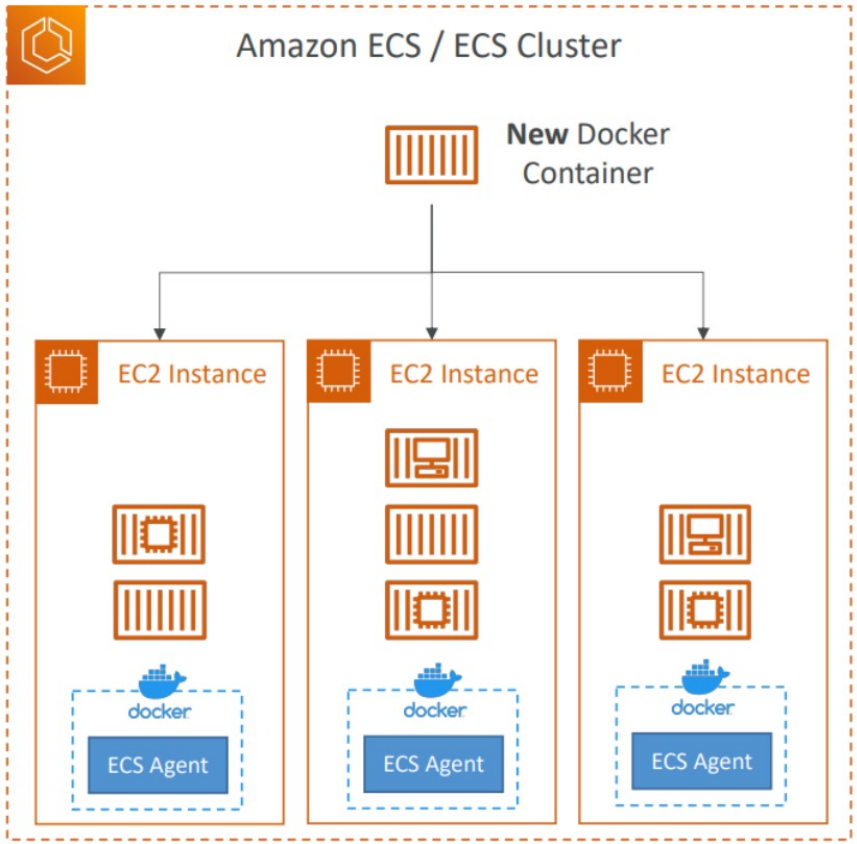


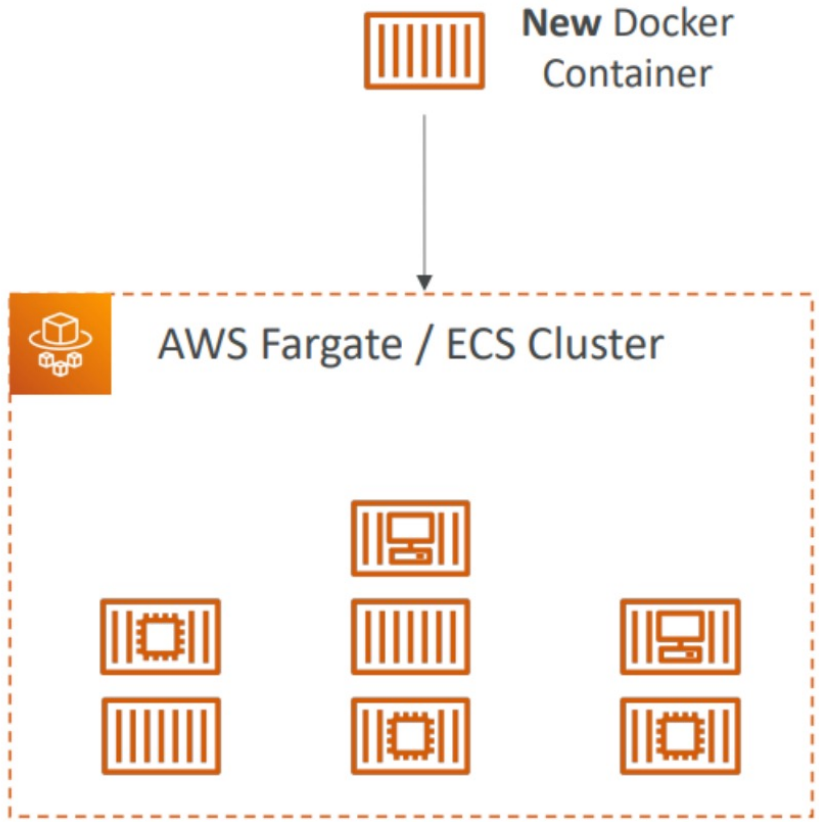
Ref: https://aws.amazon.com/blogs/hpc/understanding-aws-batch-using-amazon-cloudwatch-events/

Amazon ECS

❖ ECS Cluster

- An **ECS Cluster** is a logical group that **runs containers, services, and tasks**.
- A cluster contains:
  - Tasks (running containers)
  - Services (long-running tasks with scaling and load balancing)





Requirement

You must create an ECS Cluster first before running any tasks or services.

Amazon ECS

❖ ECS Task

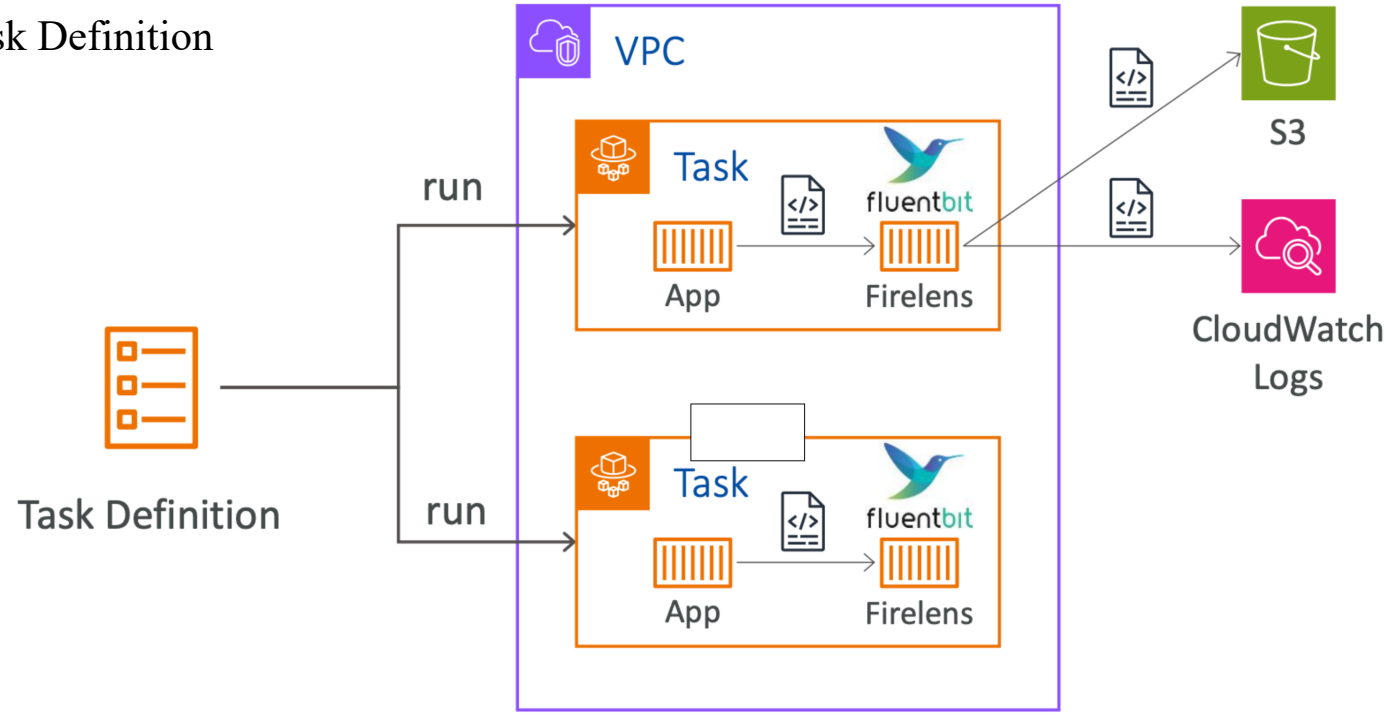
- ❑ Basic unit of work in Amazon ECS
- ❑ Represents one running instance of a task definition
- ❑ A task can run one or more containers
- ❑ Containers run according to settings defined in a Task Definition

Container Configuration

Task Definition specifies:
- Container images
- CPU and memory
- Networking
- Environment variables

Platform for running task

ECS currently offers 3 ways to run your task:
- EC2 launch type
- Fargate launch type
- External launch type



Amazon ECS

❖ EC2 Launch Type

❑ ECS tasks run on EC2 instances within an ECS Cluster

Infrastructure Management

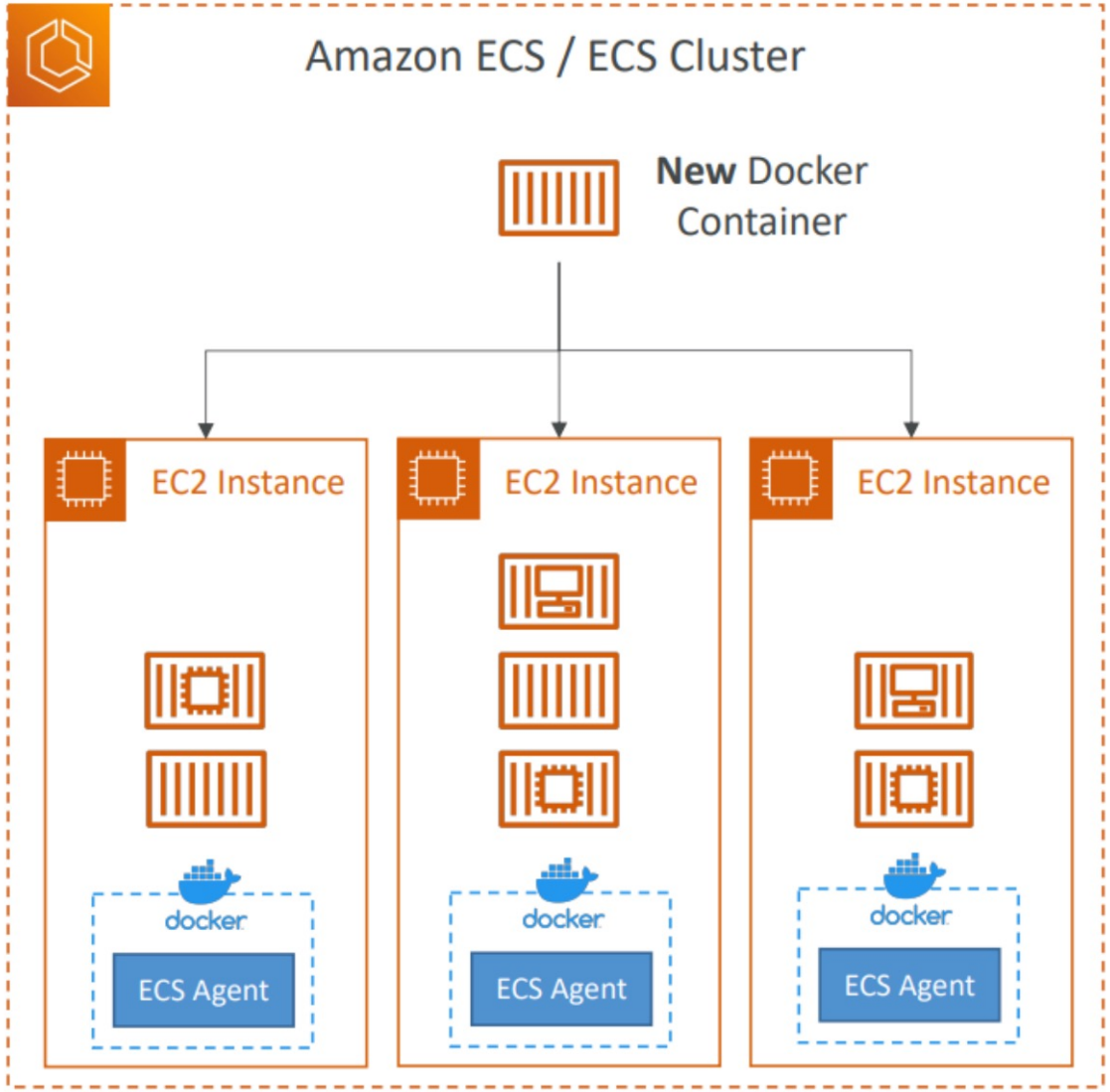
- You must provision and manage the EC2 instances
- ECS cannot launch tasks if the cluster lacks EC2 resources

ECS Agent

- Each EC2 instance must run the ECS Agent
- The agent allows the instance to register and communicate with the ECS Cluster

Container Management

- ECS manages starting and stopping containers
- Infrastructure management (EC2 scaling, updates) remains your responsibility.



Amazon ECS

❖ Fargate Launch Type

❑ ECS tasks run on AWS Fargate, a serverless compute engine for containers

Infrastructure Management

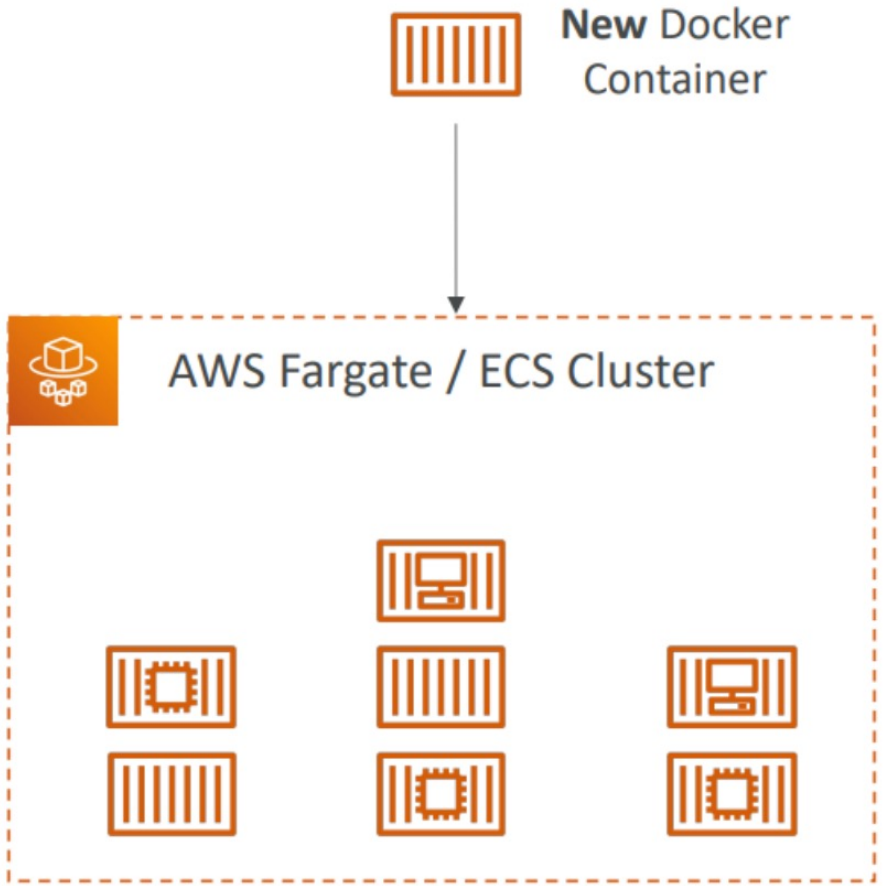
- No EC2 instances to provision or manage
- AWS fully manages the underlying infrastructure

Deployment Model

- You only need to define
- Task Definitions Specify CPU and memory requirements
- AWS automatically runs the containers

Scalability

- Scaling is done by increasing or decreasing the number of tasks
- No need to manage or scale EC2 instances



Amazon ECS

❖ External Launch Type

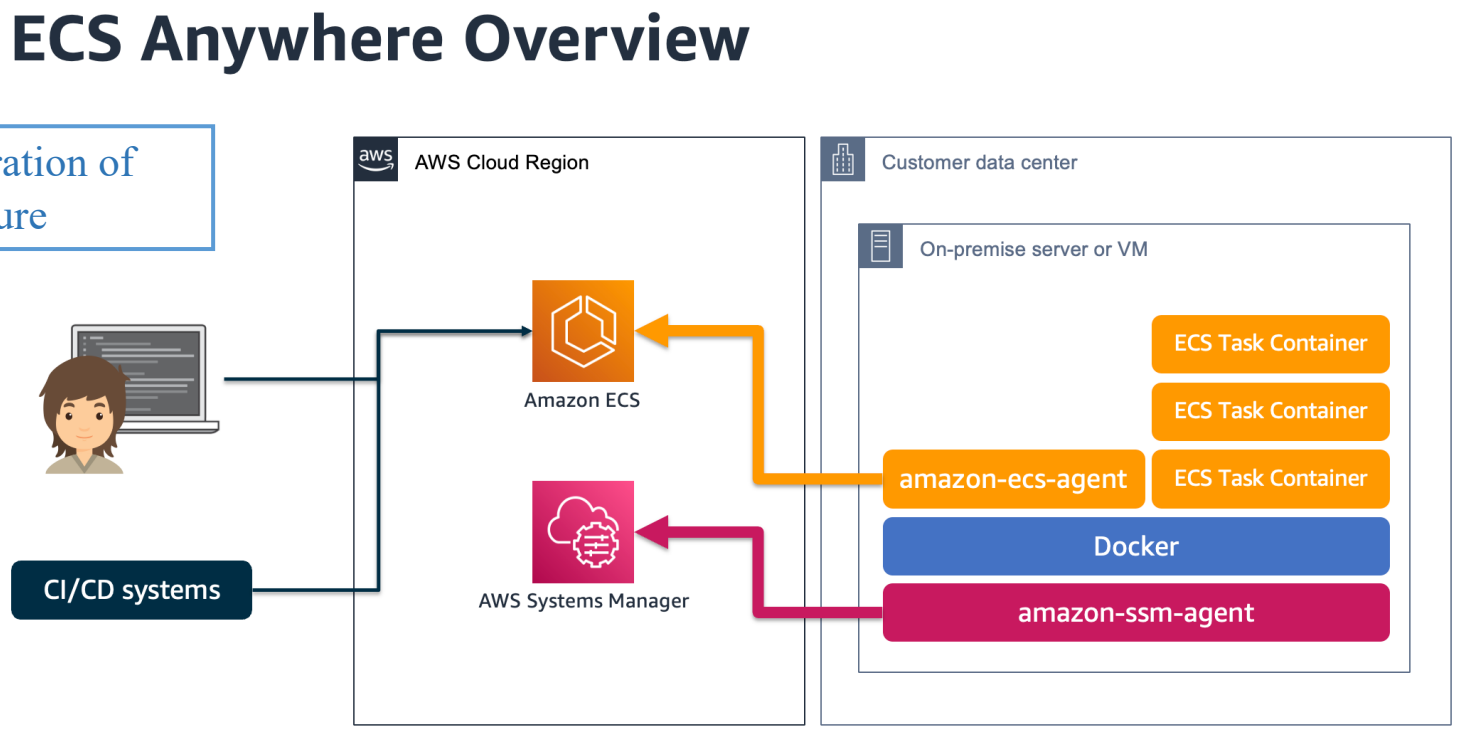
❑ Allows ECS tasks to run on external infrastructure (e.g., on-premises servers or external VMs)

Infrastructure Management

- You must provision and maintain the external machines
- These instances must connect to the ECS cluster

Task Management

- AWS ECS manages the deployment and orchestration of tasks Containers run on your external infrastructure

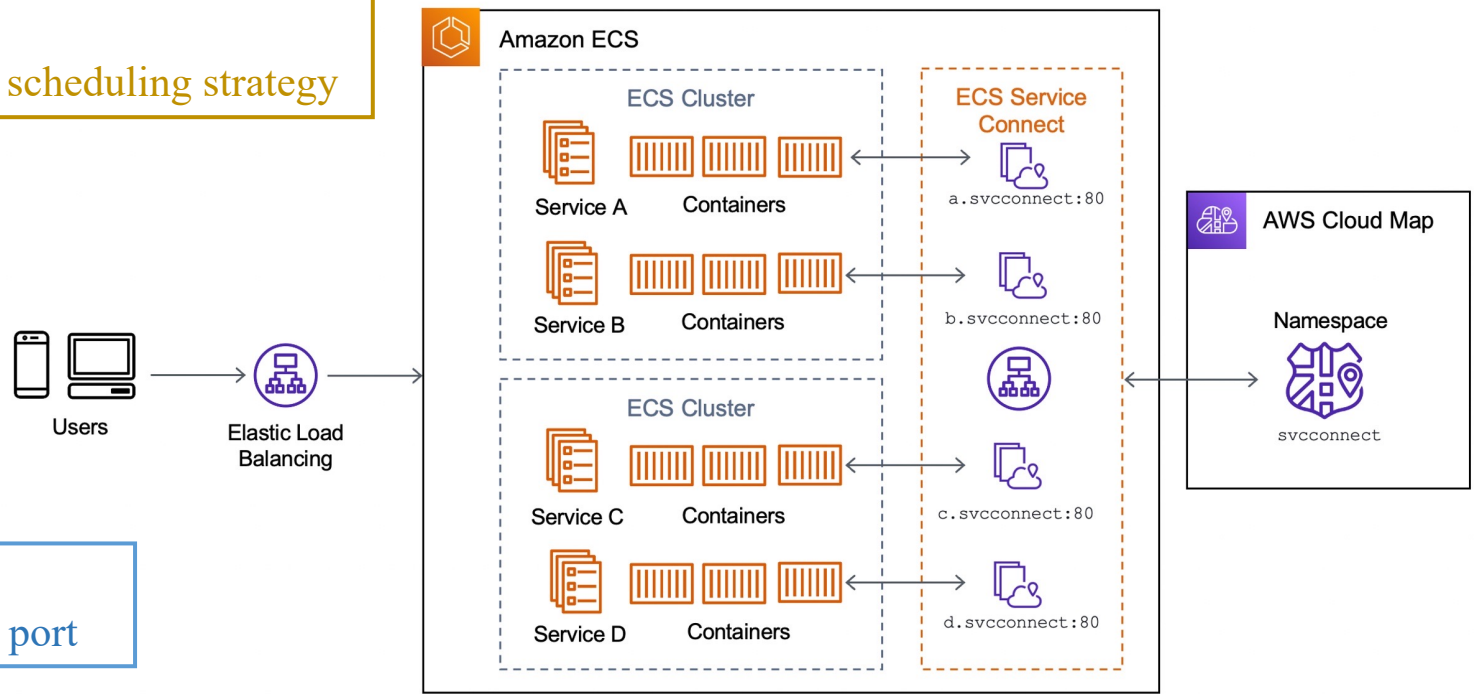


Amazon ECS

❖ ECS Service

❑ An ECS Service manages and maintains long-running ECS tasks

- Ensures tasks remain running according to the desired count
- Automatically restarts tasks if they fail
- Launches or stops tasks to maintain the defined scheduling strategy



Networking

- Can expose tasks to external clients
- Routes traffic to the correct container host and port

Ref: https://docs.aws.amazon.com/AmazonECS/latest/developerguide/ecs_services.html

Amazon ECS

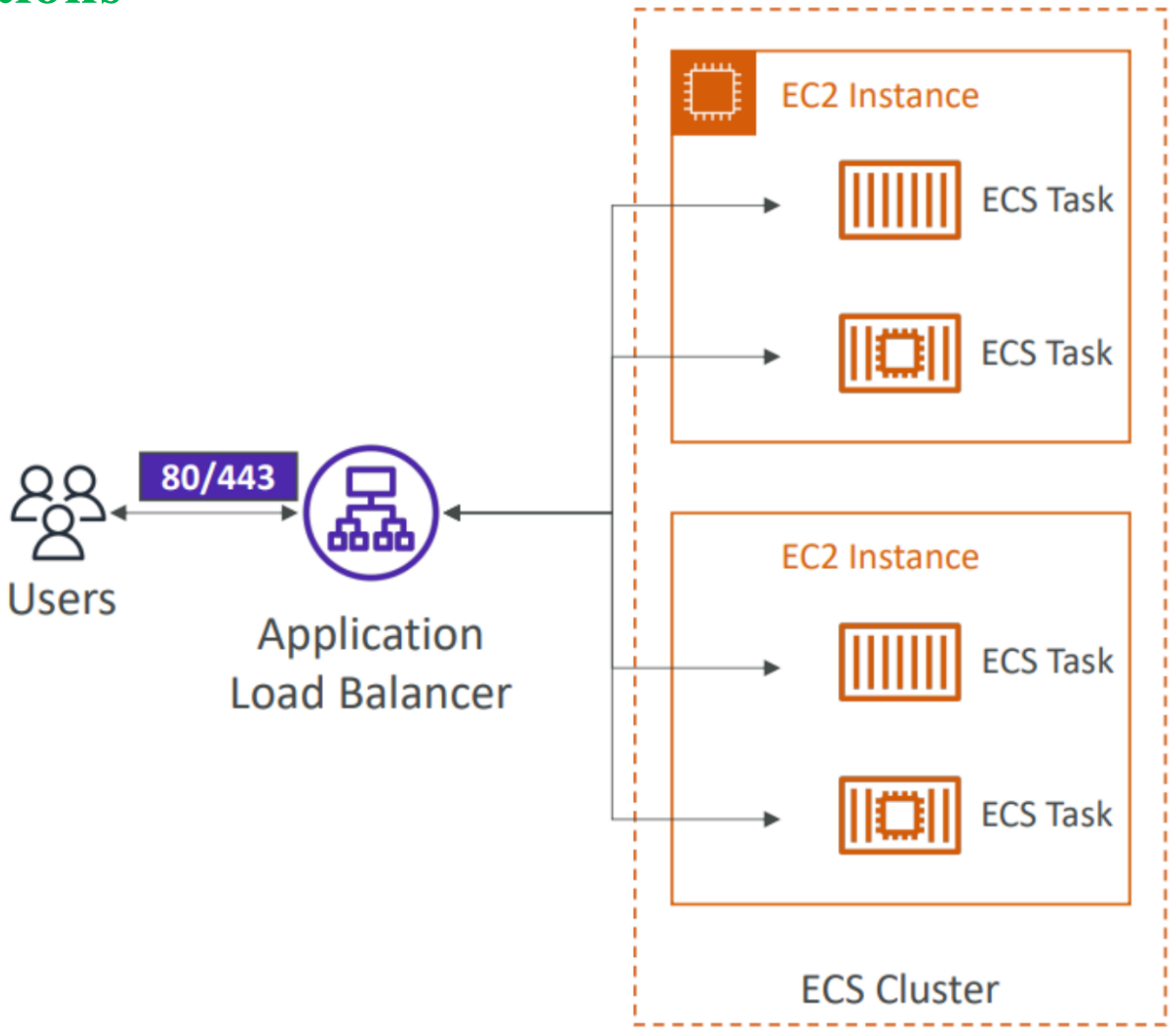
❖ Amazon ECS – Load Balancer Integrations

Application Load Balancer

- Recommended for most ECS workloads
- Supports path-based and host-based routing
- Works with dynamic port mapping
- Compatible with EC2 and Fargate launch types

Network Load Balancer

- Designed for high throughput and low latency
- Suitable for high-performance applications
- Often used with AWS PrivateLink



Amazon ECS

❖ Amazon ECS – Data Volumes (EFS)

- ❑ ECS tasks can mount Amazon EFS file systems as shared storage
- ❑ Works with both EC2 Launch Type and Fargate Launch Type

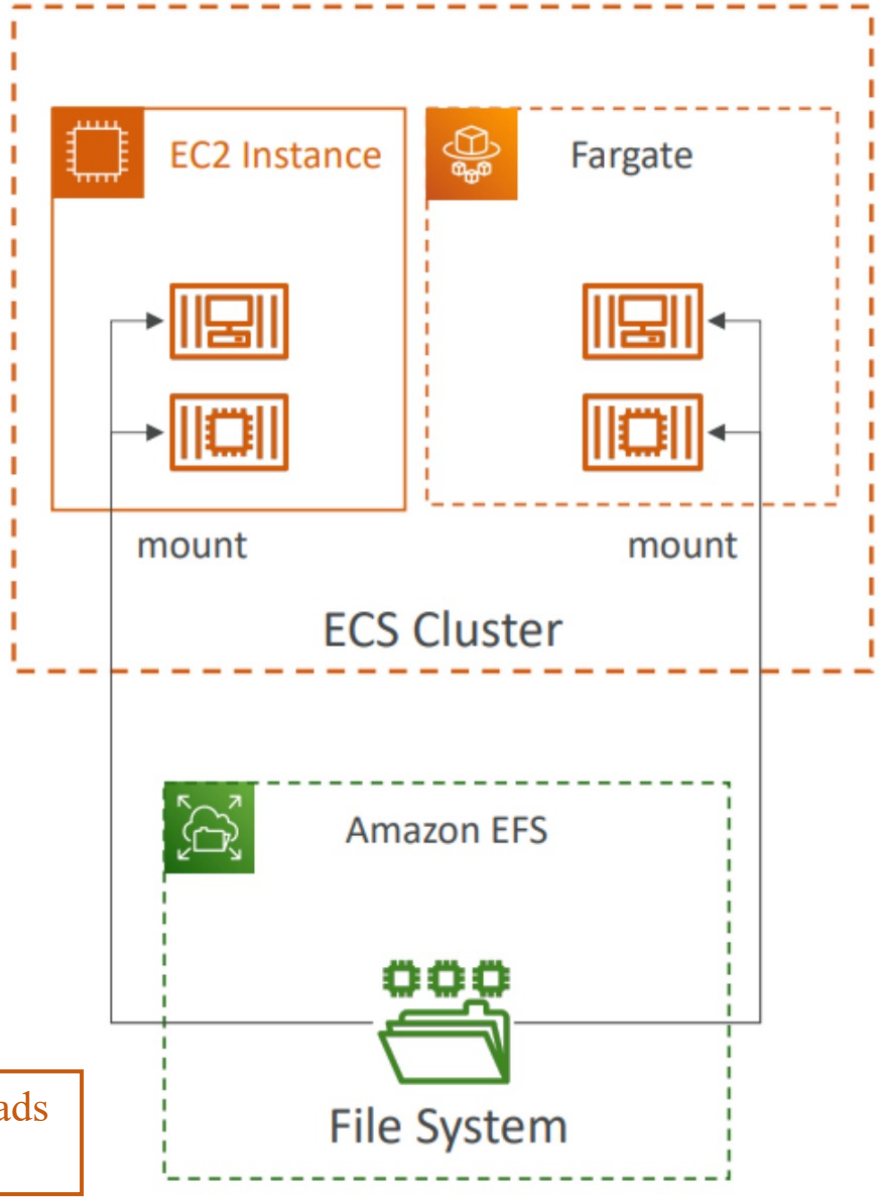
- Containers running in different Availability Zones can access the same shared data
- Provides persistent storage for containers

Use cases

- Shared storage across multiple containers
- Persistent data for containerized applications
- Multi-AZ storage for high availability

Serverless Storage

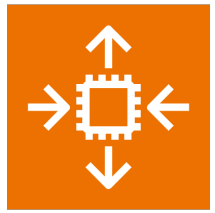
- Fargate + EFS enables fully serverless container workloads with persistent storage



Amazon ECS

❖ ECS Service Auto Scaling

- ❑ Automatically **increase or decrease the number of ECS tasks** based on workload.
- ❑ Uses **AWS Application Auto Scaling** to manage scaling policies.



AWS Auto Scaling Group

Common Scaling Metrics

- Average CPU Utilization
- Average Memory Utilization
- ALB Request Count per Target (metric from Application Load Balancer)

Important Note

ECS Service Auto Scaling scales the number of tasks EC2 Auto Scaling scales the number of EC2 instances

Scaling Policies

- Target Tracking
  - Maintain a specific target value for a CloudWatch metric.
- Step Scaling
  - Scale tasks when a CloudWatch Alarm threshold is reached.
- Scheduled Scaling
  - Adjust capacity based on specific times or predictable traffic patterns.

Ref: https://docs.aws.amazon.com/AmazonECS/latest/developerguide/service-auto-scaling.html

Amazon ECS

❖ EC2 Launch Type – Auto Scaling EC2 Instances

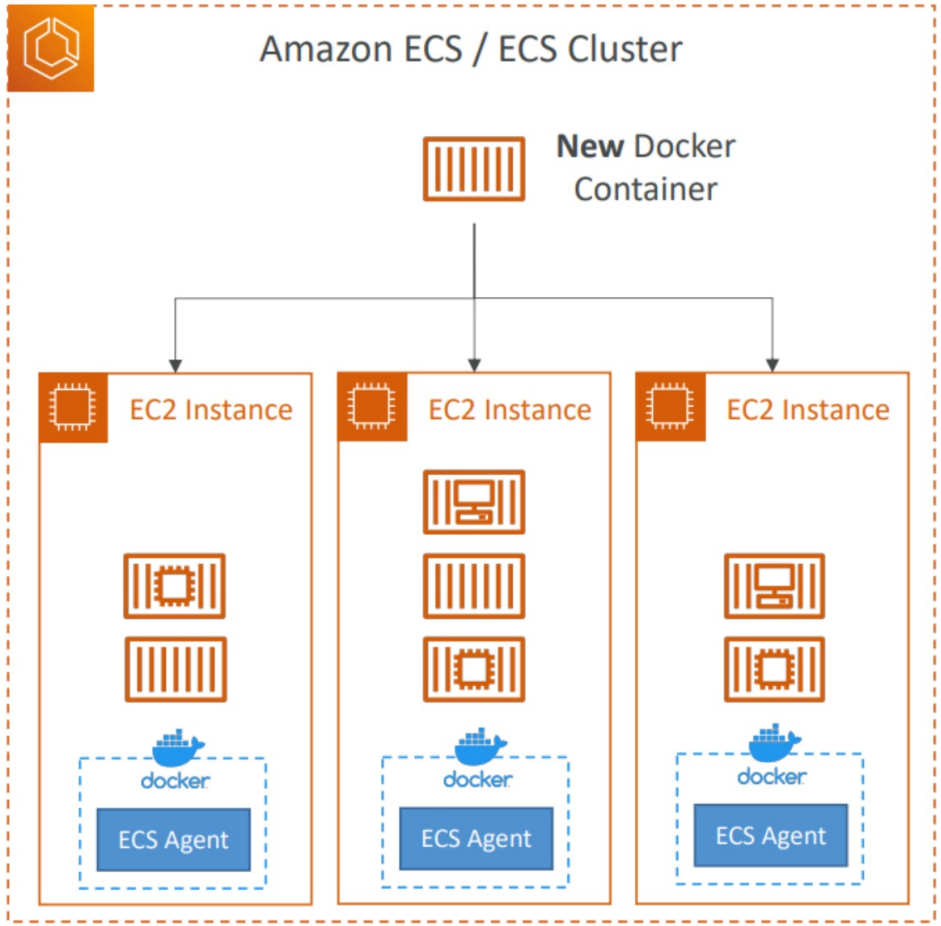
❑ Used to add EC2 instances automatically when ECS tasks require more capacity.

Auto Scaling Group (ASG)

- EC2 instances are managed using an Auto Scaling Group
- The ASG can scale based on metrics such as CPU utilization
- New EC2 instances are launched when capacity increases

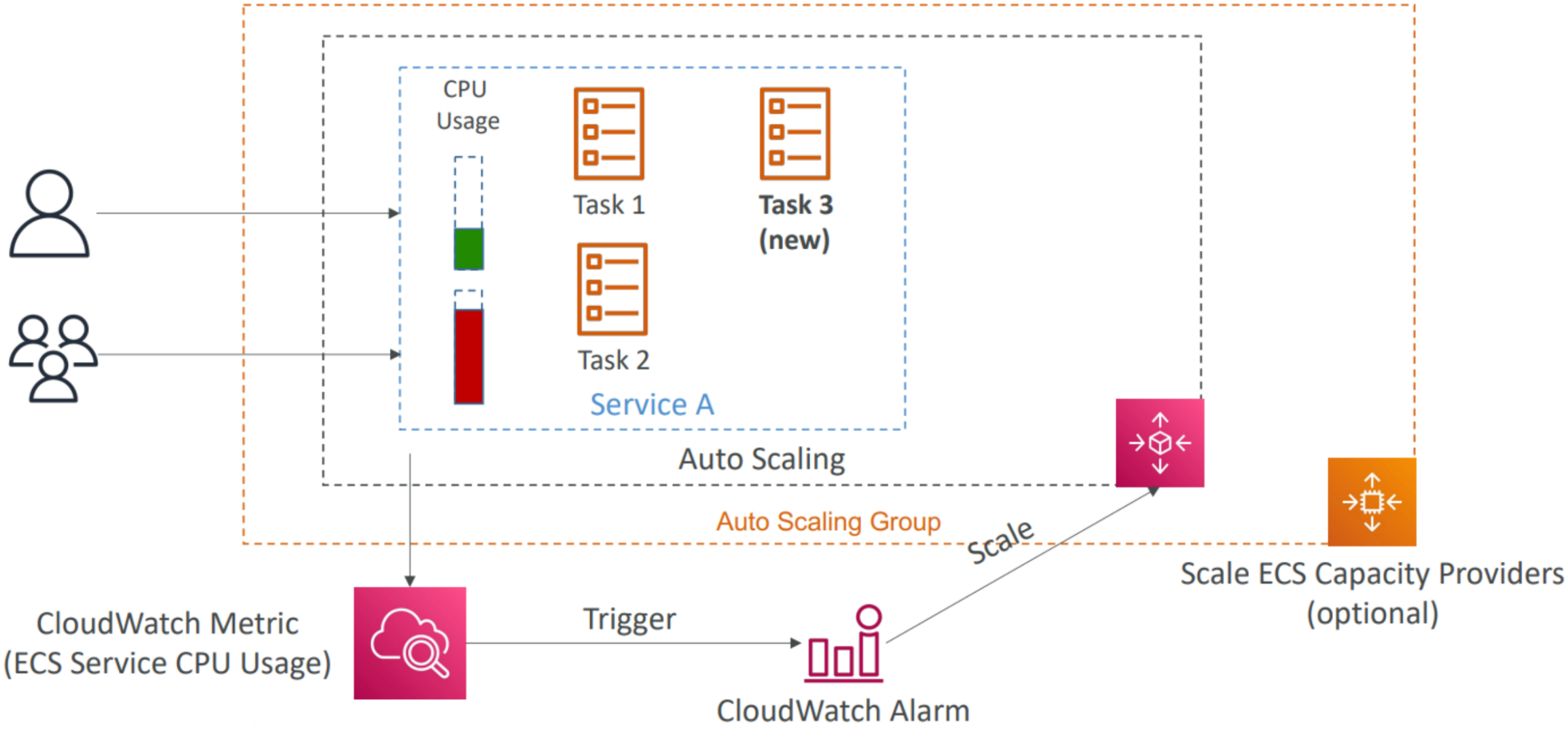
ECS Cluster Capacity Provider

- Automatically provisions and scales EC2 infrastructure for ECS tasks
- Works together with an Auto Scaling Group
- Launches additional EC2 instances when cluster resources (CPU, RAM) are insufficient



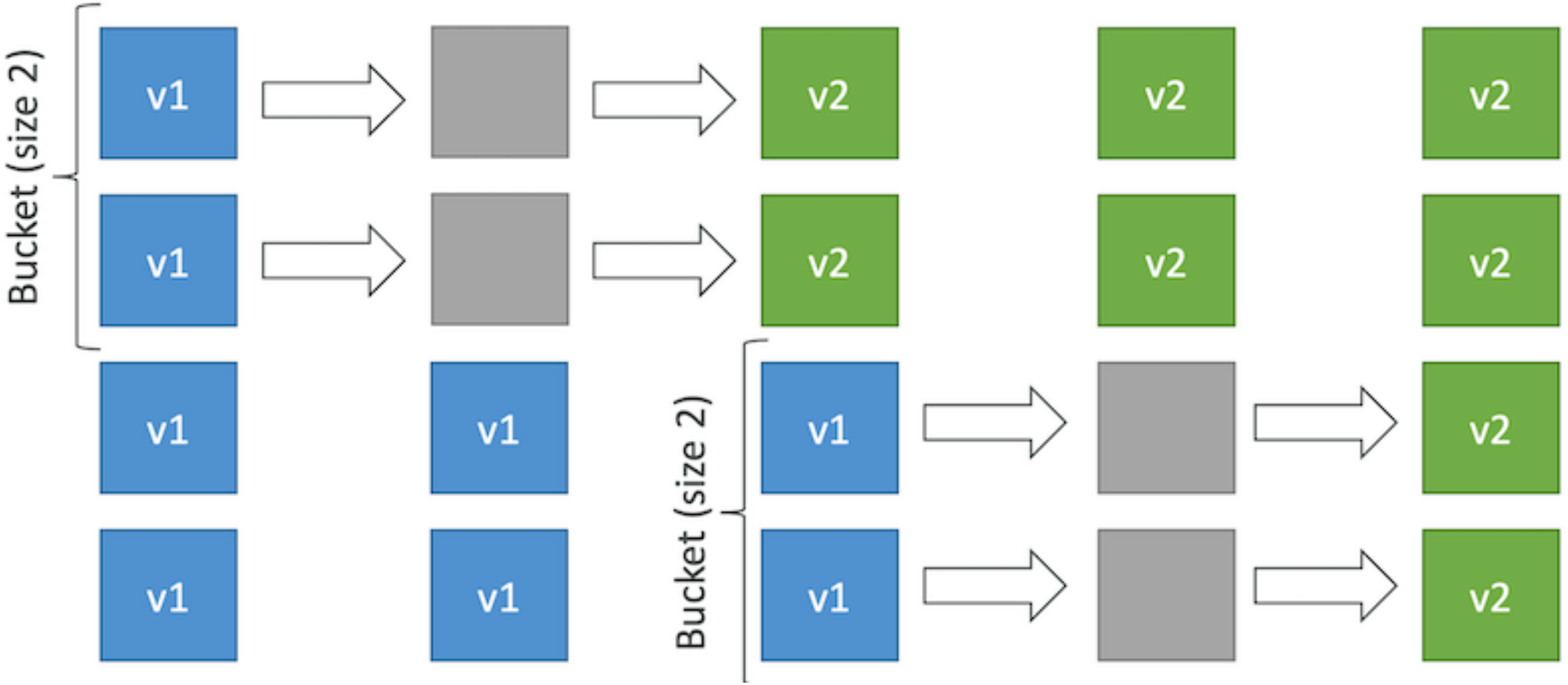
Amazon ECS

❖ ECS Scaling – Service CPU Usage Example



Amazon ECS

❖ ECS Rolling Updates



Ref: https://docs.aws.amazon.com/AmazonECS/latest/developerguide/deployment-type-ecs.html

Amazon ECS

❖ Amazon ECS – Load Balancing (EC2 Launch Type)

Dynamic Host Port Mapping

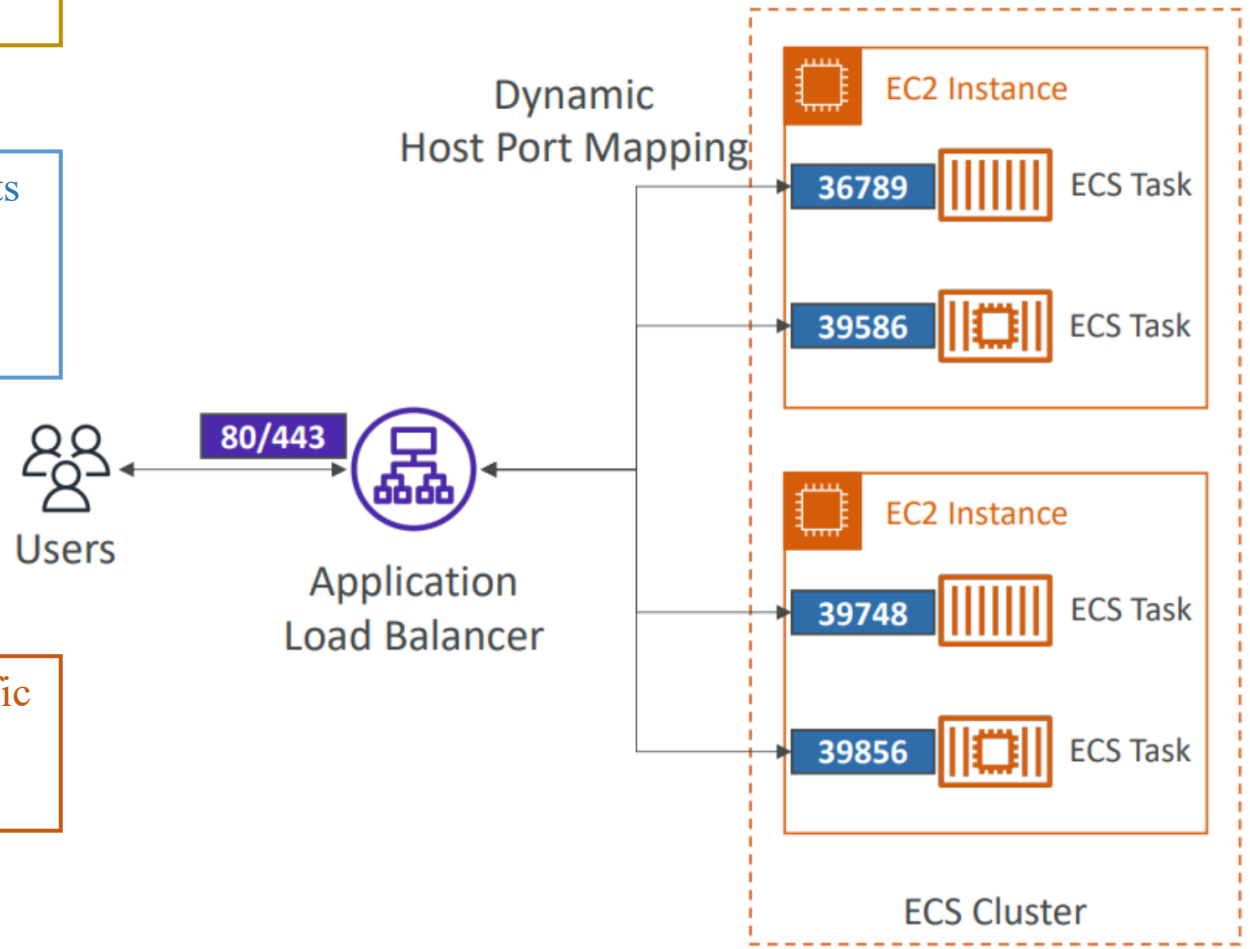
If only the container port is defined in the task definition, ECS assigns a dynamic host port on the EC2 instance.

Load Balancer Routing

- The Application Load Balancer (ALB) automatically detects the correct EC2 instance and port where the container is running.
- ALB routes traffic to the correct container port mapping.

Security Group Configuration

- The EC2 instance security group must allow incoming traffic from the ALB security group
- Allow all ports used by dynamic port mapping



Amazon ECS

❖ Amazon ECS – Load Balancing (Fargate)

Networking Model

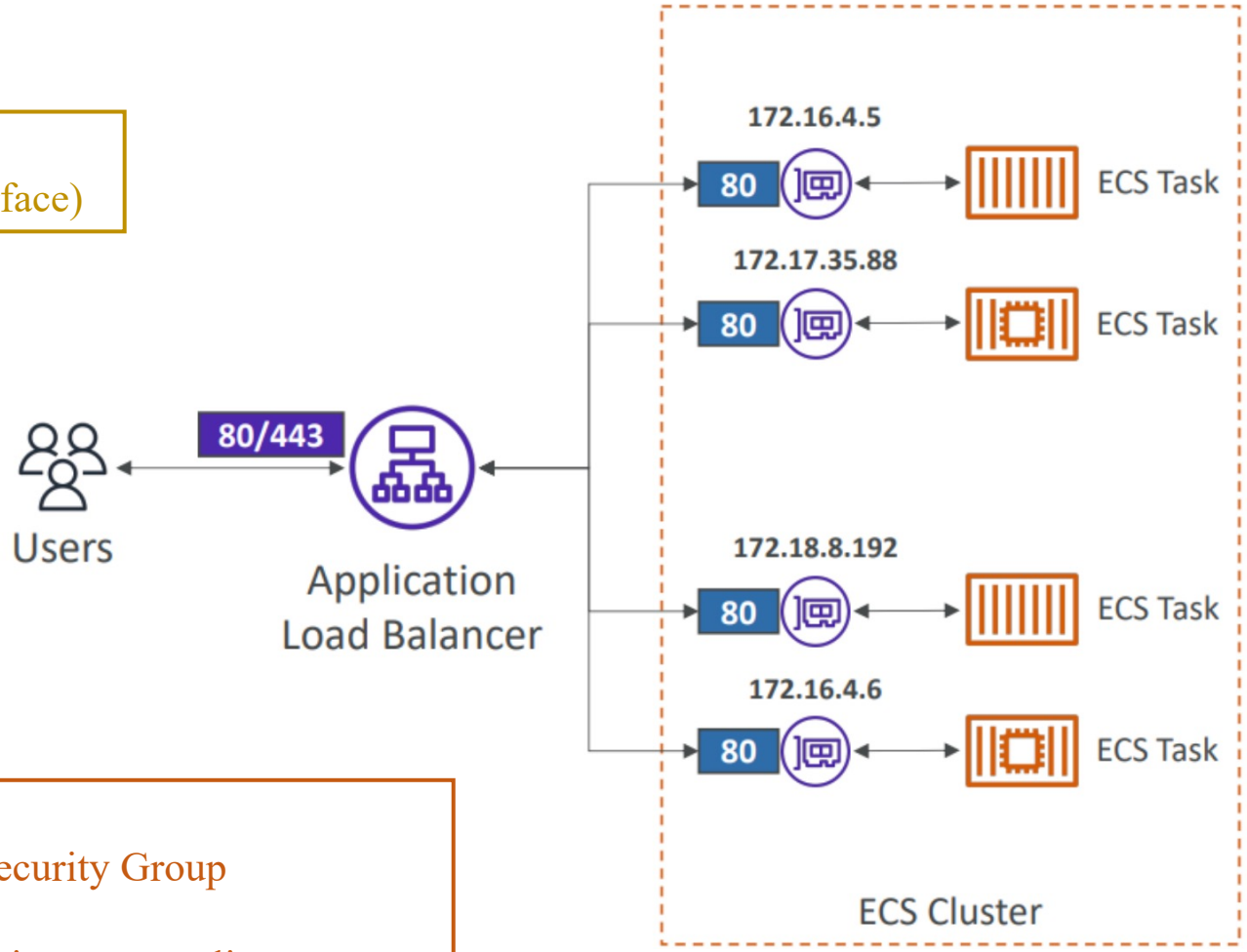
- Each Fargate task receives its own private IP address
- Uses AWS VPC networking (ENI – Elastic Network Interface)

Port Configuration

- Only the container port is defined
- Host port is not used because each task has its own network interface

Security Group Configuration

- ECS Task (ENI) Security Group
  - Allow container port (e.g., port 80) from the ALB Security Group
- ALB Security Group
  - Allow HTTP/HTTPS traffic (ports 80/443) from the internet or client sources



Amazon ECR

Amazon ECR

❖ Kinesis Data Streams

❑ ECR (Elastic Container Registry) is a service to store and manage Docker container images on AWS

Repository Types

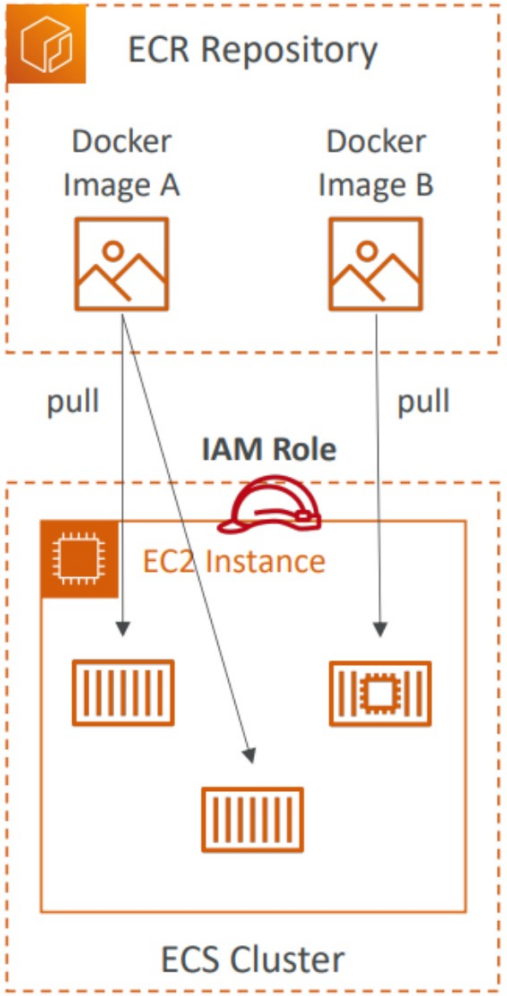
- Private repositories for internal use
- Public repositories via the Amazon ECR Public Gallery

Integration

- Fully integrated with Amazon ECS
- Container images are stored and backed by Amazon S3

Security & Access

- Access controlled through AWS IAM policies
- Permission issues usually relate to IAM policy configuration



Ref: https://aws.amazon.com/kinesis/

Amazon Kinesis

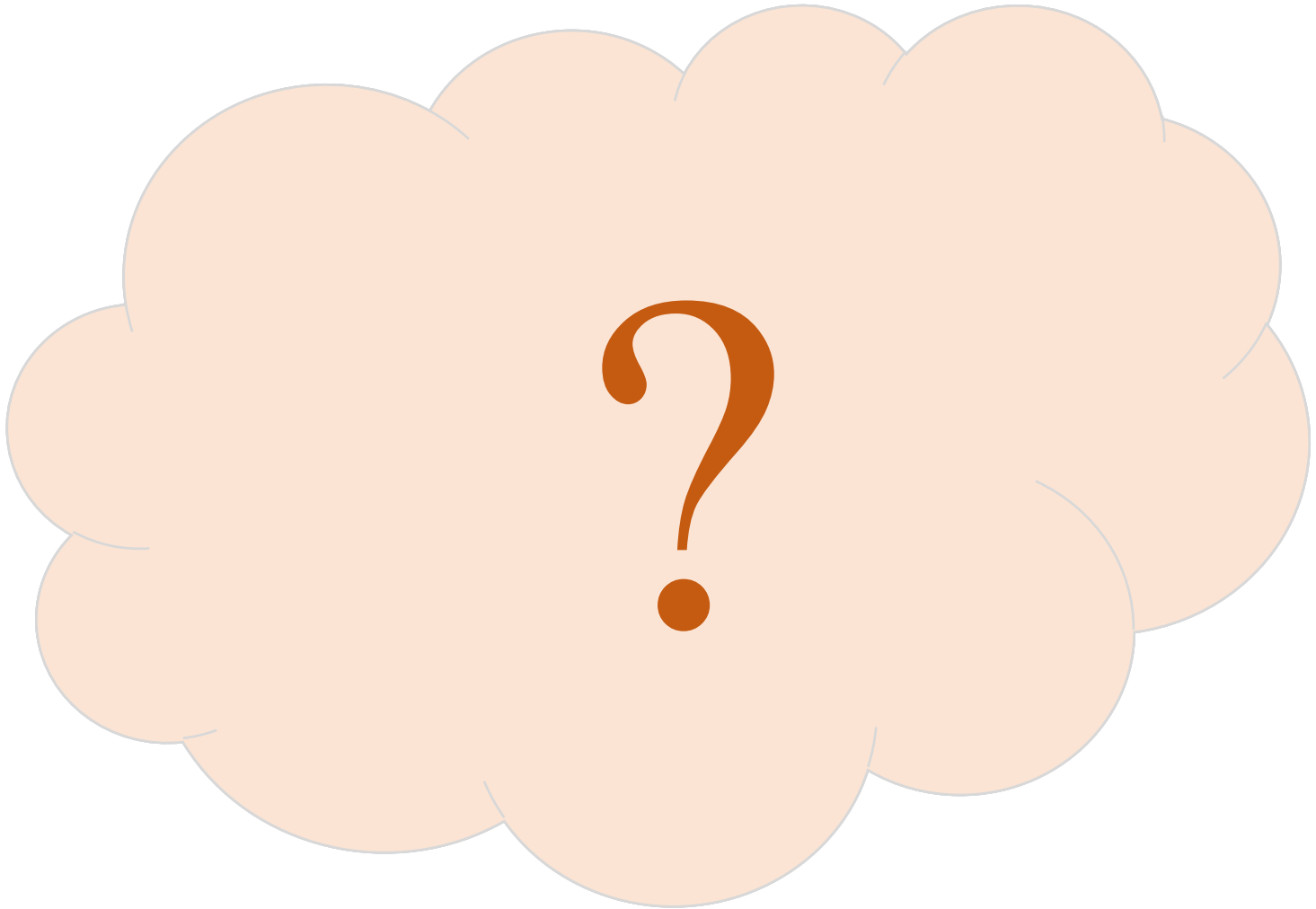
Amazon Kinesis

❖ Kinesis Data Streams

❑ Collect and store streaming data in real-time.

Ref: https://aws.amazon.com/kinesis/

Question



Thank you!